When Does Crime Enter House Prices? Evidence from Chicago

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Abstract

I estimate how crime affects housing prices in Chicago. Although homebuyers presumably care about crime, I find that most offenses have a near-zero effect on sale prices or rents. I argue that this may be due to limited information: crime only changes buyers' valuations if they learn of an incident and revise their expectations about future exposure. First, price effects are hyperlocal even though the predictive value of crimes is not. Second, price effects strengthen with real estate agents' local experience. Third, even within an offense, the effect depends on where at the property the incident occurred. These patterns are consistent with buyers often being unaware of incidents, and not responding when they are unaware.

JEL classification: R31, K42, D83, L85

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1 Introduction

A home is most homeowners' largest asset and an important consumption good. Neighborhood crime threatens both dimensions: it undermines the safety of the environment in which families live and may reduce the value of the asset they hold. In a 2025 Redfin survey, personal safety and a low crime rate were the two most cited non-negotiables among prospective homebuyers.[1]

This paper studies which crimes affect house prices, why some do and others do not, and how crime changes whether and how quickly homes sell.

I answer these questions using data from Chicago. I link 530,395 single-family, condominium, and residential-income transactions on 337,461 properties from the Chicago Multiple Listing Service, spanning 2007 to 2025, to 5.59 million geocoded crime incidents, analyzing twenty-two offense

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[1]A 2025 Redfin/Ipsos survey of 1,224 prospective homebuyers finds that personal safety (78%) and a low crime rate (74%) are the two most frequently cited "must-have" attributes among eighteen home and neighborhood features (Anderson, 2025).

categories. The listing data track properties from initial listing to sale or withdrawal, so I observe the asking price, the final sale price, time on market, and withdrawals. The rental listings provide asking rents and listing outcomes for leases.

My identification strategy compares the same property across sales. Parcel fixed effects absorb time-invariant differences across properties, and tract-by-month fixed effects absorb time-varying shocks shared by the surrounding neighborhood, such as local price trends. Within this comparison, I contrast crime within 0.05 miles of the property, roughly the same block, with crime 0.20 to 0.30 miles away in the same neighborhood; nonparametric distance profiles motivate these bands (Section 4). Moving the comparison ring out to 0.25–0.35 or 0.30–0.40 miles leaves the estimates nearly unchanged (Table A6). I estimate this design for all twenty-two offenses and assess it using post-sale crime placebos, which cannot causally affect a price already set.

Most offenses show no detectable sale-price discount. Only four of the twenty-two carry discounts across multiple exposure windows (homicide, burglary, narcotics, and vandalism), and two more are statistically significant at a single long-horizon window (disorderly conduct and sex offenses). One of the rarest offenses carries the largest discount, about three percent per incident (homicide), while several common or widely feared offenses show no detectable discount (theft, motor-vehicle theft, assault, and robbery). An offense can therefore be severe, frequent, or widely feared and still show no detectable discount.

Why, then, are effects confined to particular offenses, when buyers are presumably averse to crime in general? A simple framework helps us think about this question. For an incident to affect the house price, three conditions must hold: (a) buyers must learn of the incident; (b) learning of it must raise their expectation of future crime at the property; and (c) buyers must be willing to pay to avert that crime at the property. The first condition concerns whether information arrives, the condition Gibbons (2004) conjectures explains why hard-to-observe crimes have weak price effects; the second, what it implies about future crime; and the third, what avoiding that crime is worth: a buyer without a car who observes a car stolen from the curb may expect more car thefts on the block and still pay no less for the house.

Three patterns inform this interpretation. The first concerns how buyers use crime information. Price effects are hyperlocal, whereas the predictive content of crime extends beyond the immediate vicinity of the property. Crime a few blocks away predicts future crime near the property for nearly every offense, yet it does not lower prices. Nor do discounts rise with how well an offense predicts its own recurrence: the most predictable offense carries a small discount (narcotics), and an offense more predictable than burglary carries none (non-domestic assault). The second concerns observability. For the same offense, incidents at a home's frontage lower prices, while incidents inside the dwelling do not (burglary and narcotics; Table 3). Discounts are significant for incidents on the street but not at residential addresses, although the two estimates cannot be statistically distinguished (homicide and narcotics; Table A7).[2] Third, real estate agents may help this information

[2]Homicide is also the most heavily reported offense: it receives roughly fifteen times as many news stories per incident as the next-highest offense in Media Cloud and the largest article volume in GDELT (Section 7). Incidents in public space are more widely observed, although I do not measure coverage by incident location.

travel: the discount is larger when the buyer's agent has more experience in the neighborhood, even after controlling for total experience (homicide; Tables A18 and A19). Agents are not randomly assigned to transactions, so this relationship is descriptive.

The transaction results show that sale-price discounts generally emerge after listing rather than in the initial asking price. Asking prices respond little on average, while closing prices show discounts. Other margins vary by offense: some offenses also lower the sale-to-list ratio (narcotics); there is some evidence that some lengthen time on market, by roughly 12 to 17 percent, or about two to three days against a median of 16 in the estimation sample, under parcel-by-era fixed effects (homicide); and some increase withdrawals despite showing no consistent closing-price discount (weapons violations). For ten of the twenty-two offenses, including all four that are priced, nearby crime raises the probability that a home is listed more than the probability that it sells. Crime therefore affects whether and how quickly a home sells as well as its price, so closing prices alone give an incomplete account of the market's response.

The estimates are robust to time-varying block characteristics such as building permits, vacant buildings, business licenses, tree canopy, and total 311 service requests. Inference is also robust to Conley spatial standard errors (Conley, 1999).

This paper makes three contributions.

First, I show that house prices reflect crime selectively, estimating offense-specific price responses to nearby incidents across twenty-two categories under a common empirical design. Earlier work measures either many offenses at aggregate geographies, where estimates depend on the spatial unit (the modifiable areal unit problem, MAUP) and cannot separate crime on a property's own block from broader neighborhood conditions (Thaler, 1978; Schwartz et al., 2003; Gibbons, 2004; Ihlanfeldt and Mayock, 2010), or a single crime at point resolution (Linden and Rockoff, 2008; Pope, 2008a; Congdon-Hohman, 2013; Tang and Le, 2023; Bhattacharya et al., 2021). Braakmann (2017) studies six offense categories at the street level using street fixed effects; I extend this to twenty-two offenses at the incident level with repeat sales.[3]

Second, I provide evidence on the role of observability by comparing where incidents of the same offense occur. These comparisons test implications of Gibbons's (2004) conjecture that hard-to-observe crimes have weak price effects because buyers rarely learn of them. To my knowledge, this is the first test of that conjecture that holds the offense fixed and compares where incidents occur (Table 3). Moreno-Medina (2026) finds that news coverage of nearby homicides lowers house prices, identifying one channel through which buyers learn about crime. I examine a complementary dimension of observability: the location of an incident within a given offense category.

Third, I show that crime affects whether and how quickly a home sells, not only its closing price, by tracing each stage of the transaction. I examine asking prices, time on market, withdrawal, and the sale-to-list price ratio, allowing me to distinguish whether crime is incorporated into the seller's initial asking price or emerges through subsequent market adjustment (Table 4). Boggess et al. (2013) find that higher crime is followed by higher housing turnover, Wentland et al. (2014) estimate

[3]Table A5 summarizes prior estimates.

price and time-on-market effects of registered sex offenders from MLS data, and Dentler and Rossi (2024) show that the density of sales changes within 30 to 60 days of an incident. I extend this analysis to each stage of the transaction, holding the property fixed. Crime can affect whether a sale proceeds, not only its price: weapons violations increase withdrawals despite showing no consistent closing-price discount, and homicide-related closing discounts are larger when the buyer's agent has more local experience (Table 5).

I also extend the design to rental listings, a market studied far less in this literature. Most rent studies use aggregate crime rates (Rizzo, 1979; Koirala et al., 2023; Donovan et al., 2024), and property-level studies examine one or two offenses (Carrillo and Yezer, 2021; Klimova and Lee, 2014). I estimate effects of all offenses on asking rents, time on market, withdrawal, and relisting under the same design as the sale estimates (Table A22).

Economists often use house prices to infer how residents value neighborhood safety (Thaler, 1978; Gibbons, 2004; Linden and Rockoff, 2008). Whether prices reflect nearby crime therefore matters for policy. If some crime-related costs are not reflected in prices, valuations based on housing markets may miss part of the benefits of crime reduction: the absence of a price discount does not imply the absence of a cost. Whether these costs are reflected in prices depends in part on whether buyers learn of nearby crime.

Section 2 develops the conceptual framework, Section 3 describes the data, and Section 4 presents the empirical strategy. Section 5 reports the results, Section 6 assesses robustness, Section 7 discusses interpretation and implications, and Section 8 concludes.

2 Conceptual framework

The observed crime discount is the price consequence of a change in expected crime exposure. To make this explicit, suppose first that a household values occupancy at a property over a fixed horizon (thereby allowing us to ignore resale value). Let q_k denote the buyer's subjective expected number of future incidents of offense k near the property over that horizon, and let w_k denote willingness to pay to avoid one expected incident, expressed as a fraction of the property's value before the incident. Expected crime exposure therefore carries value $w_k q_k$.

Buyers enter the market with prior beliefs about q_k, formed from information already available about the location. A newly realized incident can affect valuation through two additional margins. First, information about the incident must reach the relevant market participant. Let $S_k \in [0, 1]$ denote the probability of information arrival: $S_k = 0$ means that the participant never learns of the incident, while $S_k = 1$ means that information about it reaches the participant with certainty. Second, conditional on information arrival, the incident may change the participant's subjective expectation of future exposure. Define

$$b_k(S_k) \equiv \frac{\partial q_k}{\partial C_k^o}, \tag{1}$$

where C_k denotes realized incidents of offense k and C_k^o those whose information reaches the participant. Thus $b_k(S_k)$ measures the change in expected future incidents generated by one additional observed incident. Writing b_k as a function of S_k allows the update per observed incident to differ across offenses with how often such incidents are learned of; if buyers know they rarely observe a type of crime, this may affect their update based on a single observation. Housing markets price what buyers can observe: asymmetric information about a property's condition and its surroundings shows up in prices and in who buys (Garmaise and Moskowitz, 2004; Kurlat and Stroebel, 2015; Stroebel, 2016), and disclosure of a hazard moves prices only once buyers learn of it (Pope, 2008b; Bernstein et al., 2019; Hino and Burke, 2021; Bakkensen and Barrage, 2022; Gourevitch et al., 2023). In the crime setting, Gibbons (2004) argues that visible offenses such as vandalism are priced because they are hard to conceal from buyers, while information about other crimes stays with sellers and neighbors. Where crime information is put in front of buyers, prices respond: house prices fall when a sex offender's address enters a public registry (Linden and Rockoff, 2008) and rebound when the offender moves away (Pope, 2008a), fall near registered gun offenders without any rise in crime (Tang and Le, 2023), and carry large, persistent discounts where deaths in a home must be disclosed (Bhattacharya et al., 2021). Most crime, by contrast, is publicly recorded but not shown where buyers search for homes. The availability of information does not ensure that it enters a decision. Chetty et al. (2009) show that consumers respond more strongly to taxes included in posted prices, even when they know that additional taxes apply. Bordalo et al. (2012, 2013) show how salient payoffs and attributes receive disproportionate weight in choice. These mechanisms motivate distinguishing information arrival from the response to information, but salience need not operate through belief updating alone. The comparisons below do not separately identify information arrival, belief updating, and valuation.

Unlike S_k, $b_k(S_k)$ is not bounded between zero and one. If $b_k(S_k) = 0$, the participant learns about the incident but does not revise expected future exposure. A value between zero and one means that one additional realized incident raises expected future exposure by less than one incident; a value of one implies a one-for-one increase; and a value greater than one means that one realized incident leads the participant to expect more than one additional future incident over the relevant horizon. Importantly, this is a subjective belief update, not the objective recurrence rate of the offense.

Following Rosen (1974), define the per-incident discount as the implicit price of an incident: the proportional change in the value of the property, P, per additional realized incident of offense k nearby, C_k,

$$\kappa_k \equiv -\frac{\partial \ln P}{\partial C_k}. \tag{2}$$

The update concerns counts, not value. A buyer can expect more incidents and attach no value to avoiding them, so $b_k > 0$ with $w_k = 0$ and $\kappa_k = 0$. This is why relevance and updating enter as separate terms.

This is the capitalization effect in the terminology of Kuminoff and Pope (2014).[4] The frame-

[4] κ_k is the marginal effect of one additional incident at the property's current exposure. The linear specification

work assumes that it equals the product of three terms,

$$\kappa_k = \underbrace{w_k}_{\text{WTP to avoid one expected incident}} \times \underbrace{S_k}_{\text{information arrival}} \times \underbrace{b_k(S_k)}_{\text{expected incidents per observed incident}}, \quad (3)$$

where w_k captures the location-specific value of avoiding expected exposure to offense k. An offense may impose substantial costs on its victims yet carry little location-specific valuation if those costs are not associated with occupying or owning property at that location.

Written in units, w_k is a fractional price change per expected future incident, S_k is the share of realized incidents that reach the buyer, and $b_k(S_k)$ is expected future incidents per observed incident, so the three terms multiply to the log-price discount per realized incident estimated empirically:

$$\underbrace{\frac{\Delta \ln P}{\text{expected incident}}}_{w_k} \times \underbrace{\frac{\text{incidents observed}}{\text{incidents realized}}}_{S_k} \times \underbrace{\frac{\text{expected incidents}}{\text{incidents observed}}}_{b_k(S_k)} = \frac{\Delta \ln P}{\text{realized incident}}. \quad (4)$$

Consider a buyer who would pay 1 percent of a property's value to avoid one expected future burglary, so $w_k = 0.01$. If one realized burglary reaches the buyer ($S_k = 1$) and raises expected future burglaries over the horizon from 0.5 to 2.5, the belief update is $b_k(S_k) = 2$ and, since the incident is observed ($S_k = 1$), the implied discount is $\kappa_k = w_k\, b_k(S_k) = 0.01 \times 2 = 0.02$, or 2 percent per realized burglary. The framework expresses κ_k as a positive magnitude. The estimate δ_k in the empirical sections is a signed price change, so a 2 percent discount appears as $\delta_k = -2$. A value of $b_k(S_k)$ above one simply means one realized burglary leads the buyer to expect more than one additional future burglary; it is not a probability. Had the incident not reached the buyer, or reached the buyer without changing expected exposure, this channel would generate no price response.

A zero observed discount can therefore arise through three distinct channels: willingness to pay to reduce expected exposure may be zero ($w_k = 0$); information about the realized incident may not reach the participant ($S_k = 0$); or the incident may be fully observed but leave expected future exposure unchanged ($b_k(S_k) = 0$). The last case is particularly important. A buyer may place a high value on avoiding future homicide and learn that a homicide occurred nearby, yet view the incident as isolated and therefore not revise the expected risk of future homicide. The price response is then zero despite high willingness to pay and full information. Thus, full information ensures that the incident is known, but not that it changes beliefs about future exposure; a zero capitalization estimate therefore need not imply low willingness to pay.

Prediction 1: Observability. An incident is capitalized only if it reaches the buyer ($S_k > 0$). Holding the offense fixed, incidents that are more likely to be observed should carry larger discounts

in Section 4 estimates a regression-weighted average of this effect across counts, with weights set by the residual variation in exposure; Table OA6 reports how it varies with the count.

than incidents that go unobserved.[5] I test this by splitting incidents of the same offense by where they occurred and whether they are visible from outside the dwelling, and, for vandalism at the frontage, by whether the damage leaves a trace at the property.

Prediction 2: Recurrence. The belief update $b_k(S_k)$ is subjective and is not observed directly. I therefore ask whether capitalization tracks how well each offense predicts future crime. Holding valuation and information arrival fixed, stronger predictive content should generate a larger discount; across offenses these other components may differ, so recurrence alone need not determine the ranking of discounts. I use a forecasting regression of whether current crime predicts recurrence to measure what an incident objectively implies about future crime, not what buyers believe it implies. If discounts do not rise with recurrence, recurrence alone does not determine which offenses are priced.

Prediction 3: Price Formation. The framework describes a change in buyers' valuation but does not say where in the transaction that change appears. If sellers anticipate the effect on buyers' willingness to pay, asking prices should adjust. If they do not, the valuation change can still appear after listing: in a lower sale-to-list ratio, or in non-price margins such as a longer time on the market or a greater chance the listing is withdrawn. I use these margins to trace where capitalization enters the transaction.

3 Data

The analysis combines three sources: residential listings from the Chicago Multiple Listing Service, incident-level crime reports from the Chicago Police Department, and measures of neighborhood conditions matched to each parcel. This section describes each source and the construction of the estimation sample.

Chicago is well suited for this analysis because it provides variation along both dimensions the design requires. First, the volume and breadth of crime incidents allow effects to be estimated separately across the 22 offense categories reported here: the data contain 10,766 homicides alongside more than 1.3 million thefts over the sample period. Second, crime varies substantially at fine spatial scales: roughly 55 percent of the variation in crime exposure within 0.05 miles occurs within tract-months, against 14 percent for crime in the surrounding 0.20–0.30 mile ring. Near-ring crime therefore provides substantially more within-neighborhood variation than crime in the surrounding ring.

Housing transactions. The housing data are the universe of Chicago Multiple Listing Service (MLS) closed transactions in single-family, condominium, and residential-income properties. I

[5] Interpreting this difference as evidence of observability assumes that interior and frontage burglaries signal similar future risks and that buyers value avoiding them similarly. The most evident difference between them works against the result rather than producing it: an intrusion into the dwelling is more harmful, so buyers who know of it should demand a larger discount, yet interior burglaries carry no detectable discount.

keep transactions closing between 2007 and 2025 with a recorded close price between $10,000 and $10,000,000, and parcels whose recorded parcel count is the same at every sale, which excludes properties that were split or consolidated between sales. The resulting sample is 530,395 sales on 337,461 parcels, of which 136,866 parcels sell at least twice (329,800 sales) and so identify the within-property comparison. Single-family homes make up 88 percent of these repeat sales, two-to-four-unit buildings 11 percent, and condominiums 1 percent. Table OA1 reports the construction. Each record carries the property identifier, close date, and close price and the full listing history: original and revised list prices, days on market, and whether the listing ended in sale or withdrawal. The sale-price sample consists of closed transactions; the withdrawal analysis instead uses all MLS listings, whether or not they ended in a sale. That history lets me trace how crime affects asking prices, price revisions, time on market, and eventual transaction outcomes.

For the quantity margin, I build an annual panel of all Chicago dwelling parcels in the Cook County Assessor file, excluding vacant land, garages, and minor improvements, 2007–2025. The *listing rate* is the share of parcels listed in the MLS in a given year, and the *turnover rate* is the share that sell. At the parcel level, each is an indicator for whether the parcel is listed, or sells, that year; their sample means are 6.0 and 3.8 percent. Near- and far-ring crime counts are measured over the same calendar year.

Lease listings. The rental analysis uses MLS lease listings subject to the same property-type, location, and parcel-count filters as the sale sample: single-family, condominium, and residential-income units in Chicago, on parcels with a constant parcel count, listed between 2007 and 2025 with monthly asking rents of $500 to $10,000. These restrictions yield 191,155 lease listings. Only 0.1 percent record a closing rent, so the rental price outcome is the asking rent, with crime exposure measured relative to the list date. The design otherwise mirrors the sale analysis, including days on market and withdrawal. Applying the same near–far comparison to leases allows me to estimate how nearby crime affects asking rents.

Property and listing characteristics. Table A1 summarizes both samples. The median sale property has three bedrooms and two bathrooms and closes at $279,000; the median lease is a smaller, two-bedroom unit with a monthly asking rent of $1,950. Physical characteristics (living area, bedrooms, bathrooms, and year built) come from standardized listing fields and are available for the large majority of transactions. Two-thirds of sale listings record at least one asking-price reduction before sale.

Crime incidents. Incident-level crime data come from the Chicago Police Department's public "Crimes, 2001 to present" file. Over 2007–2025, 5,657,404 offenses were reported citywide, of which 98.8 percent carry coordinates for the location of the incident; coverage varies by offense, from 78.8 percent for embezzlement to above 99 percent for most offenses (Table 1). Panel B of Figure 1 maps their density across the city, alongside residential sales in Panel A. Incident locations are released at block-face resolution: the published point is shifted from the actual location for partial

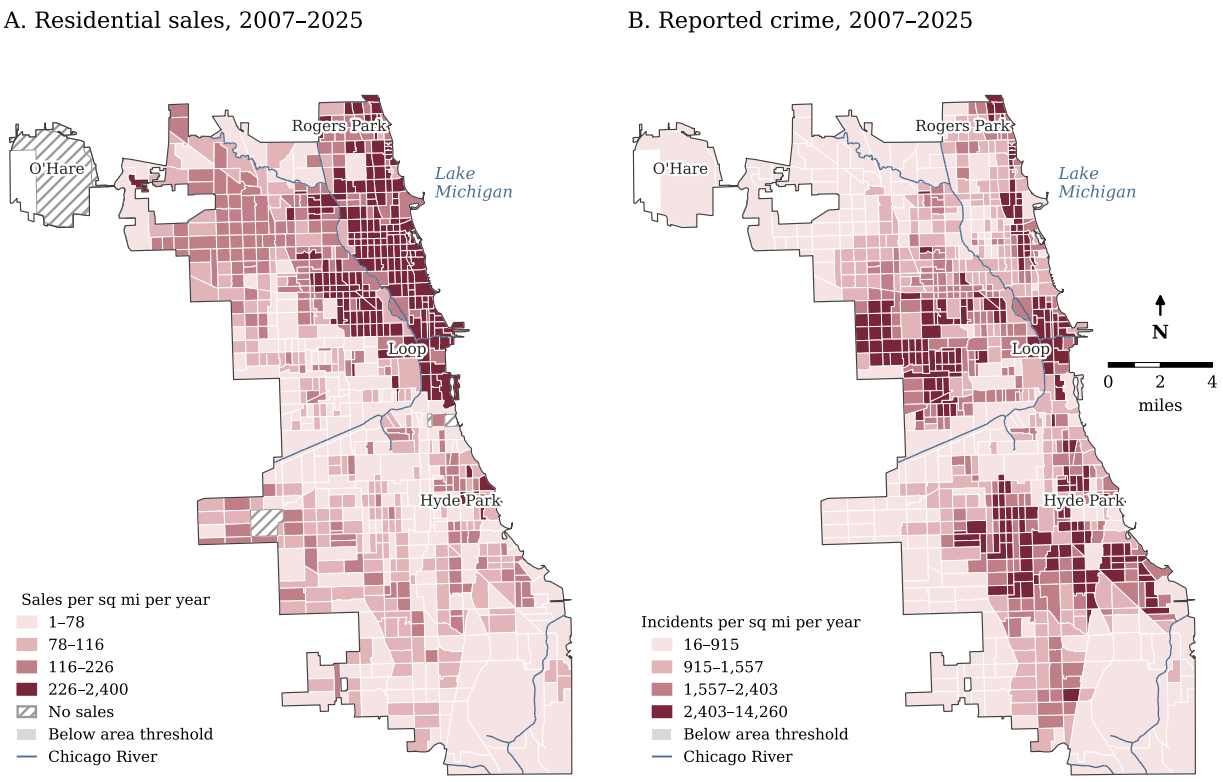


Figure 1: Residential sales and reported crime, Chicago, 2007–2025.

Notes. Census-tract densities are expressed as annual averages over 2007–2025 per square mile of land within the city boundary. Panel A shows closed MLS sales in the analysis sample (529,449 of 530,395 sales counted; 30 lack coordinates and 916 lie outside the city boundary); it uses the full sales data rather than the repeat-sales estimation sample. Panel B shows all reported Chicago Police Department incidents with coordinates inside the city boundary (5,575,987 of 5,657,404). Shading denotes quartiles of tract density, calculated separately for each panel; hatched tracts have no sales. Tract portions smaller than 0.01 square miles are shown in grey and excluded from the density classification. Tracts are clipped to the city's land boundary.

redaction but falls on the same block, so distance between a parcel and an incident is measured with error of up to a block face. For homicide the file records one row per victim rather than per incident, so homicide counts, and the per-incident homicide estimates below, are per victim record.[6]

I report estimates for twenty-two offenses, each at every window in Table A3; Appendix A.1 describes the remaining offense groups, which are too sparse or too imprecise to be informative. Assault combines the department's assault codes (threats without physical contact) and battery codes (physical contact), and is split into domestic and non-domestic incidents using the department's

[6]To protect victim privacy, the Chicago Police Department releases each incident's address "at the block level only" and does not identify the specific location (City of Chicago Data Portal, *Crimes – 2001 to Present*); the recorded point is placed on the correct block, so it can differ from the true location by up to a block face. Chicago blocks are about 330 feet east–west and 660 feet north–south, against a near-ring radius of about 264 feet, so the shift can move an incident into or out of the near ring. Location masking can assign incidents to the wrong distance ring, introducing measurement error into the crime counts. The direction of the resulting bias is not determined a priori.

domestic flag.

For each listing I count incidents of each offense type within concentric rings of the listing's location over pre-transaction windows of one to twelve months, matching incidents to properties by distance,[7] and time the counts to the listing's close date (to its list date on the lease side). Chicago's largely regular street grid makes straight-line distance a useful approximation to local proximity, although it does not capture barriers or differences in street connectivity.[8] Table 1 reports, for each offense, its citywide count, geocoding rate, and near exposure, together with the raw cross-sectional price gap between listings with and without a nearby incident. These raw gaps are large, exceeding 40 percent for homicide and narcotics, and reflect differences between blocks with and without nearby crime rather than the effect of crime itself. Listings with an incident within 0.05 miles in the three or six months before sale are also cheaper, smaller, and slower to sell (Table OA2).

Neighborhood conditions. To guard against the discount proxying for physical decline or disinvestment rather than crime, I assemble five predetermined measures of neighborhood condition at the parcel: tree canopy, active business licenses, 311 service requests, vacant buildings, and building permits.[9] Each enters with the same near-ring and far-ring structure as crime, differenced within parcel and tract×month, but as a cumulative stock of events recorded more than twelve months before the close, that is, before every crime-exposure window used, so it is predetermined relative to the crime and cannot be an outcome of it. Permits, licenses, 311 requests, and vacant buildings are time-varying, counted as of that pre-crime cutoff for each listing; tree canopy is measured from the land-cover raster two years before the sale. Foreclosures lower nearby prices and raise nearby crime (Campbell et al., 2011; Ellen et al., 2013; Anenberg and Kung, 2014), so I control for the predetermined vacancy stock near and far. I also construct five lagged measures of local housing-market conditions from the MLS: the mean repeat-sale price change of neighboring homes, the share of neighboring sales that were distressed (bank-owned real estate, short sale, or foreclosure), the mean sale-to-list ratio, mean days on market, and the number of new sale listings. Each is measured in the near and far rings over the 12 to 24 months before the close, before every crime-exposure window, and excludes the sale's own building; a ring with no neighboring sale receives a missing-value indicator, so the sample is unchanged.

[7] Distances are great-circle distances from each listing's MLS coordinates.

[8] Distance-based rings do not account for physical barriers such as expressways, rail lines, or rivers. Across-barrier incidents constitute approximately 1–2% of near-ring exposure for the four priced offenses. I do not separately estimate their effects.

[9] Tree canopy is from the U.S. Geological Survey National Land Cover Database; business licenses, 311 service requests, building permits, and vacant-building reports are from the City of Chicago Data Portal. Appendix A.2 describes their construction.

Table 1: Crime Incidence and Near Exposure

Offense	Incidents (city)	Geocoded (%)	Mean near count	% any near	Raw price gap (%)
Theft	1,364,912	97.8	5.56	83.2	13.7
Assault (non-domestic)	739,961	99.7	1.92	56.3	-20.9
Assault (domestic)	670,887	99.7	1.71	55.5	-35.6
Criminal damage/vandalism	632,166	99.6	1.88	67.1	-18.2
Narcotics	439,526	97.9	0.83	25.3	-45.5
Other offenses	360,851	99.0	1.09	49.5	-24.1
Burglary	296,724	99.6	0.97	47.1	-15.5
Motor vehicle theft	292,290	99.6	0.87	46.5	-9.9
Robbery	213,364	99.8	0.55	32.3	-12.2
Trespass	138,862	99.7	0.41	23.8	-1.3
Fraud	136,957	97.5	0.71	33.4	18.0
Weapons	114,818	99.6	0.23	14.4	-36.1
Sex offenses	63,549	89.2	0.18	14.6	-9.8
Disorderly conduct	39,056	99.7	0.09	7.4	-24.2
Forgery	32,711	97.1	0.11	8.9	15.9
Prostitution	31,173	99.6	0.07	2.2	-14.9
Family (nonviolent)	29,131	99.0	0.07	5.9	-35.2
Homicide	10,766	100.0	0.02	2.0	-42.9
Arson	9,488	99.7	0.02	2.1	-35.8
Bad checks	9,424	97.3	0.03	3.0	12.8
Gambling	8,105	99.4	0.02	1.2	-72.6
Liquor violations	7,941	99.5	0.03	2.2	2.5
Kidnapping	5,921	99.5	0.01	1.4	-34.0
Extortion	3,281	98.7	0.01	1.1	4.4
Child pornography/obscenity	2,677	89.8	0.01	0.6	-4.3
Stolen property	1,586	98.6	0.00	0.4	1.3
Embezzlement	1,055	78.8	0.00	0.4	45.4

Notes: Offense groups ordered by citywide incident count (Chicago Police Department, 2007–2025). Near exposure: 0–0.05 miles, twelve months before sale (N = 530,395). Raw gap: cross-sectional log-price difference (percent) between listings with and without a near incident; descriptive only.

4 Empirical Strategy

4.1 Design and Identification

The design is a spatial difference-in-differences with two-way fixed effects and continuous treatment. Crime exposure within a narrow ring around the property constitutes the near treatment, while exposure in a far ring around the same property provides the spatial comparison. Identification combines three comparisons: within parcel across transactions, within tract and month, and near relative to farther crime exposure (Figure 2). Parcel fixed effects absorb permanent differences across properties, and tract-by-month fixed effects absorb contemporaneous shocks shared by the surrounding neighborhood. This division of variation is borne out in the data: only 14 percent of far-ring crime exposure varies within tract-months, so the far ring is almost entirely a neighborhood-level quantity that the fixed effects absorb nearly in full, whereas a majority of near-ring exposure survives within tract-months (Section 3). The near-relative-to-far contrast is therefore identified from within-neighborhood differences in exposure. The parameter of interest is the differential price response,

$$\delta = \beta_{\text{near}} - \beta_{\text{far}}.$$

The estimate δ_k corresponds to $-\kappa_k$ in Equation (3) under the linear approximation and the far-ring restriction: it is a regression-weighted average of the per-incident effect across counts.

For property i transacting at date t, let Near_{it} be the count of incidents of a given offense in the near ring, within 0.05 miles of the parcel, in the m months before the sale, and Far_{it} the count in the far ring, 0.20–0.30 miles, over the same window, the same comparison ring as Tang and Le (2023); Table A6 reports alternative far rings. Stale counts SNear_{it} and SFar_{it}, near- and far-ring incidents in the part of a twelve-month lookback that falls *beyond* the m-month window, and therefore non-overlapping with Near_{it} and Far_{it}, absorb slower-moving local conditions; they are empty when $m = 12$. The window length m is not fixed a priori; I estimate across $m = 1, \ldots, 12$ months and examine the choice in Section 4.2. I report estimates for all twelve exposure windows to show how the results vary with the period over which nearby crime is measured. The estimating equation is

$$\ln P_{it} = \beta_{\text{near}}\text{Near}_{it} + \beta_{\text{far}}\text{Far}_{it} + \gamma_n\text{SNear}_{it} + \gamma_f\text{SFar}_{it} + \mu_i + \lambda_{c(i)t} + \varepsilon_{it}, \tag{5}$$

where μ_i is a parcel fixed effect and $\lambda_{c(i)t}$ a tract×month fixed effect.

Identification requires that, after accounting for parcel and tract-by-month fixed effects and the included controls, the difference in crime exposure between the near ring and the far ring is unrelated to other factors that affect the sale price. This does not require near and far rings to be alike in levels, or unconditional parallel trends across high- and low-crime properties. Parcel fixed effects absorb time-invariant property differences, and tract-by-month fixed effects absorb shocks shared within a tract and month. What remains is block-level variation across a property's own

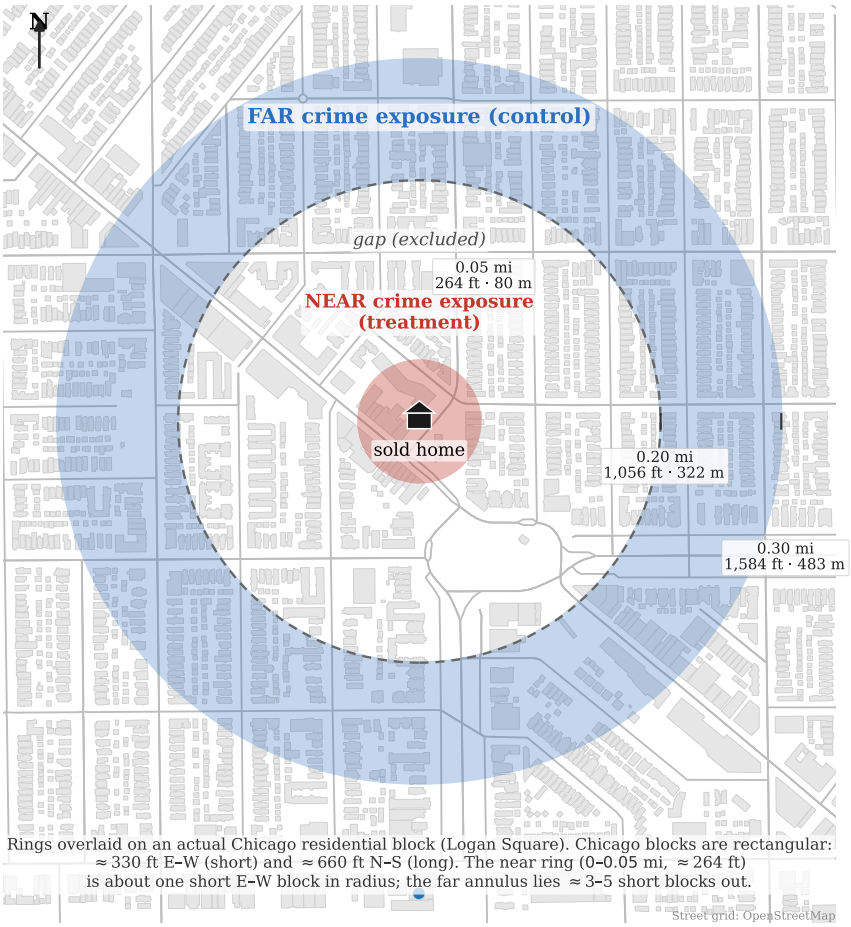


Figure 2: The near–far design. Crime within the near ring (0–0.05 miles, about 260 feet, roughly one short east–west Chicago block) is the treatment; crime in the far ring (0.20–0.30 miles, about three to five short blocks out) is the spatial comparison, separated from the near ring by a gap so the two rings are non-adjacent. The near–far difference $\delta = \beta_{\text{near}} - \beta_{\text{far}}$ differences out conditions common to the surrounding neighborhood. Near ring in red, far ring in blue; incidents in the gap and beyond the far ring (grey) are excluded.

transactions. Formally, let $\mathbf{X}_{it} = (\text{Near}_{it}, \text{Far}_{it})'$. Identification requires

$$E\left[\widetilde{\mathbf{X}}_{it}\,\widetilde{\varepsilon}_{it}\right] = \mathbf{0}, \tag{6}$$

where tildes denote residualization with respect to the fixed effects and controls in Equation (5). Under this condition β_{near} and β_{far} are identified, and so is $\delta = \beta_{\text{near}} - \beta_{\text{far}}$.

Two threats to this assumption, and the interpretation of δ, are addressed directly.

1. *No omitted shocks correlated with the near–far crime difference.* A local event, such as a neighboring building becoming vacant, could raise crime on the block relative to the far ring while independently lowering the property's price. The near–far comparison removes shocks shared equally by the two rings, but not those confined to the block, so the estimate could attribute such an event's price effect to crime. I assess sensitivity to these conditions by adding predetermined neighborhood and lagged housing-market controls (Section 6). These controls capture observable pre-existing conditions but cannot rule out unobserved shocks during the exposure period.

2. *No reverse causality or confounding price dynamics.* Crime could respond to earlier prices, or both could reflect a changing block: a temporary decline in local property values could coincide with rising crime, so the estimate would capture that trajectory rather than the effect of an incident. Measuring crime before the sale establishes temporal order but does not rule this out. I assess the concern with a mirror placebo for every estimate, replacing pre-sale crime with post-sale crime. A significant placebo raises concerns about confounding dynamics or anticipated risk; a null is consistent with the identifying assumption but does not establish it.

The interpretation requires that the far ring is beyond the effect's reach. δ compares marginal responses to near and far exposure and equals the near-ring effect only under the far-ring restriction, that crime a fifth to three tenths of a mile out has no price effect of its own. Crime at that distance is the benchmark for what would have happened on the block anyway; if an incident there still moved the price, the contrast would recover a relative effect rather than the full near-ring effect. If $\beta_{\text{far}} < 0$, subtracting it makes δ less negative than β_{near} alone. The spatial decay results are consistent with the restriction: the discount is no longer detectable at the adjacent block.

The design does not assume that crime is randomly distributed across the city, across neighborhoods, or over time; only that the residual near–far variation is unrelated to unobserved price shocks. Section 6 reports each check.

The parcel fixed effect restricts identification to the 136,866 parcels that transact more than once (329,800 sales). The tract-by-month fixed effect then drops any transaction that is the only sale in its cell, together with the singletons this creates, leaving the 242,815 transactions that identify the two-way model and form the sample in the results tables (Table OA1).

Enforcement intensity affects recorded crime directly (Levitt, 1998) and can bias the estimate in either direction. Where enforcement deters crime and improves the block, detection may still raise recorded counts, so a block that is getting safer shows more incidents and the discount is attenuated. Where enforcement is aimed at blocks that are already deteriorating, recorded counts rise alongside falling prices and the discount is inflated. Tract-by-month fixed effects absorb enforcement changes shared within each tract and month.[10] Uneven deployment within tracts, including enforcement targeted at particular addresses, remains a concern. This matters for offenses recorded through enforcement rather than victim complaint, which here means narcotics, prostitution, gambling, and liquor violations, and I describe those measures as enforcement-recorded offenses rather than as a census of crime.

4.2 Choice of Spatial Bands and Exposure Windows

The rings and windows in (5) are chosen from the data rather than imposed. I estimate the price effect as a smooth function of distance and time since incident by a tensor-product spline[11] (Wood, 2017; Korolev, 2026). For burglary and narcotics, the distance profile is significantly negative within 0.05 miles and not significantly different from zero at the adjacent-block ring. This pattern guides the choice of the 0–0.05 mile near ring. The 0.20–0.30 mile far ring serves as the reference region for the spline (Figure 3; Online Appendix J). Margaretic and Sosa (2025) find that estimated crime spillovers depend on the spatial scale of aggregation. Administrative boundaries need not match the area over which households experience crime. Estimates may therefore depend on the geographic unit chosen, the modifiable areal unit problem (MAUP). I report estimates across nine near–far ring definitions (Table A6) and all one-to-twelve-month windows. Summary tables use the main window among those with significant discounts; Table 2 reports estimates across all twelve windows. Significant-window counts are descriptive; the windows are nested, not independent.

Section 6 reports four robustness checks. A post-transaction placebo addresses reverse causality and mean reversion; predetermined near–far measures of neighborhood conditions address hyper-local confounding; a full-universe model of sale propensity and a complementary selection correction assess endogenous transaction selection; and a mass-point diagnostic following Caetano (2015) and Caetano et al. (2021) addresses selection into zero crime exposure. The full set of diagnostics is reported in the robustness section. The comparisons that follow use incident location and recurrence to distinguish among the mechanisms in the conceptual framework.

The parameter of interest is the effect of a recent incident on the property's block, relative to the

[10] A Chicago police beat averages about 0.8 square miles and spans roughly three census tracts; the 791 tracts in the estimation area average about 0.29 square miles. Beats nest within sectors and sectors within the city's roughly two dozen districts. Beat and tract boundaries cross rather than nest, so beat-level deployment is absorbed only in part.

[11] A tensor-product spline models a function of two variables as a weighted sum of products of one-dimensional basis functions, here a basis in distance from the parcel and a basis in months since the incident. Each product term lets the profile in one dimension vary with the other, so the distance profile can differ across lags without imposing a parametric form. The basis sizes are $c(6, 12)$, six in distance and twelve in lag, and the smoothness penalty is chosen by restricted maximum likelihood (Wood, 2017).

surrounding neighborhood in the same month. The spatial scale is not imposed: the distance profile in Figure 3, estimated without rings, locates the response within 0.05 miles, and the rings follow it. The estimates do not capture the long-run co-movement of crime and prices across neighborhoods.

Main window. I report estimates for all twelve exposure windows. When a table requires one estimate per offense, I use the median window among those with a negative estimate significant at five percent, choosing the lower median when their number is even. This gives ten months for homicide and narcotics, eight for burglary, and six for vandalism. For offenses with no significant discount, I report the twelve-month estimate.

5 Results

5.1 Sale Prices

Four of twenty-two offenses carry a sale-price discount: homicide, burglary, narcotics, and vandalism (Table 2). Based on my subjective assessment of the estimates across all twelve windows, I classify offenses into four broad groups: *priced*, *suggestive*, *precise nulls*, and *inconclusive*. An offense is priced if its discount is significant at five percent in two or more consecutive windows. It is suggestive if the discount is significant at five percent in exactly one window and significant at ten percent in an adjacent window. Among offenses with no discount significant at five percent in any window, precise nulls are those whose 95 percent confidence intervals rule out a one-percent discount in at least nine of twelve windows; the rest are inconclusive. The table is ordered accordingly, with column definitions in the notes. Table A3 reports full results for every offense at all twelve windows.

Disorderly conduct and sex offenses are suggestive. Twelve offenses are precise nulls. For larceny, motor vehicle theft, and non-domestic assault, the intervals also rule out, in at least ten of twelve windows, a discount as large as burglary's eight-month estimate (0.24 percent per incident). The estimates for bad checks and family offenses are positive. Tables OA8 and OA9 split assault and the other offenses by domestic status. Weapons, arson, gambling, and stolen property are inconclusive.

The homicide discount of about 3 percent per incident is similar in magnitude to estimates from other studies of individual murders (Table A5). Klimova and Lee (2014) find a 3.9 percent decline in sale prices within 0.2 miles of a murder in Sydney during the following year, while Gautier et al. (2009) find a decline of about 3 percent in asking prices in Amsterdam neighborhoods with large Muslim populations following the van Gogh murder. Discounts are substantially larger for the property where the incident occurs: in Hong Kong, such units sell for roughly a third less, with discounts of about 19 percent for other units on the same floor and 10 percent for units elsewhere in the building, and no statistically significant recovery over time (Bhattacharya et al., 2021). At the neighborhood level, Ihlanfeldt and Mayock (2010) find no detectable effect of murder on house prices, while effects of aggravated assault and robbery emerge with a lag of two to four years. The contrast suggests that the geographic scale of crime measurement may matter for the estimated

magnitude and timing of capitalization, although differences in settings and research designs also limit direct comparisons.

Fraud, forgery, and bad checks impose direct monetary losses, yet none carries a detectable discount in the pooled estimates; embezzlement has too few incidents to estimate (Appendix B.1).

I next examine how discounts vary with the number of nearby incidents, replacing the near-ring count with indicators for one, two, three, and four or more incidents, relative to zero (Table OA6; Figure OF1). Four or more burglaries and vandalism incidents are associated with discounts of 1.85 and 1.05 percent at eight and six months, respectively. No priced offense shows a significant discount for exactly one incident at any window. A joint test of linearity over zero to three nearby incidents rejects the restriction for burglary at eight months ($p = 0.02$), but not for the other priced offenses at their main windows.

Spatial decay. For burglary and narcotics, estimated discounts fall sharply beyond 0.05 miles and remain near zero through 0.20 miles (Table A6; Appendix C). Widening the burglary ring to 0–0.10 miles retains a significant discount, but the 0.05–0.10 mile ring alone does not at eight months, consistent with the wider-ring result being driven primarily by the nearest incidents. Homicide discounts also appear in the 0.10–0.15 mile ring, although the estimates do not decline smoothly with distance; the spatial falsification tests in Section 6 report additional distance-band specifications.

Spline estimates. A tensor-product spline in distance and lag, estimated as a fixed-effects signal regression following Korolev (2026), shows a similar decay (Figure 3). For burglary and narcotics, the effect is significantly negative within 0.05 miles and not significantly different from zero at the adjacent-block ring. Vandalism's distance profile is not significantly different from zero within 0.05 miles, consistent with its small per-incident discount (Table 2). Homicide's confidence band is roughly ten times wider than those of the other three offenses and includes zero within 0.05 miles, consistent with sparse homicide exposure (Online Appendix J). In the time dimension, the burglary effect is concentrated among incidents about three to seven months before the sale and the vandalism effect three to six months before, while the narcotics profile shows no significant pattern; Appendix C.1 compares recent with earlier exposure.

Table 2: Sale Prices

Offense	Identifying parcels	Sig. discount windows	Min δ (%) (s.e.) [mo]	Max δ (%) (s.e.) [mo]	"Median" δ (%) (s.e.) [mo]
Priced					
Homicide	5,701	2/12	-5.86^{*} (3.35) [1]	-1.24 (1.43) [8]	-2.23 (2.60) [2]
Narcotics	53,844	5/12	-0.93 (0.75) [1]	-0.18 (0.19) [6]	-0.33^{***} (0.11) [12]
Burglary	88,094	6/12	-0.34^{**} (0.14) [6]	-0.18^{*} (0.09) [12]	-0.22^{**} (0.10) [11]
Vandalism	104,661	2/12	-0.17^{**} (0.08) [6]	0.26 (0.22) [1]	-0.10 (0.07) [9]
Suggestive					
Disorderly conduct	19,550	1/12	-1.33^{*} (0.78) [2]	-0.46 (0.54) [4]	-0.65 (0.63) [3]
Sex offenses	36,393	1/12	-0.47^{**} (0.23) [12]	0.11 (0.35) [3]	-0.26 (0.25) [9]
Precise nulls					
Larceny/theft	115,548	0/12	0.01 (0.02) [12]	0.08^{***} (0.03) [1]	0.02 (0.01) [8]
Motor vehicle theft	85,781	0/12	-0.02 (0.09) [12]	0.51^{*} (0.30) [1]	0.07 (0.09) [11]
Assault (non-dom.)	91,783	0/12	-0.05 (0.06) [7]	0.01 (0.11) [3]	-0.03 (0.09) [4]
Bad checks	8,919	0/12	0.66^{*} (0.35) [11]	2.20^{**} (1.03) [1]	1.08^{**} (0.43) [7]
Family (non-violent)	15,928	0/12	1.00^{*} (0.51) [12]	2.16^{***} (0.82) [5]	1.49^{**} (0.60) [9]
Fraud	67,030	0/12	-0.36 (0.23) [1]	-0.04 (0.09) [6]	-0.08 (0.08) [10]
Assault (domestic)	90,500	0/12	-0.19^{*} (0.10) [12]	0.02 (0.15) [5]	-0.03 (0.12) [7]
Trespass	54,025	0/12	-0.15 (0.14) [6]	0.18 (0.22) [3]	-0.09 (0.38) [1]
Robbery	65,876	0/12	-0.06 (0.14) [9]	0.30 (0.42) [1]	0.01 (0.13) [11]
Forgery	23,614	0/12	-0.07 (0.59) [1]	0.47 (0.45) [2]	0.23 (0.18) [11]
Liquor law	6,393	0/12	0.08 (1.20) [1]	1.34 (0.91) [2]	0.51 (0.36) [10]
Prostitution	6,326	0/12	-1.00 (0.82) [1]	0.08 (0.12) [12]	-0.18 (0.35) [5]
Inconclusive					
Weapons	32,294	0/12	-0.62 (0.47) [5]	0.11 (0.58) [3]	-0.17 (0.35) [9]
Arson	6,116	0/12	-0.20 (1.06) [11]	1.27 (3.58) [1]	0.07 (1.63) [6]
Gambling	3,125	0/12	-4.77 (4.33) [3]	1.99 (8.32) [1]	-1.79 (1.77) [11]
Stolen property	1,262	0/12	-2.23 (2.63) [2]	0.81 (1.20) [8]	-0.67 (1.75) [6]

Notes. $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$: percent change in sale price per incident within 0–0.05 miles, relative to 0.20–0.30 miles. Parcel and tract × year-month fixed effects; standard errors clustered by tract; $N = 242{,}815$ repeat sales, 2007–2025. Estimated at each cumulative window, one to twelve months. Sig. windows: negative and significant at 5%. "Median" δ: the sixth-lowest of the twelve window estimates (one of the two medians); Min and Max: most negative and most positive estimates across the twelve windows (standard error; window in brackets). Identifying parcels: exposure varies within parcel, counted at the twelve-month window. Priced: a discount significant at 5% in two or more consecutive windows. Suggestive: a discount significant at 5% in exactly one window and at 10% in an adjacent window. Precise nulls: no discount significant at 5% in any window, and the 95 percent intervals rule out a one-percent discount in at least nine of twelve windows; for larceny, motor vehicle theft, non-domestic assault, bad checks, and family offenses they also rule out a discount as large as burglary's main-window estimate (0.24 percent, eight months) in at least nine of twelve windows, for the last two because their estimates are positive. Inconclusive: the remaining offenses with no discount significant at 5% in any window. Kidnapping, embezzlement, extortion, child pornography, and other offenses have too few incidents to estimate.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

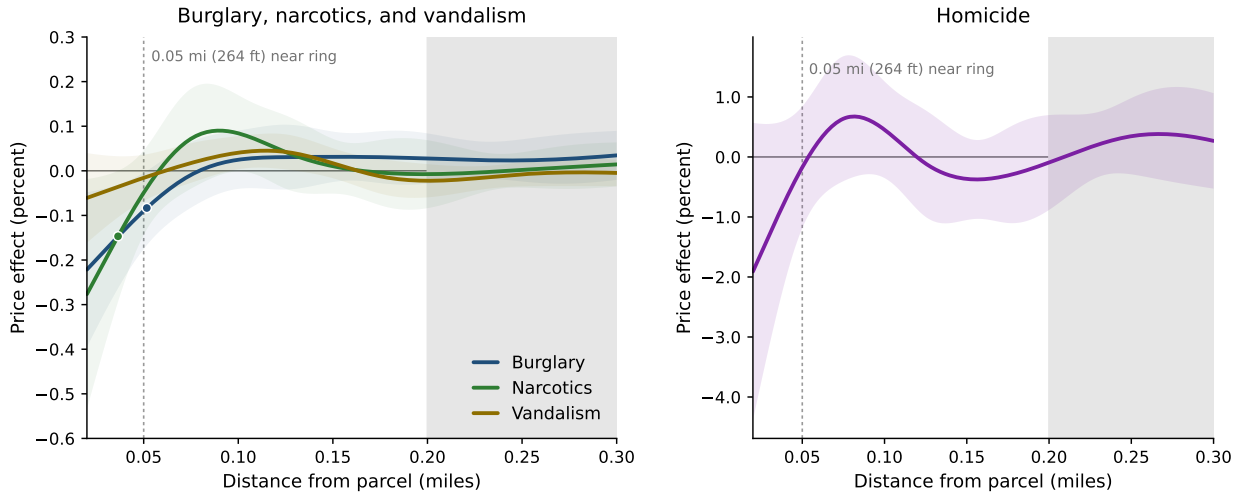


Figure 3: Price response by distance from the parcel. Distance profiles from a tensor-product spline in distance and lag, estimated as a fixed-effects signal regression in `mgcv` following Korolev (2026), fit on 0–0.30 miles and centered on their mean, averaged over the one-to-twelve-month windows; MLS close-date sample, 2007–2025, with 95 percent cluster-robust bands. *Left*: burglary, narcotics, and vandalism. *Right*: homicide, whose band is wider because incidents are rare and includes zero within 0.05 miles. The dotted line marks the 0.05-mile near-ring edge; the shaded band is the 0.20–0.30 mile comparison ring. The profile locates where the response is negative and where it is flat; its level is not the near–far discount in Table 2. Estimation details in Online Appendix J.

5.2 Mechanisms

Sixteen of twenty-two offenses carry no detectable discount, and two more (disorderly conduct and sex offenses) are significant only at a single long-horizon window.

Location of the incident Prediction 1 says that, holding the offense fixed, incidents more likely to be observed carry larger discounts. I split incidents of the same offense in two ways: (1) by the type of place where they were recorded (a residential address, the street, or a commercial premises), and (2) among incidents at residential addresses, by whether they occurred inside the dwelling or at its frontage (the garage, porch, yard, driveway, or entryway).[12]

The homicide and narcotics discounts are significant for incidents on the street (in 4 and 5 of 12 windows) but not for incidents at residential addresses, although the two estimates cannot be statistically distinguished; the vandalism discount is concentrated at residential addresses (8 of 12 windows) and differs significantly from street vandalism in 7 of 12 (Tables A7 and A8).[13] At its main six-month window, residential vandalism is associated with a 0.56 percent lower sale price per incident (significant at one percent), compared with an estimate of −0.01 percent for street vandalism.

[12]Interior incidents are those whose recorded location is a residence, apartment, Chicago Housing Authority apartment, house, coach house, rooming house, or trailer; frontage incidents are those recorded at a residential garage, porch, yard, driveway, or entryway, or in a hallway or common area (Table OA13). The residence and apartment codes record that an incident occurred at the dwelling without a more specific exterior location, so the interior category may include some incidents outside the living space.

[13]The pooled residential burglary estimate has a comparable post-sale placebo estimate (Table 3 note), so the frontage–interior split, whose placebos are clean, provides the relevant test for burglary.

The interior–frontage split compares incidents of the same offense recorded at residential addresses but in locations that differ in their likely visibility from the street.[14] For burglary, frontage incidents carry a discount (−0.74 percent, s.e. 0.33), significant in 7 of 12 windows, while interior incidents have no detectable effect at any window (Table 3); the difference is significant at twelve months ($p = 0.030$) but not at the main window ($p = 0.109$). For narcotics, frontage discounts are significant in seven of twelve windows, while no significant discount is detected for interior incidents. The frontage–interior difference is significant in six of twelve windows. For vandalism, the discount is concentrated instead in interior incidents (−0.73 percent, s.e. 0.25; 9 of 12 windows).[15] As a comparison, larceny shows no detectable discount at either location, and the difference is not statistically significant (Table OA14).

Target of the incident Among frontage incidents, damage to property can leave a visible trace at the site, whereas damage to a car can leave with the vehicle. Detailed vandalism codes distinguish property damage from vehicle damage.[16] Among frontage vandalism, property-damage incidents carry significant discounts in six of twelve windows, while vehicle-damage incidents show no significant discount at any window. The difference is significant at the ten-percent level at ten and twelve months (Table 3; Online Appendix D).[17]

Risk of future crime Prediction 2 asks whether discounts track what an incident reveals about future crime at the property—the updating term $b_k(S_k)$ in the framework (Section 2). These tests measure predictability in recorded crime, not buyers' subjective beliefs, which are unobserved. I use two measures: how well crime 0.20–0.30 miles away forecasts future crime within 0.05 miles, and how strongly crime within 0.05 miles predicts its own recurrence near the same parcel.

Neither measure shows a consistent correspondence with the size of the discounts. At the six-month horizon, fraud, weapons violations, domestic assault, and motor vehicle theft are forecast at least as well as burglary and vandalism but carry no detectable discount (Table A13). Homicide carries the largest discount yet is among the least forecastable offenses and has a small, negative recurrence coefficient ($\rho = -0.026$, s.e. 0.012). Narcotics is the most self-predicting offense in Table A12 ($\rho = 0.40$), yet carries a much smaller discount. Homicide's weak recurrence is consistent

[14] Bhattacharya et al. (2021) similarly find that deaths of residents occurring outside the home carry no discount, while deaths inside the unit do. Their comparison varies whether the event is attached to the property; mine varies whether an incident near the property is observable from the street. Frontage also includes porch, hallway, and common-area codes, which make up 39 percent of frontage-coded incidents across all offenses (Table OA13), so frontage is an imperfect proxy for street visibility.

[15] CPD masks each incident to the hundred-block, so an interior record cannot be separated between the sold parcel and a neighbor's. For burglary this matters little, since the loss is the stolen contents and any entry damage is minor; for vandalism an interior record is damage to the property itself, which lowers its value as repair cost rather than through information. For vandalism, the location split therefore separates damage to the dwelling from damage outside it rather than visible from hidden incidents.

[16] Vandalism is the only priced offense for which detailed police offense codes distinguish property damage from vehicle damage; homicide targets a person, while burglary and narcotics codes record the method of entry or the drug, not the target.

[17] Because the police records mask each incident to the hundred-block, a frontage damage-to-property record can be the sold parcel itself, so this contrast carries an own-property component.

Table 3: Within-Offense Comparisons by Incident Location and Target

	Exposure window (months before sale)											
	1	2	3	4	5	6	7	8	9	10	11	12
Panel A. Observability: residential incidents, frontage versus interior												
Burglary												
Frontage (visible)	−0.37	−0.45	−0.54	−0.68	−0.54	−0.74**	−0.76**	−0.74**	−0.62**	−0.70**	−0.71***	−0.73***
	(0.77)	(0.59)	(0.49)	(0.46)	(0.41)	(0.36)	(0.35)	(0.33)	(0.29)	(0.27)	(0.25)	(0.24)
Interior (hidden)	−0.17	−0.27	−0.12	−0.11	−0.20	−0.30	−0.22	−0.14	−0.18	−0.19	−0.20	−0.13
	(0.47)	(0.30)	(0.26)	(0.22)	(0.20)	(0.19)	(0.18)	(0.16)	(0.15)	(0.13)	(0.13)	(0.13)
Frontage = interior (p)	0.824	0.796	0.457	0.258	0.462	0.299	0.183	0.109	0.182	0.098	0.081	0.030
Narcotics												
Frontage (visible)	−7.45	−3.68	−7.08**	−3.94	−4.63**	−4.63**	−3.17*	−3.54**	−2.59*	−2.92**	−2.86**	−2.35**
	(6.31)	(3.57)	(2.97)	(2.69)	(2.07)	(1.93)	(1.83)	(1.70)	(1.52)	(1.29)	(1.12)	(1.01)
Interior (hidden)	−1.20	−0.44	0.42	0.72	0.39	0.92	0.76	0.73	0.31	0.37	0.27	−0.28
	(2.74)	(1.94)	(1.41)	(1.20)	(1.06)	(1.00)	(0.88)	(0.80)	(0.78)	(0.75)	(0.74)	(0.72)
Frontage = interior (p)	0.344	0.424	0.023	0.118	0.035	0.014	0.063	0.025	0.087	0.029	0.025	0.111
Vandalism												
Frontage (visible)	0.76	1.02*	0.70*	0.32	0.15	−0.14	−0.45	−0.54*	−0.54*	−0.60**	−0.54**	−0.48**
	(0.74)	(0.55)	(0.42)	(0.37)	(0.33)	(0.33)	(0.30)	(0.30)	(0.29)	(0.28)	(0.26)	(0.24)
Interior (hidden)	−0.34	−0.70	−0.79*	−0.66**	−0.70**	−0.73***	−0.67***	−0.62***	−0.50**	−0.45**	−0.38**	−0.44**
	(0.67)	(0.51)	(0.40)	(0.31)	(0.28)	(0.25)	(0.23)	(0.23)	(0.22)	(0.20)	(0.19)	(0.18)
Frontage = interior (p)	0.246	0.024	0.010	0.047	0.056	0.170	0.557	0.847	0.892	0.671	0.626	0.883
Panel B. Vandalism, frontage incidents, by target												
Damage to property	0.85	1.06	0.80	0.22	0.03	−0.49	−0.89**	−0.95**	−0.91**	−1.03***	−0.92**	−0.89***
	(1.12)	(0.83)	(0.63)	(0.58)	(0.51)	(0.47)	(0.45)	(0.43)	(0.41)	(0.39)	(0.36)	(0.34)
Damage to vehicle	0.70	0.89	0.50	0.33	0.20	0.21	0.05	−0.07	−0.13	−0.08	−0.09	0.02
	(0.93)	(0.67)	(0.56)	(0.53)	(0.48)	(0.45)	(0.42)	(0.40)	(0.40)	(0.38)	(0.37)	(0.34)
Property = vehicle (p)	0.916	0.874	0.724	0.896	0.818	0.298	0.143	0.142	0.168	0.078	0.102	0.057

Notes. δ as in Table 2, at each window; standard errors below. Panel A splits residential burglary, narcotics, and vandalism incidents into frontage (garage, porch, yard, driveway, entryway, porch/hallway and common areas; Table OA13) and interior incidents. Panel B splits frontage vandalism by target. Last row of each panel: Wald p for equal effects. Post-sale placebos for the frontage components are significant in no window for any of the three offenses, and for the interior components in at most one window, except narcotics interior (six windows), which carries no discount; the pooled residential burglary coefficient, by contrast, is matched window-for-window by post-sale crime (Online Appendix C); both fixed-effect designs in Online Appendix D. The burglary frontage estimate rests on about 45,000 incidents; its post-sale placebo is not significant at five percent in any window.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

with Carrillo and Yezer (2021), who find it less persistent from year to year than other violent crime. Caetano and Maheshri (2018) document offense-specific persistence over shorter horizons. These measures of predictability therefore do not, on their own, explain the pattern of discounts. Appendix D reports the forecasting and recurrence specifications and additional results.

5.3 Price Formation

Prediction 3 asks where in the transaction the discount appears. I examine asking prices, mid-listing price cuts, the sale-to-list ratio, time on market, and withdrawal to distinguish price responses present at listing from those that emerge later in the transaction. The housing-market microstructure literature studies asking prices, price revisions, selling time, and transaction prices as jointly determined outcomes of search and bargaining (Knight, 2002; Merlo and Ortalo-Magné, 2004; Han and Strange, 2015).

Table 4: Transaction Outcomes: From Asking Price to Sale

Offense	Ask Price %	Sale-to-list %	Price cut pp	Days on market %	Withdrawal pp	Close %
Homicide	0.38	0.53	$8.24^{**\ddagger}$	14.54^{**}	2.24^{**}	-3.02^{**}
	(0.57)	(0.45)	(3.30)	(6.85)	(0.96)	(1.38)
Narcotics	0.21	-0.16^{**}	0.43	−0.78	0.04	-0.29^{**}
	(0.17)	(0.07)	(0.35)	(0.98)	(0.04)	(0.14)
Burglary	−0.15	−0.01	−0.33	0.23	−0.10	-0.24^{**}
	(0.12)	(0.05)	(0.33)	(1.25)	(0.10)	(0.12)
Vandalism	0.05	0.00	−0.12	−0.10	0.00	-0.17^{**}
	(0.07)	(0.03)	(0.21)	(0.69)	(0.05)	(0.08)
Disorderly conduct	-0.43^{**}	-0.28^{**}	2.55^{***}	−1.97	0.10	-0.65^{**}
	(0.21)	(0.12)	(0.95)	(3.86)	(0.23)	(0.29)
Weapons violations	−0.14	0.11	1.35^{**}	−0.33	0.59^{***}	0.10
	(0.20)	(0.25)	(0.53)	(3.52)	(0.21)	(0.28)
Parcel FE	Yes	Yes	Yes	Yes	Yes	Yes
Tract × month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Near–far estimates as in Table 2, pooled 2007–2025. Columns: log asking price, log sale-to-list ratio, mid-listing price cut (percentage points), log days on market, withdrawal (percentage points), and log closing price. Closing price at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6, disorderly conduct 12, weapons violations 12 months); listing margins at the median significant window. Days on market uses parcel × era fixed effects; withdrawal uses all listings.

‡The price cut compares the last asking price with the original asking price of the same listing. It does not record when the cut was made, so it cannot show whether the cut followed the crime. The price-cut column uses crime during the listing, with fixed effects for the number of months listed.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Where the discount appears The discount is detectable in closing prices but not in asking prices. All four priced offenses show significant closing-price discounts, and none shows a statistically significant asking-price response, although the burglary estimate is too imprecise to rule out a decline in asking prices as large as its closing discount (Table 4). For narcotics, the sale-to-list ratio shows the gap directly: homes close further below their asking price (−0.16, s.e. 0.07). A nearby

homicide during the listing is associated with a higher probability that the home is cut below its original asking price (8.24 percentage points, s.e. 3.30; Table 4), although the cut may precede the incident within the listing. Together, these findings are consistent with discounts that emerge after listing rather than in the initial asking price.

Market frictions The response to nearby crime can also appear outside the price: crime may affect how quickly homes sell and whether listings reach a sale at all. The estimates suggest that nearby homicide raises time on market by about 15 percent at eight months, equivalent to roughly 2.4 days at the sample median of 16 days; this estimate is sensitive to specification, with statistical significance confined to the seven-to-nine-month window under parcel-by-era fixed effects (Table A15). Homicide withdrawal estimates are consistently positive but significant only at two months. Weapons violations show the more consistent withdrawal response—positive in all twelve windows under both specifications and significant in several—despite no consistent closing-price discount (Table 4; Table A16). Withdrawals also select the sample of closing prices: if the listings that withdraw would have faced the largest discounts, the price estimates are biased toward zero (Section 6). More broadly, price estimates alone may understate the response to crime, part of which operates through withdrawals and longer time on market. Appendix E reports additional details.

Real estate agent experience Prior work shows that real estate agents' incentives and information shape transaction outcomes (Rutherford et al., 2005; Levitt and Syverson, 2008; Hendel et al., 2009; Han and Hong, 2011; Barwick and Pathak, 2015; Gilbukh and Goldsmith-Pinkham, 2024). I ask whether agents with more local experience, measured by the number of listings the agent has previously handled in the same census tract (in logs), respond more to nearby crime (Table 5). For homicide, they do on both sides of the transaction. Asking prices fall more when the listing agent has more local experience, significantly in seven of twelve windows. Closing prices fall more when the buyer's agent has more local experience: comparing the same agent across transactions, a one-standard-deviation increase in the buyer's agent's local experience deepens the homicide discount by 1.14 percentage points (s.e. 0.53; significant in six of twelve windows; Table A17), and this result holds when the agent's total experience is controlled for. For narcotics, the pattern is weaker: asking prices fall more with the listing agent's local experience in seven of twelve windows, but the difference disappears once total experience is controlled for (Appendix E.1). Since real estate agents are not randomly assigned to listings, these estimates are associations rather than causal effects.

Table 5: Agent Local Experience: Ask (Listing) and Close (Buyer)

Offense	Most negative (s.e.)	Most positive (s.e.)	Sig. windows
Panel A. Listing agent, asking price (ln list)			
Homicide	-1.16^{***} (0.40)	-0.04 (0.62)	7/12
Burglary	-0.02 (0.04)	0.08 (0.06)	0/12
Narcotics	-0.13^{**} (0.06)	-0.06 (0.10)	7/12
Vandalism	-0.07^{**} (0.03)	-0.02 (0.08)	3/12
Panel B. Buyer agent, closing price (ln close)			
Homicide	-4.45^{**} (1.85)	-0.68 (0.63)	6/12
Burglary	0.02 (0.10)	0.38^{**} (0.19)	0/12
Narcotics	-0.04 (0.33)	0.27^{*} (0.15)	0/12
Vandalism	-0.03 (0.10)	0.10^{**} (0.04)	0/12

Notes. Near–far effect interacted with agent experience, $E = \ln(1 +$ prior same-agent same-tract listings). Panel A: listing agent, asking price. Panel B: buyer's agent, closing price. Parcel, tract × month, and agent fixed effects; clustered by tract. Gradient: change in the near–far effect per standard deviation of E (s.d. 1.20 for listing agents and 0.71 for buyer's agents, each on its estimation sample) (negative: experienced agents price crime more), shown as the most negative and most positive estimates across the twelve windows and the number of significantly negative windows. Table A17 reports the same specification at the windows given in its notes.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The quantity margin Using the annual parcel panel, I estimate each offense's effect on the listing rate and the turnover rate, with parcel and tract × year fixed effects (Online Appendix Table OA17). Ten of the twenty-two offenses, including all four priced offenses, raise the listing rate significantly more than the turnover rate; for domestic assault and weapons offenses, this quantity response appears without a detectable price response. Since crime and listings are measured within the same calendar year, these estimates do not establish that crime precedes the listing.[18]

5.4 Heterogeneity

Housing-market conditions changed sharply over the sample, so I test whether the discounts differ between the 2009–2013 foreclosure crisis and the 2014–2025 recovery. I interact near–far exposure with era indicators, with 2007–2008 as the base, and estimate each era's discount in one regression. The point estimates differ: burglary and vandalism discounts are larger in the crisis, and homicide's in the recovery. For none of the four priced offenses, however, does a Wald test reject equal discounts across the two eras (p between 0.10 and 0.63; Appendix H).

The discounts also vary with neighborhood socioeconomic status. When I interact the near–far term with each of ten standardized, predetermined tract characteristics, one at a time and controlling for the tract's predetermined crime level, all four priced offenses carry larger discounts in lower-income and less-educated tracts (Table OA19). A one-standard-deviation increase in tract income is associated with a 2.28-percentage-point smaller homicide discount (s.e. 1.07), with smaller

[18]The turnover rate is the parcel-level counterpart of the tract-level measure studied by Boggess et al. (2013), who find that higher neighborhood crime is followed by more housing sales, with sales inhibited only where crime and vacancy are already high.

differences for burglary, narcotics, and vandalism. The effect of nearby crime varies with tract income for several offenses with no average discount as well, such as trespass and fraud, so this variation is not specific to the priced offenses. The effect of nearby crime varies with tract income for several offenses with no average discount as well, such as trespass and fraud, so this variation is not specific to the priced offenses. Income, poverty, education, home values, and rents are strongly correlated. The results suggest a common socioeconomic gradient rather than distinct effects of each characteristic. Joint specifications do not consistently separate income from racial composition or segregation; detailed results appear in Online Appendix G.

6 Robustness

6.1 Threats to Identification

Reverse Causality and Mean Reversion Prices could drive crime rather than the reverse, or repeat-sales properties transacting at temporarily depressed prices could appreciate mechanically. I test both using a post-date placebo that replaces pre-transaction crime with crime occurring after the transaction, which cannot affect a price already set. The priced offenses pass this placebo at their main windows and in almost all specifications: post-sale crime is significant in none of the 63 specifications for homicide and vandalism, one for narcotics, and four for burglary (bottom panels of Figure 4 and Figures A3–A5). The exception is burglary, where post-sale crime is significant at six months (these are the placebo hits in Figure A5); six months is also where burglary's pre-sale estimate is largest (-0.34, s.e. 0.14; Table A3), so the placebo does not fully separate burglary's peak discount from post-sale crime.

Predetermined Neighborhood Controls The estimated discounts remain similar after controlling for measured pre-existing neighborhood conditions (Table A20). I add five neighborhood measures—building permits, vacant buildings, business licenses, tree canopy, and total 311 service requests—each entered separately for the near and far rings. Tree canopy is measured two years before the sale; the other four measures are cumulative counts of events recorded more than twelve months before closing. I also add five lagged housing-market measures from the MLS: neighboring repeat-sale price changes, the distressed-sale share, the sale-to-list ratio, days on market, and the number of new listings. These enter separately for the two rings and are measured 12–24 months before closing. The controls vary across sales of the same parcel and therefore are not absorbed by parcel fixed effects. All controls precede the twelve-month crime-exposure window. This timing avoids conditioning on contemporaneous neighborhood conditions that may themselves respond to crime, while accounting for measured differences in neighborhood conditions before the exposure period.

Point estimates change little with the additional controls (Table A20), although precision falls: burglary and narcotics are significant only at the ten-percent level when the five neighborhood controls are included jointly or when all ten controls are included. Housing-market controls alone

leave all four estimates essentially unchanged and significant at five percent.

Cross-Crime Confounding A discount attributed to one offense could reflect another offense that occurs nearby. Homicide is nearly uncorrelated with other offenses in the near ring (correlations of at most 0.18); burglary and narcotics are moderately correlated with them (at most 0.47 and 0.43). Entering all twenty-two offenses' near–far terms jointly at a common twelve-month window leaves their discounts nearly unchanged relative to separate estimates at that window (Online Appendix K; Table OA25). Vandalism is more strongly correlated with other offenses, especially domestic and non-domestic assault (0.55 and 0.51). Its joint estimate is −0.02 percent, but its separate twelve-month estimate is already small and insignificant (−0.05 percent, s.e. 0.06). This comparison therefore does not establish whether its six-month discount survives adjustment for other offenses. Disorderly conduct retains a discount in the joint specification (−0.58 percent, significant at five percent).

Sample Selection The repeat-sales estimates may be affected by selection if crime changes which properties sell and those changes are related to unobserved determinants of sale prices (Online Appendix H.1 examines distressed sales). I examine this in three ways. First, nearby crime barely changes whether a parcel sells: among the priced offenses, only burglary has a statistically significant positive effect on the annual sale probability, by 0.02 percentage points against a base rate of 3.8 percent (Table OA17). This does not rule out changes in the composition of properties sold. Second, a Heckman correction, which uses mortgage-rate lock-in to predict whether an owner sells, leaves the crime coefficients largely unchanged (Online Appendix L). Third, a cross-sectional hedonic regression on all sales, including properties that sell only once, replaces the parcel fixed effect with observed characteristics and keeps the four priced discounts negative and significant (Table A21).

The hedonic estimates are larger than the repeat-sales estimates for burglary, narcotics, and vandalism and smaller for homicide. Robbery, domestic and non-domestic assault, and motor-vehicle theft also have significant negative estimates in the hedonic specification, but these shrink substantially under repeat sales, with comparable or smaller standard errors. This pattern is consistent with these offenses being concentrated near properties with persistently lower values, differences that parcel fixed effects absorb but observed hedonic controls may not fully capture. The hedonic sample also includes properties that sell only once, so part of the difference may reflect sample composition.

Reporting Endogeneity Recorded incidents reflect both underlying crime and reporting or detection. The near–far comparison mitigates reporting differences shared by nearby locations, and the largest discount is for homicide, one of the most reliably measured crimes (Fox and Zawitz, 2003). Estimates remain nearly unchanged after controlling for predetermined 311 requests (Table A20), while post-sale placebos provide additional reassurance (Section 6.3). Bias could remain if reporting or enforcement changes not captured by the controls differ between the rings and covary with unobserved determinants of sale prices.

6.2 Exposure Specification and Inference

Selection into Zero Exposure Properties with no nearby crime could differ from exposed properties in unobserved ways. I apply the zero-exposure test of Caetano (2015) and Caetano et al. (2021) to the three priced offenses with sufficient observations at positive exposure levels to estimate the dose–response relationship. About 98 percent of sales have no nearby homicide within the ten-month window. For narcotics and vandalism, the raw discontinuity is no longer statistically significant after including the baseline fixed effects. Adding an indicator for zero exposure leaves the narcotics estimate nearly unchanged and makes the vandalism estimate more negative. For burglary, a discontinuity appears once parcel fixed effects are included (-0.98, s.e. 0.32), and adding an indicator for zero exposure moves the discount from -0.24 (s.e. 0.12) to -0.55 (s.e. 0.17). With a wider 0–0.10 mile near ring, the test does not reject and the burglary discount remains significant (-0.14, s.e. 0.07; Table A6; Online Appendix M).

Functional Form I compare three ways of measuring crime exposure: raw counts, an indicator for any incident, and the inverse hyperbolic sine (IHS) transformation (Table OA24). Homicide's estimated discount remains significant under all three at its main window. For burglary, narcotics, and vandalism, only the raw-count estimates are significant at five percent, although narcotics remains negative under all three forms. The count-indicator specification allows a more flexible relationship and shows significant discounts for four or more burglaries and vandalism incidents (Table OA6). Online Appendix I provides further details.

Spatial Standard Errors Baseline standard errors are clustered by census tract (Cameron and Miller, 2015). Conley standard errors that allow correlation across tract boundaries, with cutoffs of 0.5 and 1.0 km (Conley, 1999), give similar inference: homicide, burglary, and vandalism remain significant at five percent under both cutoffs, and narcotics at five percent at 0.5 km and ten percent at 1.0 km (Table OA22). Clustering instead by the 77 community areas, a much coarser unit, leaves homicide and burglary significant at five percent and vandalism at ten percent, while narcotics is not significant at ten percent ($p = 0.12$).

6.3 Specification Curves and Spatial Checks

Specification curves: sale prices Figure 4 reports the sale-price specification curve for vandalism (narcotics, homicide, and burglary appear in Figures A3–A5; theft, an offense with no detectable discount, appears in Figure 5 as a comparison: 0 of 63 specifications are significant at five percent); each curve sorts the near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ across 63 specifications. The specifications vary the exposure window, the fixed effects, the comparison ring, the functional form, and the predetermined controls; the notes to Figure 4 list the full grid. The window begins at six months because shorter windows fall inside the listing and under-contract period, where measured pre-sale exposure is sparse and partly contemporaneous with the sale itself.

Nearly all specifications are negative for the four priced offenses; the exceptions are the presence-indicator estimates for burglary (0.54, s.e. 0.98) and vandalism (about 0.46), both imprecise. The discount is significant at five percent in 33 of 63 specifications for vandalism, 29 for narcotics, 29 for burglary, and 15 for homicide, whose point estimates range from about −3.6 percent to about zero with wide intervals. The post-sale placebo is significant in none of the 63 for homicide and vandalism, one for narcotics (the presence indicator at ten months), and four for burglary (the linear count at six months under each of the three far rings, and the inverse hyperbolic sine at eight months).[19] The full curve over all one-to-twelve-month exposure windows (103 specifications) is reported in Figures OF4–OF7.

Spatial Falsification Moving the comparison ring out to 0.25–0.35 or 0.30–0.40 miles leaves the estimates nearly unchanged (Table A6). A falsification test that moves the treatment band outward to 0.05–0.20 leaves narcotics, burglary, and vandalism significant in at most 1 of 48 specifications, indicating effects concentrated within 0.05 miles.

Homicide behaves differently. It is significant in 11 of 48 specifications outside the near ring, all in the 0.10–0.15 band, with a median estimate (−2.72) essentially identical to the near ring's (−2.78),[20] while the intervening 0.05–0.10 band is null. Homicide is therefore robust at the doorstep but, unlike the other three offenses, not localized to it. This is consistent with Section 5.2, where the homicide discount is significant for incidents on the street but not at residential addresses, although the two estimates cannot be statistically distinguished. Homicide's estimated price effects extend beyond the near ring, but the comparison-ring coefficient is small and statistically insignificant at the main window (0.20, s.e. 0.35; Table OA5). Omitting the far-ring control leaves the estimated discount similar: −2.83 percent, compared with the baseline near–far estimate of −3.02 percent.

Significantly negative placebo estimates outside the near ring (3 of 48 for narcotics, 5 of 48 for burglary and vandalism) modestly exceed the 5 percent expected by chance and are concentrated in the wider bands, consistent with the contamination this falsification test is designed to detect rather than with a near-ring effect.

7 Discussion

7.1 Why most offenses are not priced

Section 2 notes that house prices respond to localized signals of risk, such as a registered offender moving nearby, even without a new crime. My findings suggest the converse: realized crime need not lower prices, even when it predicts future incidents. Together, these results imply that capitalization depends on how risk becomes observable and relevant to buyers.

[19] The specification curve shows how stable the estimates are across modeling choices. The specifications share data and most modeling choices. Their estimates are therefore dependent, and the number of significant estimates is descriptive rather than a formal test.

[20] The −2.78 and −2.72 medians are taken over the significant estimates in the band-varying close specification grid (window × fixed effects × far ring), the object common to both bands; the larger near-ring significant count (15) additionally varies the predetermined controls and the functional form.

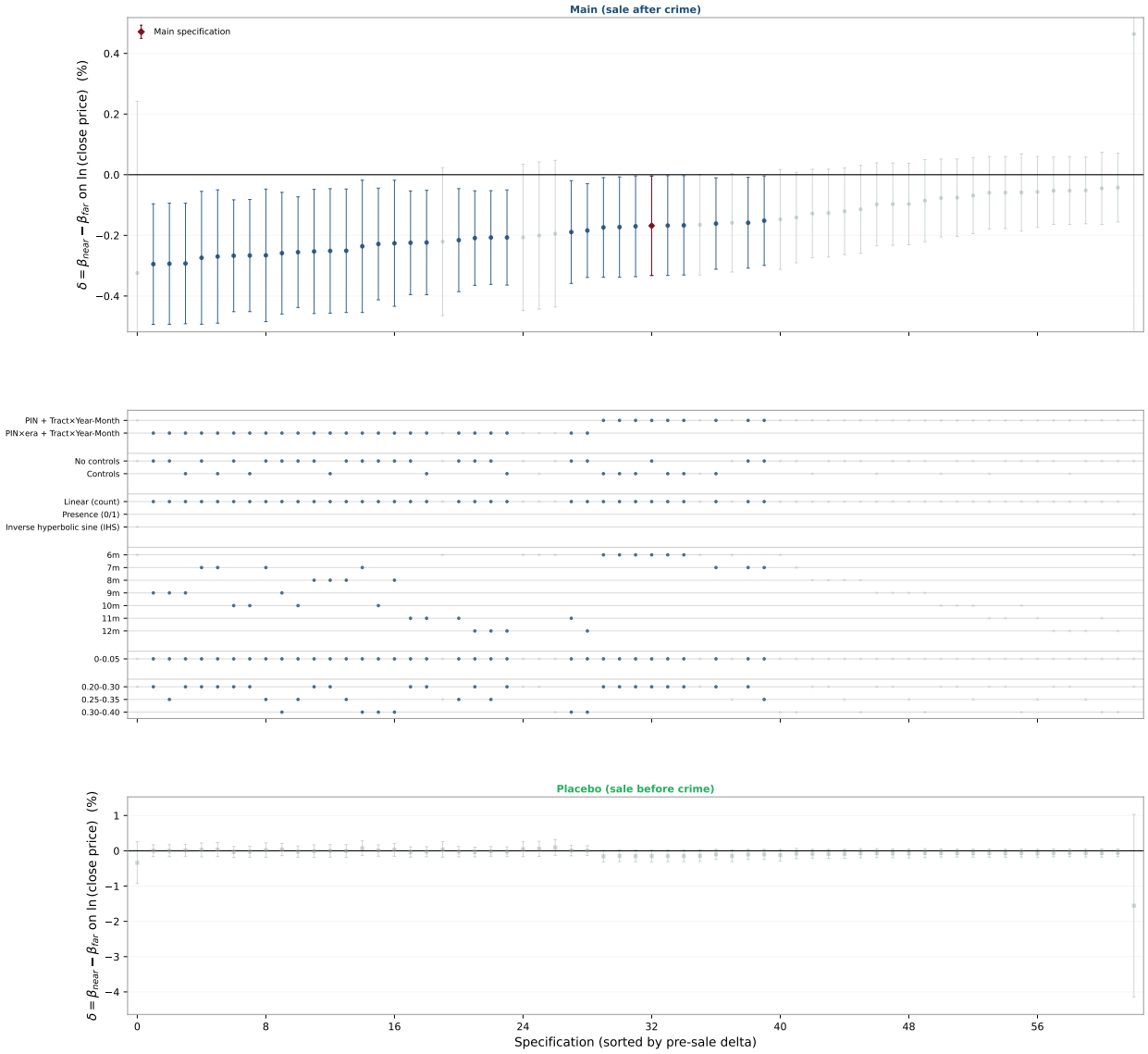


Figure 4: Vandalism: Sale-Price Discount Across Specifications

Notes. Common to this figure, Figure 5, and Figures A3–A5. The 63 specifications vary the exposure window from six to twelve months, the fixed-effect design (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, 0.30–0.40 mile), the functional form (linear count, presence indicator, inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time). Top panel: δ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, and no controls, at each offense's main window: ten months for homicide and narcotics, eight for burglary, and six for vandalism. Middle panel: specification choices. Bottom panel: the post-sale placebo, which replaces pre-sale crime with crime after the sale. All years pooled (2007–2025); standard errors clustered by tract. Vandalism: 33/63 significant at five percent; placebo 0/63.

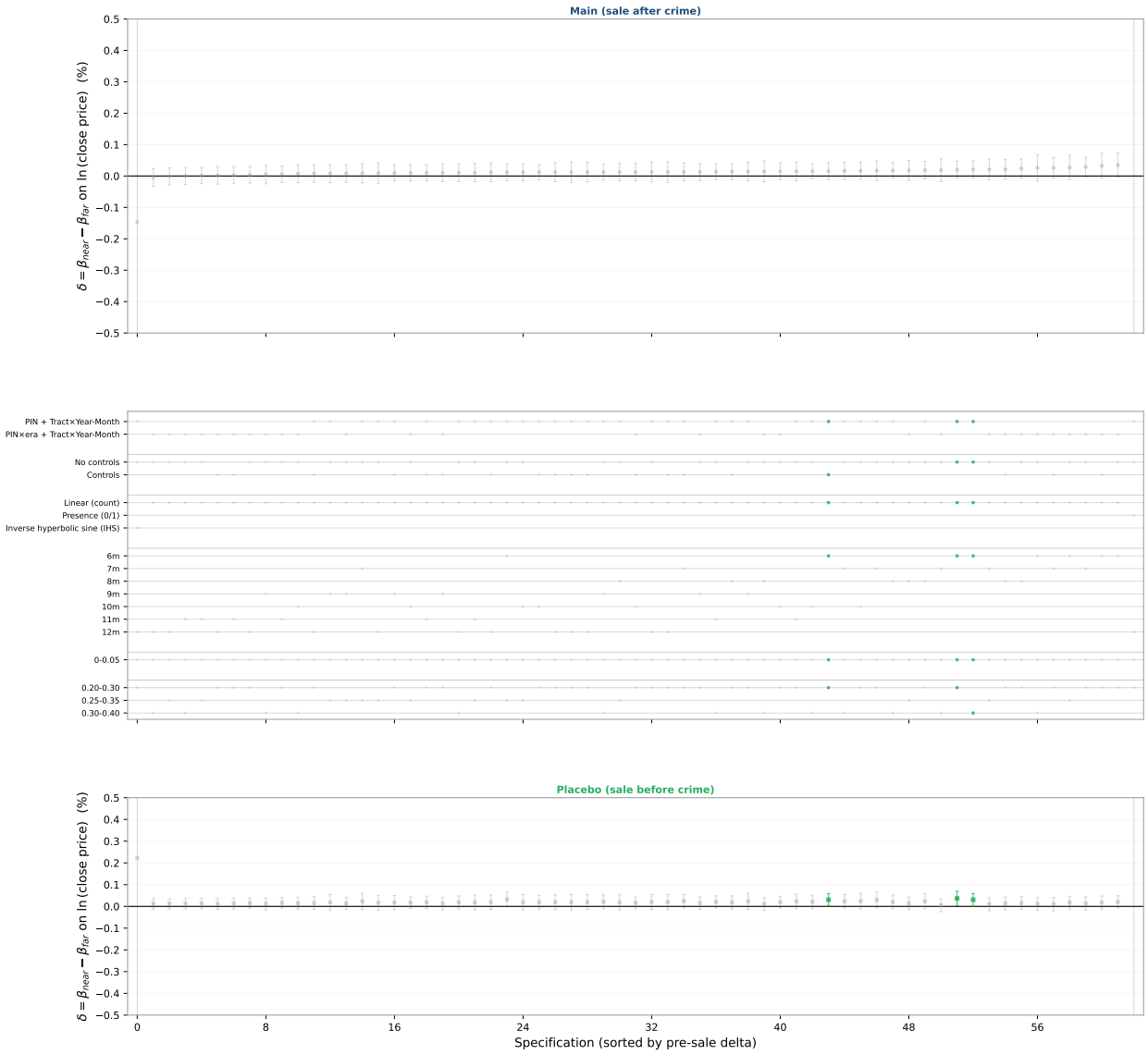


Figure 5: Theft: Sale-Price Discount Across Specifications

Notes. The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: δ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and a twelve-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Theft: 0/63 significant at 5%, placebo 3/63. The vertical axis is cut at ±0.5 percent.

Neither frequency nor severity alone explains which offenses are priced. Homicide is rare and carries the largest discount, while larceny, motor vehicle theft, and non-domestic assault rule out discounts as large as burglary's. Severity alone does not account for the pattern either: domestic assault is a serious offense, yet it has no detectable effect on sale prices (Table A3). One possible explanation is limited buyer awareness of incidents occurring inside homes between people who know each other. The distinction between offenses anchored to an address, which threaten the house or the place, and offenses that harm a person, money, or a movable item offers one interpretation (Section 2, Table OA15). All four priced offenses are anchored to the address, while among the eleven that harm a person, money, or a movable item, only sex offenses show a suggestive discount. This grouping is my own, however, and the pattern depends on it. Under the FBI's NIBRS classification, which groups offenses by victim type, the priced offenses fall in all three categories: homicide among crimes against persons, burglary and vandalism among crimes against property, and narcotics among crimes against society (Table A4). The relevant distinction is therefore not the type of victim but whether the harm stays with the location a buyer is purchasing. The groups also differ in severity, frequency, and visibility. The pattern is therefore consistent with a role for relevance, but does not isolate it from these other characteristics.

Measured predictability does not explain the pattern either. Crime on surrounding blocks predicts subsequent crime near the property but carries no detectable discount, and the most predictive offenses are not those that are priced (Section 5.2). These findings are consistent with selective attention to crime information (Bordalo et al., 2012; Gabaix, 2014), although recorded-crime forecasts need not match buyers' expectations. Within burglary, frontage incidents lower prices while interior incidents have no detectable effect, with a significant difference at the longest exposure windows (Table 3). This supports a role for observability and may help explain the contrast with Gibbons (2004), whose burglary measure pools incidents within 250 meters across locations that may differ in visibility.

The evidence therefore points to relevance and observability rather than predictive content, without separately identifying their contributions. An undetected discount may reflect limited awareness, little relevance to the property, unchanged expectations, or limited precision. Changes in crime information displayed by real estate platforms, discussed in Section 8, would provide a more direct test of the information channel.

7.2 Why asking prices respond little

The transaction margins point to information frictions: the asking price, set before the buyer arrives, rarely moves, while the discount appears at closing. A nearby homicide also increases time on market, significantly so at seven to nine months under parcel-by-era fixed effects. The absence of a detectable average asking-price response to homicide, alongside a roughly three-percent closing-price discount, is consistent with the framework's fixed seller reservation value, although asking prices do not directly measure it (Table 4). Several explanations may account for the muted response. Sellers may anchor on their purchase price or a previous valuation: Genesove and Mayer

(2001) show that owners facing prospective nominal losses set higher asking prices and sell more slowly. Alternatively, comparative market analyses may rely on nearby sales that predate the incident. The more negative asking-price responses among experienced listing agents are consistent with information or expertise beyond these comparables, but the data do not distinguish between the explanations. Sample composition may also mute the average asking-price response. Among listings that sell, the homicide asking-price estimate is negative in every window, though significant only at one month (Table A14). If sellers near a homicide who list high and do not adjust are more likely to withdraw, consistent with withdrawal estimates that are positive in every window, the all-listings asking price would understate the response among completed sales. The data do not separate this channel from the others. The weapons-violation results suggest that housing-market adjustment can occur through listing withdrawal as well as transaction prices. A weak closing-price response need not imply an absence of market adjustment. Recorded weapons violations reflect both weapons activity and police enforcement, and may affect withdrawals through the incident itself, visible police activity, or what either reveals about neighborhood safety; the estimates do not distinguish among these channels or establish what buyers and sellers observed.

Closing-price discounts could also reflect changes in sellers' bargaining positions. For burglary, narcotics, and vandalism, discounts do not differ detectably between soft and tight markets ($p > 0.4$ at each offense's main window; Table OA28); homicide estimates are too imprecise for an informative comparison. These results do not rule out bargaining effects, and the contract-timing and distressed-sale tests in Online Appendix N also have limited power.

7.3 Who carries the information

Publicly recorded crime need not become known to buyers: finding and interpreting incident information takes effort (Stigler, 1961). The agent results distinguish local knowledge from general experience. For homicide, the buyer-agent gradient in closing prices remains after controlling for total transactions, while total experience has no detectable gradient. The listing-agent gradient survives this control at only one of twelve windows. For narcotics, the asking-price gradient instead reflects listing agents' total experience, suggesting general pricing expertise. These patterns relate to evidence that agents' information advantages affect transaction outcomes (Levitt and Syverson, 2008), but nonrandom agent assignment prevents a causal interpretation of the gradients.

News coverage may also contribute to information reaching buyers. Homicide receives roughly fifteen times as many stories per incident as the next-highest offense in Media Cloud[21] and the largest article volume in GDELT,[22] and it carries the largest discount. The location split points the same way: the homicide discount is significant only for incidents on the street, in public view, although the residential estimate is too imprecise to rule out a similar effect (Tables A7 and A8).

[21] Media Cloud, an open-source platform for large-scale quantitative news analysis (`https://mediacloud.org`); the measure is the count of news articles mentioning an offense category over the sample period, divided by the number of recorded incidents in that category.

[22] The GDELT Project, a global open database that monitors and indexes news coverage worldwide (`https://www.gdeltproject.org`).

Yet category-level coverage does not measure whether buyers learn of a particular incident, and narcotics is priced despite far less coverage. Klimova and Lee (2014) find no additional house-price effect from more heavily covered murders in Sydney, whereas Moreno-Medina (2026), exploiting variation in local news coverage driven by sports schedules, finds that news coverage of nearby homicides lowers house prices. Coverage can therefore matter, but my data do not distinguish the relative roles of media, agents, and local contacts.

Neighborhood reputation may influence how buyers interpret an incident. Vandalism discounts are larger in more segregated tracts even after controlling for income (Table OA20). This parallels Sampson and Raudenbush (2004), who find greater perceived disorder in neighborhoods with more Black residents, holding observed disorder constant. The result is consistent with stigma amplifying the response to visible disorder, but does not establish that mechanism.

7.4 Welfare and policy

The estimates measure price responses among completed transactions, not average household willingness to pay to avoid crime. Buyers most averse to crime may withdraw from consideration, leaving a selected group of purchasers. Negotiated prices also reflect seller valuations and bargaining positions, while buyers may consider resale consequences as well as current occupancy. The transaction data do not separate these influences.

Evaluations of crime-reduction policies based only on sale prices may therefore omit important benefits. Longer marketing periods can entail carrying costs and search effort; withdrawals may represent forgone moves, postponement, or alternative uses of the property. Their welfare implications depend on what follows. Nor do price effects rank offenses by victim costs (McCollister et al., 2010). A small or undetected discount need not imply a small benefit from preventing an offense, particularly when buyers rarely learn of it.

Information disclosure raises a separate policy question. If observability contributes to price responses, crime maps or disclosure rules may affect property values without changing underlying crime. Illinois imposes no duty to disclose that a property was the site of a crime (225 ILCS 454/15-20), whereas in Hong Kong sellers must disclose unnatural deaths, which carry large and persistent price discounts (Bhattacharya et al., 2021). A resulting price decline need not indicate a welfare loss: buyers may simply be better informed about an existing risk. The analysis motivates this distinction but does not estimate the effects of disclosure on prices or welfare.

7.5 Scope

For offenses recorded through enforcement, price responses may reflect policing as well as underlying crime. Enforcement affects recorded offense counts (Levitt, 1998) and can itself affect local well-being (Ang, 2021; Legewie and Fagan, 2019). If buyers view concentrated drug or vice enforcement as a disamenity, the design cannot separate that response from the response to crime. Among the four priced offenses, this concern applies most directly to narcotics.

The estimates summarize the price effect per additional incident within 0.05 miles, relative to an incident 0.20–0.30 miles away. The distance-based design can be applied elsewhere, although the same radius may encompass different numbers of properties and blocks. Whether the magnitudes generalize depends on local density, street layout, housing-market conditions, and how buyers learn about nearby incidents. Responses also vary with neighborhood income within Chicago (Section 5.4), making the local income distribution relevant to comparisons across settings.

The rental and sale estimates differ, in some cases in sign as well as size, for several reasons. The smaller lease sample generally yields less precise estimates and consists of formal, MLS-listed units concentrated in condominiums (Table A1). Differences in property composition may therefore contribute to differences in treatment effects. Renters and buyers also differ demographically and may value safety differently. A lease buys occupancy, whereas a purchase also exposes the owner to future buyers' assessments through resale. Prices also form differently: asking rents reflect landlords' pricing decisions and anticipated demand, while sale prices reflect negotiation between buyers and sellers. These differences prevent attributing the sale–rent gap solely to resale exposure. The rental findings describe posted rents in the formal MLS market, which underrepresents the informal and lower-income rentals where crime exposure is highest, and need not generalize to them.

8 Conclusion

Only about one in five prospective homebuyers is willing to compromise on personal safety to afford a home (Anderson, 2025), yet I find that most types of crime do not measurably lower house prices. In repeat sales of the same property, with crime on its block compared against crime a few blocks away in the same neighborhood and month, sale-price discounts appear for only four of twenty-two offense categories. Neither severity nor measured recurrence alone explains which offenses are priced.

Three patterns inform this interpretation of why only some offenses are priced. First, price effects are more localized than the predictive relationships in the crime data: crime a few blocks away predicts subsequent crime near the property, yet detectable price effects are confined to the immediate block for burglary and narcotics and extend only a few blocks for homicide. Second, price responses tend to be larger when agents have local experience, most clearly for homicide, where closing discounts are larger when the buyer's agent has more experience in the neighborhood. Third, price effects vary with where an incident occurs: for burglary and narcotics, incidents at the frontage, visible from the street, are associated with lower prices, while those inside the dwelling have no detectable effect; the homicide discount is significant only for street incidents, although residential estimates are too imprecise to rule out a similar effect, and the vandalism discount is significant only for incidents at residential addresses. These patterns are consistent with buyers responding to incidents they learn about and regard as relevant to the property, with where and how incidents become observable differing across offenses.

The response to crime also differs across transaction outcomes. Asking prices respond little on average, while closing prices show discounts; there is some evidence that a nearby homicide lengthens time on market; and weapons violations increase withdrawals despite showing no consistent closing-price discount. Estimates based on completed sales alone can therefore miss part of the market's response.

Taken together, the results are consistent with price responses depending on whether buyers learn of an incident, regard it as relevant to the property, and revise their expectations of future exposure. An absent price discount may therefore reflect limited awareness, limited relevance to the property, no revision in expected exposure, adjustment on other transaction margins, or an effect too small to detect. Under the proposed framework, observed discounts need not equal willingness to pay to avoid crime, and property values need not rank offenses by the harm they impose. Static hedonic estimates of willingness to pay can also be biased when households anticipate changes in neighborhood crime (Bishop and Murphy, 2011). Measures of the benefits of crime reduction based on house prices therefore need not capture all of those benefits.

The addition or removal of crime maps from real estate platforms changes buyers' access to crime information. These changes offer a setting to test whether information affects how nearby crime is capitalized into house prices, while accounting for changes in underlying crime.[23]

[23] Zhang et al. (2022) propose a related design based on the introduction of public crime maps.

References

Dana Anderson. One in five house hunters is willing to compromise personal safety to afford a home. Redfin News, Aug 2025. Survey fielded May 2025. `https://www.redfin.com/news/survey-homebuyers-compromise-safety/`.

Elliot Anenberg and Edward Kung. Estimates of the size and source of price declines due to nearby foreclosures. *American Economic Review*, 104(8):2527–2551, 2014. doi: 10.1257/aer.104.8.2527.

Desmond Ang. The effects of police violence on inner-city students. *The Quarterly Journal of Economics*, 136(1):115–168, 2021.

Laura A. Bakkensen and Lint Barrage. Going underwater? Flood risk belief heterogeneity and coastal home price dynamics. *The Review of Financial Studies*, 35(8):3666–3709, 2022. doi: 10.1093/rfs/hhab122.

Panle Jia Barwick and Parag A. Pathak. The costs of free entry: An empirical study of real estate agents in Greater Boston. *RAND Journal of Economics*, 46(1):103–145, 2015. doi: 10.1111/1756-2171.12082.

Asaf Bernstein, Matthew T. Gustafson, and Ryan Lewis. Disaster on the horizon: The price effect of sea level rise. *Journal of Financial Economics*, 134(2):253–272, 2019. doi: 10.1016/j.jfineco.2019.03.013.

Utpal Bhattacharya, Daisy Huang, and Kasper Meisner Nielsen. Spillovers in prices: The curious case of haunted houses. *Review of Finance*, 25(3):903–935, 2021.

Kelly C. Bishop and Alvin D. Murphy. Estimating the willingness to pay to avoid violent crime: A dynamic approach. *American Economic Review: Papers and Proceedings*, 101(3):625–629, 2011. doi: 10.1257/aer.101.3.625.

Lyndsay N. Boggess, Robert T. Greenbaum, and George E. Tita. Does crime drive housing sales? Evidence from Los Angeles. *Journal of Crime and Justice*, 36(3):299–318, 2013. doi: 10.1080/0735648X.2013.812976.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Salience theory of choice under risk. *Quarterly Journal of Economics*, 127(3):1243–1285, 2012. doi: 10.1093/qje/qjs018.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Salience and consumer choice. *Journal of Political Economy*, 121(5):803–843, 2013. doi: 10.1086/673885.

Nils Braakmann. The link between crime risk and property prices in England and Wales: Evidence from street-level data. *Urban Studies*, 54(8):1990–2007, 2017. doi: 10.1177/0042098016634611.

Carolina Caetano. A test of exogeneity without instrumental variables in models with bunching. *Econometrica*, 83(4):1581–1600, 2015. doi: 10.3982/ECTA11231.

Carolina Caetano, Gregorio Caetano, Hao Fe, and Eric Nielsen. A dummy test of identification in models with bunching. Finance and Economics Discussion Series 2021-068, Board of Governors of the Federal Reserve System, 2021.

Gregorio Caetano and Vikram Maheshri. Identifying dynamic spillovers of crime with a causal approach to model selection. *Quantitative Economics*, 9(1):343–394, 2018. doi: 10.3982/QE756.

A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.

John Y. Campbell, Stefano Giglio, and Parag Pathak. Forced sales and house prices. *American Economic Review*, 101(5):2108–2131, 2011. doi: 10.1257/aer.101.5.2108.

Paul E. Carrillo and Anthony M. Yezer. What do hedonic house price estimates tell us when capitalization rates vary? IIEP Working Paper 2021-03, Institute for International Economic Policy, George Washington University, 2021.

Raj Chetty, Adam Looney, and Kory Kroft. Salience and taxation: Theory and evidence. *American Economic Review*, 99(4):1145–1177, 2009. doi: 10.1257/aer.99.4.1145.

Joshua M. Congdon-Hohman. The lasting effects of crime: The relationship of discovered methamphetamine laboratories and home values. *Regional Science and Urban Economics*, 43(1):31–41, 2013.

Timothy G. Conley. GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45, 1999.

Alexander Dentler and Enzo Rossi. Residents' willingness to pay to avoid crime. *Journal of Housing Economics*, 2024. Earlier version: Swiss National Bank Working Paper 2021-20, "Shooting Up Liquidity: The Effect of Crime on Real Estate".

Stuart Donovan, Thomas de Graaff, Henri L. F. de Groot, and Aaron Schiff. An urban overhead? Crime, agglomeration, and amenity. *Journal of Housing Economics*, 64:101994, 2024. doi: 10.1016/j.jhe.2024.101994.

Ingrid Gould Ellen, Johanna Lacoe, and Claudia Ayanna Sharygin. Do foreclosures cause crime? *Journal of Urban Economics*, 74:59–70, 2013. doi: 10.1016/j.jue.2012.09.003.

James Alan Fox and Marianne W. Zawitz. Homicide trends in the United States: 2000 update. Crime Data Brief NCJ 197471, Bureau of Justice Statistics, U.S. Department of Justice, 2003.

Xavier Gabaix. A sparsity-based model of bounded rationality. *Quarterly Journal of Economics*, 129(4):1661–1710, 2014. doi: 10.1093/qje/qju024.

Mark J. Garmaise and Tobias J. Moskowitz. Confronting information asymmetries: Evidence from real estate markets. *The Review of Financial Studies*, 17(2):405–437, 2004. doi: 10.1093/rfs/hhg037.

Pieter A. Gautier, Arjen Siegmann, and Aico van Vuuren. Terrorism and attitudes towards minorities: The effect of the Theo van Gogh murder on house prices in Amsterdam. *Journal of Urban Economics*, 65(2):113–126, 2009.

David Genesove and Christopher Mayer. Loss aversion and seller behavior: Evidence from the housing market. *Quarterly Journal of Economics*, 116(4):1233–1260, 2001.

Stephen Gibbons. The costs of urban property crime. *Economic Journal*, 114:F441–F463, 2004.

Sonia Gilbukh and Paul Goldsmith-Pinkham. Heterogeneous real estate agents and the housing cycle. *The Review of Financial Studies*, 37(11):3431–3489, 2024. doi: 10.1093/rfs/hhae048.

Jesse D. Gourevitch, Carolyn Kousky, Yanjun Liao, Christoph Nolte, Adam B. Pollack, Jeremy R. Porter, and Joakim A. Weill. Unpriced climate risk and the potential consequences of overvaluation in US housing markets. *Nature Climate Change*, 13(3):250–257, 2023. doi: 10.1038/s41558-023-01594-8.

Lu Han and Seung-Hyun Hong. Testing cost inefficiency under free entry in the real estate brokerage industry. *Journal of Business & Economic Statistics*, 29(4):564–578, 2011. doi: 10.1198/jbes.2011.08314.

Lu Han and William C. Strange. The microstructure of housing markets: Search, bargaining, and brokerage. In *Handbook of Regional and Urban Economics*, volume 5, pages 813–886. Elsevier, 2015.

Igal Hendel, Aviv Nevo, and François Ortalo-Magné. The relative performance of real estate marketing platforms: MLS versus FSBOMadison.com. *American Economic Review*, 99(5):1878–1898, 2009. doi: 10.1257/aer.99.5.1878.

Miyuki Hino and Marshall Burke. The effect of information about climate risk on property values. *Proceedings of the National Academy of Sciences*, 118(17):e2003374118, 2021. doi: 10.1073/pnas.2003374118.

Keith Ihlanfeldt and Tom Mayock. Panel data estimates of the effects of different types of crime on housing prices. *Regional Science and Urban Economics*, 40(2–3):161–172, 2010.

Anastasia Klimova and Adrian D. Lee. Does a nearby murder affect housing prices and rents? The case of Sydney. *Economic Record*, 90(S1):16–40, 2014. doi: 10.1111/1475-4932.12118.

John R. Knight. Listing price, time on market, and ultimate selling price: Causes and effects of listing price changes. *Real Estate Economics*, 30(2):213–237, 2002.

Bishwa Koirala, Rik Chakraborti, and Gyan Pradhan. Urban crime and its net implicit price to North Carolina households. *American Business Review*, 26(1):83–94, 2023. doi: 10.37625/abr.26.1.83-94.

Ivan Korolev. Estimating semiparametric and nonparametric fixed effects panel data models with `mgcv`, 2026. Working paper, arXiv:2606.12739.

Nicolai V. Kuminoff and Jaren C. Pope. Do "capitalization effects" for public goods reveal the public's willingness to pay? *International Economic Review*, 55(4):1227–1250, 2014. doi: 10.1111/iere.12088.

Pablo Kurlat and Johannes Stroebel. Testing for information asymmetries in real estate markets. *The Review of Financial Studies*, 28(8):2429–2461, 2015. doi: 10.1093/rfs/hhv028.

Joscha Legewie and Jeffrey Fagan. Aggressive policing and the educational performance of minority youth. *American Sociological Review*, 84(2):220–247, 2019.

Steven D. Levitt. Why do increased arrest rates appear to reduce crime? Deterrence, incapacitation, or measurement error? *Economic Inquiry*, 36(3):353–372, 1998.

Steven D. Levitt and Chad Syverson. Market distortions when agents are better informed: The value of information in real estate transactions. *The Review of Economics and Statistics*, 90(4):599–611, 2008.

Leigh Linden and Jonah E. Rockoff. Estimates of the impact of crime risk on property values from Megan's Laws. *American Economic Review*, 98(3):1103–1127, 2008.

Paula Margaretic and Juan Bautista Sosa. How local is the crime effect on house prices? *The Journal of Real Estate Finance and Economics*, 70:754–793, 2025. doi: 10.1007/s11146-023-09941-4.

Kathryn E. McCollister, Michael T. French, and Hai Fang. The cost of crime to society: New crime-specific estimates for policy and program evaluation. *Drug and Alcohol Dependence*, 108(1–2):98–109, 2010.

Antonio Merlo and François Ortalo-Magné. Bargaining over residential real estate: evidence from england. *Journal of Urban Economics*, 56(2):192–216, 2004.

Jonathan Moreno-Medina. Crime news bias: Extent and effects on housing markets. Working paper, University of Texas at San Antonio, January 2026.

Jaren C. Pope. Fear of crime and housing prices: Household reactions to sex offender registries. *Journal of Urban Economics*, 64(3):601–614, 2008a.

Jaren C. Pope. Buyer information and the hedonic: The impact of a seller disclosure on the implicit price for airport noise. *Journal of Urban Economics*, 63(2):498–516, 2008b.

Mario J. Rizzo. The effect of crime on residential rents and property values. *The American Economist*, 23(1):16–21, 1979.

Sherwin Rosen. Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1):34–55, 1974.

Ronald C. Rutherford, Thomas M. Springer, and Abdullah Yavaş. Conflicts between principals and agents: Evidence from residential brokerage. *Journal of Financial Economics*, 76(3):627–665, 2005. doi: 10.1016/j.jfineco.2004.06.006.

Robert J. Sampson and Stephen W. Raudenbush. Seeing disorder: Neighborhood stigma and the social construction of "broken windows". *Social Psychology Quarterly*, 67(4):319–342, 2004.

Amy Ellen Schwartz, Scott Susin, and Ioan Voicu. Has falling crime driven New York City's real estate boom? *Journal of Housing Research*, 14(1):101–135, 2003.

George J. Stigler. The economics of information. *Journal of Political Economy*, 69(3):213–225, 1961.

Johannes Stroebel. Asymmetric information about collateral values. *The Journal of Finance*, 71 (3):1071–1112, 2016. doi: 10.1111/jofi.12288.

Cheng Keat Tang and Thao Le. Crime risk and housing values: Evidence from the gun offender registry. *Journal of Urban Economics*, 134:103526, 2023. doi: 10.1016/j.jue.2022.103526.

Richard Thaler. A note on the value of crime control: Evidence from the property market. *Journal of Urban Economics*, 5(1):137–145, 1978.

Scott A. Wentland, Bennie D. Waller, and Raymond T. Brastow. Estimating the effect of crime risk on property values and time on market: Evidence from Megan's Law in Virginia. *Real Estate Economics*, 42(1):223–251, 2014. doi: 10.1111/1540-6229.12028.

Simon N. Wood. *Generalized Additive Models: An Introduction with R*. Chapman and Hall/CRC, 2 edition, 2017.

Meng Le Zhang, Monsuru Adepeju, and Rhiannon Thomas. Estimating the effect of crime (maps) on house prices using a (un)natural experiment. SocArXiv preprint 9zupw, 2022. Study protocol.

Appendix

This appendix reports supporting tables and figures, grouped in the order of the main text. Unless a note says otherwise, each reports $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$, the percent sale-price change per incident within 0–0.05 miles of the property against 0.20–0.30 miles, with parcel and tract × year-month fixed effects and standard errors clustered by census tract; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All other supporting material is in the Online Appendix.

A Data and Offense Coverage

Table A1: Summary Statistics

	Mean	SD	p10	Median	p90	Obs.
A. Sale listings ($N = 530,395$)						
Close price ($)	366,777	382,539	74,000	279,000	700,000	530,395
Log close price	12.44	0.92	11.21	12.54	13.46	530,395
Original list price ($)	384,828	410,804	85,000	289,900	725,000	529,956
Days on market	96	147	5	42	248	529,995
Any list-price cut	0.68	0.47	0.00	1.00	1.00	530,395
Bedrooms	3.0	1.7	1.0	3.0	5.0	517,982
Bathrooms	2.1	1.1	1.0	2.0	3.0	530,395
Living area (sq. ft.)	1,175	1,115	0	1,100	2,483	454,018
Total rooms	7.1	3.2	4.0	6.0	11.0	517,753
Year built	1954	41	1902	1954	2007	453,343
Garage spaces	1.6	0.6	1.0	2.0	2.0	351,747
Fireplaces	0.9	0.7	0.0	1.0	2.0	206,694
B. Lease listings ($N = 191,155$)						
Monthly rent ($)	2,208	1,086	1,250	1,950	3,400	191,155
Log monthly rent	7.61	0.41	7.13	7.58	8.13	191,155
Days on market	43	57	6	25	102	191,122
Bedrooms	1.8	1.0	1.0	2.0	3.0	191,031
Bathrooms	1.6	0.8	1.0	1.0	2.0	191,155
Living area (sq. ft.)	883	675	0	900	1,691	191,041
Total rooms	4.7	1.4	3.0	5.0	6.0	191,032
Year built	1972	37	1912	1982	2008	150,337

Notes: Chicago MLS single-family, condominium, and residential-income listings, 2007–2025, on parcels with a constant parcel count. Panel A: closed sales ($10,000–$10,000,000). Panel B: leases (monthly rent $500–$10,000). Any list-price cut: at least one reduction before sale.

A.1 Offense Coverage

I estimate separate effects for 27 of the 29 offense groups, omitting human trafficking and nonforcible sex offenses (222 incidents combined) as too sparse.[24] Of these 27, I report 22 in the main analysis. I set aside the residual "all other offenses" category, which names no specific offense, and four offenses, kidnapping, embezzlement, extortion, and child pornography, whose estimates are too imprecise to be informative.[25]

A.2 Neighborhood Control Data

Tree canopy is percent canopy cover from the U.S. Geological Survey National Land Cover Database (NLCD) Tree Canopy Cover product (30 m resolution). Near and far canopy are averages over the 30 m cells within the 0–0.05 mile disk and the 0.20–0.30 mile ring. Because the canopy data are annual and the twelve-month crime window can extend into the previous calendar year, I use the raster from two calendar years before the sale, the latest year that ends before the crime window begins. Business licenses, 311 service requests, and building permits are geocoded records from the City of Chicago Data Portal (`https://data.cityofchicago.org`). Before the City's unified 311 system launched in December 2018, the portal releases requests in category-specific historical files; the 311 control combines those files for 2011–2018 (graffiti removal, sanitation code complaints, vacant and abandoned buildings, and abandoned vehicles) with requests of all types from the unified system for 2019–2024, so its composition changes in 2019. Vacant and abandoned buildings are "Vacant and Abandoned Buildings Reported" records from the same portal; because such complaints also enter the City's unified 311 stream from 2019 onward, this measure partly overlaps the total-311 control over the recent period.

[24] The pornography/obscene-material group is predominantly child pornography rather than adult material: about two-thirds of the Chicago Police incidents so classified are child pornography (IUCR 1582) or related offenses involving children, and roughly one-quarter is adult obscene material (obscene telephone calls are classified separately, under all other offenses). I therefore label this group "child pornography/obscenity" throughout. It is sparse, returns nulls in every market, and its estimates appear only in the appendix.

[25] Their standard errors run from roughly 0.6 to 1.3 percentage points, too wide to rule out either a null or a large discount.

B Sale-Price Estimates

Table A2: Sale-Price Capitalization, All Twenty-Two Estimable Offenses

Offense	Identifying parcels	Sig. discount windows	Min δ (s.e.) [mo]	Max δ (s.e.) [mo]
Priced				
Homicide	5,701	2/12	-5.86^{*} (3.35) [1]	-1.24 (1.43) [8]
Narcotics	53,844	5/12	-0.93 (0.75) [1]	-0.18 (0.19) [6]
Burglary	88,094	6/12	-0.34^{**} (0.14) [6]	-0.18^{*} (0.09) [12]
Vandalism	104,661	2/12	-0.17^{**} (0.08) [6]	0.26 (0.22) [1]
Suggestive				
Disorderly conduct	19,550	1/12	-1.33^{*} (0.78) [2]	-0.46 (0.54) [4]
Sex offenses	36,393	1/12	-0.47^{**} (0.23) [12]	0.11 (0.35) [3]
Precise nulls				
Theft	115,548	0/12	0.01 (0.02) [12]	0.08^{***} (0.03) [1]
Motor vehicle theft	85,781	0/12	-0.02 (0.09) [12]	0.51^{*} (0.30) [1]
Assault (non-dom.)	91,783	0/12	-0.05 (0.06) [7]	0.01 (0.11) [3]
Bad checks	8,919	0/12	0.66^{*} (0.35) [11]	2.20^{**} (1.03) [1]
Family (nonviol.)	15,928	0/12	1.00^{*} (0.51) [12]	2.16^{***} (0.82) [5]
Fraud	67,030	0/12	-0.36 (0.23) [1]	-0.04 (0.09) [6]
Assault (domestic)	90,500	0/12	-0.19^{*} (0.10) [12]	0.02 (0.15) [5]
Trespass	54,025	0/12	-0.15 (0.14) [6]	0.18 (0.22) [3]
Robbery	65,876	0/12	-0.06 (0.14) [9]	0.30 (0.42) [1]
Forgery	23,614	0/12	-0.07 (0.59) [1]	0.47 (0.45) [2]
Liquor	6,393	0/12	0.08 (1.20) [1]	1.34 (0.91) [2]
Prostitution	6,326	0/12	-1.00 (0.82) [1]	0.08 (0.12) [12]
Inconclusive				
Weapons	32,294	0/12	-0.62 (0.47) [5]	0.11 (0.58) [3]
Arson	6,116	0/12	-0.20 (1.06) [11]	1.27 (3.58) [1]
Gambling	3,125	0/12	-4.77 (4.33) [3]	1.99 (8.32) [1]
Stolen property	1,262	0/12	-2.23 (2.63) [2]	0.81 (1.20) [8]

Notes. All twenty-two estimable offenses under the specification of Table 2; kidnapping, embezzlement, extortion, child pornography, and other offenses have too few incidents to estimate and are excluded; columns as there, with Min δ and Max δ the most negative and most positive estimates across the twelve windows; offense groups as in Table 2: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Sig. discount windows counts windows with a significantly negative estimate at five percent, so positive significant estimates (bad checks, family offenses) count as zero.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A3: Sale-Price Capitalization at All Twelve Exposure Windows

	Exposure window (cumulative months)											
Offense	1	2	3	4	5	6	7	8	9	10	11	12
Priced												
Homicide	−5.86*	−2.23	−1.45	−2.04	−1.28	−1.30	−1.59	−1.24	−2.54*	−3.02**	−3.15**	−2.49*
	(3.35)	(2.60)	(2.46)	(2.05)	(1.82)	(1.77)	(1.53)	(1.43)	(1.36)	(1.38)	(1.38)	(1.36)
Narcotics	−0.93	−0.37	−0.35	−0.32	−0.28	−0.18	−0.20	−0.38**	−0.31**	−0.29**	−0.35***	−0.33***
	(0.75)	(0.42)	(0.38)	(0.27)	(0.23)	(0.19)	(0.17)	(0.18)	(0.15)	(0.14)	(0.13)	(0.11)
Burglary	−0.20	−0.23	−0.20	−0.21	−0.25*	−0.34**	−0.30**	−0.24**	−0.22**	−0.22**	−0.22**	−0.18*
	(0.31)	(0.20)	(0.17)	(0.16)	(0.15)	(0.14)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.09)
Vandalism	0.26	0.13	−0.06	−0.11	−0.16*	−0.17**	−0.16**	−0.13*	−0.10	−0.08	−0.06	−0.05
	(0.22)	(0.15)	(0.12)	(0.10)	(0.09)	(0.08)	(0.08)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)
Suggestive												
Disorderly conduct	−1.22	−1.33*	−0.65	−0.46	−0.71	−0.66	−0.58	−0.50	−0.63*	−0.60*	−0.57*	−0.65**
	(1.03)	(0.78)	(0.63)	(0.54)	(0.48)	(0.44)	(0.38)	(0.35)	(0.33)	(0.32)	(0.29)	(0.29)
Sex offenses	−0.08	0.10	0.11	−0.02	−0.29	−0.31	−0.25	−0.20	−0.26	−0.39	−0.43*	−0.47**
	(0.64)	(0.42)	(0.35)	(0.32)	(0.31)	(0.27)	(0.26)	(0.25)	(0.25)	(0.25)	(0.24)	(0.23)
Precise nulls												
Larceny/theft	0.08***	0.04*	0.04*	0.03**	0.03*	0.02	0.02	0.02	0.01	0.02	0.01	0.01
	(0.03)	(0.02)	(0.02)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)	(0.02)	(0.01)	(0.02)	(0.02)
Motor vehicle theft	0.51*	0.28	0.17	0.10	0.05	0.04	0.09	0.09	0.05	0.06	0.07	−0.02
	(0.30)	(0.22)	(0.18)	(0.16)	(0.14)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.09)	(0.09)
Assault (non-dom.)	−0.05	−0.02	0.01	−0.03	−0.04	−0.04	−0.05	−0.03	−0.02	−0.01	−0.03	−0.04
	(0.20)	(0.14)	(0.11)	(0.09)	(0.08)	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)	(0.04)	(0.04)
Bad checks	2.20**	2.00***	1.36**	1.38**	1.09**	1.18***	1.08**	0.97**	0.77*	0.72**	0.66*	0.66**
	(1.03)	(0.74)	(0.61)	(0.57)	(0.53)	(0.45)	(0.43)	(0.41)	(0.39)	(0.37)	(0.35)	(0.33)
Family (non-violent)	1.99	1.07	1.16	1.68*	2.16***	1.77**	1.89***	1.68***	1.49**	1.28**	1.07*	1.00*
	(1.87)	(1.33)	(1.09)	(0.92)	(0.82)	(0.75)	(0.66)	(0.62)	(0.60)	(0.57)	(0.55)	(0.51)
Fraud	−0.36	−0.29*	−0.11	−0.07	−0.06	−0.04	−0.05	−0.05	−0.05	−0.08	−0.11	−0.11
	(0.23)	(0.17)	(0.12)	(0.10)	(0.10)	(0.09)	(0.10)	(0.09)	(0.09)	(0.08)	(0.08)	(0.08)
Assault (domestic)	0.02	0.00	−0.02	0.02	0.02	0.01	−0.03	−0.09	−0.17	−0.13	−0.18*	−0.19*
	(0.37)	(0.26)	(0.21)	(0.17)	(0.15)	(0.13)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.10)
Trespass	−0.09	0.16	0.18	0.00	−0.10	−0.15	−0.14	−0.12	−0.06	−0.05	−0.09	−0.14
	(0.38)	(0.29)	(0.22)	(0.18)	(0.15)	(0.14)	(0.13)	(0.12)	(0.11)	(0.10)	(0.10)	(0.10)
Robbery	0.30	0.15	0.09	0.08	0.04	0.02	−0.02	−0.03	−0.06	−0.01	0.01	−0.03
	(0.42)	(0.30)	(0.24)	(0.22)	(0.19)	(0.18)	(0.16)	(0.15)	(0.14)	(0.13)	(0.13)	(0.13)
Forgery	−0.07	0.47	0.12	0.15	0.05	0.38	0.44*	0.39*	0.30	0.30*	0.23	0.21
	(0.59)	(0.45)	(0.38)	(0.32)	(0.28)	(0.24)	(0.22)	(0.20)	(0.19)	(0.18)	(0.18)	(0.17)
Liquor law	0.08	1.34	0.71	0.72	0.47	0.39	0.61	0.65	0.64*	0.51	0.36	0.36
	(1.20)	(0.91)	(0.81)	(0.60)	(0.52)	(0.48)	(0.43)	(0.40)	(0.38)	(0.36)	(0.35)	(0.35)
Prostitution	−1.00	−0.43	−0.23	−0.23	−0.18	−0.28	−0.14	−0.04	0.01	0.01	0.05	0.08
	(0.82)	(0.51)	(0.48)	(0.39)	(0.35)	(0.33)	(0.30)	(0.27)	(0.22)	(0.18)	(0.15)	(0.12)
Inconclusive												
Weapons	−0.47	0.07	0.11	−0.51	−0.62	−0.39	−0.18	−0.12	−0.17	0.04	0.09	0.10
	(0.92)	(0.71)	(0.58)	(0.48)	(0.47)	(0.44)	(0.40)	(0.37)	(0.35)	(0.31)	(0.30)	(0.28)
Arson	1.27	0.32	0.27	0.10	−0.13	0.07	1.07	0.57	−0.15	−0.13	−0.20	0.07
	(3.58)	(2.75)	(2.33)	(2.01)	(1.78)	(1.63)	(1.49)	(1.31)	(1.22)	(1.15)	(1.06)	(1.03)
Gambling	1.99	1.69	−4.77	−1.39	−1.35	−0.76	−1.17	−3.04	−2.84	−2.91*	−1.79	−1.94
	(8.32)	(5.78)	(4.33)	(3.99)	(3.39)	(3.12)	(2.74)	(2.43)	(2.10)	(1.75)	(1.77)	(1.75)
Stolen property	−2.23	−2.23	−1.78	−2.02	−1.14	−0.67	0.26	0.81	0.08	0.19	−0.25	−0.43
	(4.06)	(2.63)	(2.18)	(1.95)	(1.83)	(1.75)	(1.59)	(1.20)	(1.06)	(1.02)	(1.05)	(1.05)

Notes. δ at every cumulative window from one to twelve months, under the specification of Table 2; offense groups as there: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Covers the twenty-two estimable categories, including the white-collar offenses, prostitution, gambling, and stolen property. Standard errors in parentheses.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Priced Offenses under Two Classifications

Offense	Priced	What is threatened	NIBRS crimes against
Homicide	Yes	Place (violent event)	Persons
Burglary	Yes	Dwelling	Property
Narcotics	Yes	Place (environment)	Society
Vandalism	Yes	Structure or land	Property
Disorderly conduct	Suggestive	Place (environment)	Group B (none)
Sex offenses	Suggestive	Person	Persons
Assault (domestic)	No	Person	Persons
Assault (non-domestic)	No	Person	Persons
Robbery	No	Person	Property
Theft	No	Portable asset	Property
Motor vehicle theft	No	Portable asset	Property
Stolen property	No	Portable asset	Property
Fraud	No	Financial	Property
Forgery	No	Financial	Property
Bad checks	No	Financial	Group B (none)
Arson	No	Structure or land	Property
Trespass	No	Structure or land	Group B (none)
Prostitution	No	Place (environment)	Society
Weapons	No	Place (environment)	Society
Gambling	No	Place (environment)	Society
Liquor law	No	Place (environment)	Group B (none)
Family offenses	No	Person	Group B (none)

Priced offenses by group
Anchored to the address: 4 of 11; harms a person, money, or a movable item: 0 of 11
NIBRS (Group A): persons 1 of 4; property 2 of 9; society 1 of 4; Group B offenses (5) have no crime-against category.

Notes. "What is threatened" follows Table OA15. "NIBRS crimes against" is the Federal Bureau of Investigation's National Incident-Based Reporting System category for each offense. "Priced" follows Table 2. NIBRS assigns a crime-against category only to Group A offenses.

B.1 Offenses with Monetary Losses

Fraud, forgery, and bad checks impose direct monetary losses, yet none carries a detectable discount in the pooled estimates (Table OA3); embezzlement has too few incidents to estimate. Bad checks instead carry a positive and significant estimate (0.66, s.e. 0.33), positive at five percent in ten of twelve windows (Table A3). Splitting each offense's incidents by recorded location (residential, street, and commercial) leaves the picture unchanged (Table A9). At the 12-month window, bad checks have no detectable effect in any location category, and forgery's only significant estimate is positive and occurs at commercial addresses.

Fraud is associated with a residential discount of 0.40 percent (s.e. 0.15) that survives the post-sale placebo. This estimate is one of sixty-six location-specific comparisons, does not survive a Bonferroni correction, and appears only at the longest exposure windows, so I treat it as suggestive.

Table A5: Effects of Crime on Housing Prices in the Prior Literature, Compared with This Paper

Study	Setting	Design	Offense	Spatial unit of crime	Estimate as reported
Sale prices, aggregate crime rates					
Thaler (1978)	Rochester, NY	Cross-sectional hedonic	Property crime	Census tract	$\approx 3\%$ per SD†
Gray & Joelson (1979)	Minneapolis	Cross-sectional hedonic, tract	Eight offense types	Tract	Only vandalism and burglary significant: +1% burglary −\$336; +1% vandalism −\$117†
Burnell (1988)	Chicago suburbs	IV, community fiscal and demographic instruments	Property crime	Community (71)	Elasticity $-0.10^{\dagger}$
Gibbons (2004)	London	Hedonic, deviations from district means, IV	Criminal damage	250 m radius	$\approx 1\%$ per 0.1 SD
Gibbons (2004)	London	Hedonic, deviations from district means, IV	Burglary	250 m radius	Null
Schwartz, Susin & Voicu (2003)	New York City	Hedonic and repeat-sales price models	Violent (murder, rape, robbery, assault)	Precinct (74)	Elasticity 0.13; $\approx$ one third of the post-1994 boom
Schwartz, Susin & Voicu (2003)	New York City	Hedonic and repeat-sales price models	Property (burglary, larceny, auto theft)	Precinct (74)	Negative, insignificant
Tita, Petras & Greenbaum (2006)	Columbus, OH	Hedonic, IV (murder rate), by neighborhood income	Violent, property, total	Tract (189)	Violent crime most costly; effects vary by income
Ihlanfeldt & Mayock (2010)	Miami-Dade	First differences of tract repeat-sales index, IV	Murder, assault, robbery	Tract	Assault −0.15, robbery −0.11 (elasticities); murder null
Ihlanfeldt & Mayock (2010)	Miami-Dade	First differences of tract repeat-sales index, IV	Property	Tract	Small or null
Bishop & Murphy (2011)	San Francisco Bay Area	Dynamic hedonic (structural)	Violent (homicide, rape, robbery, assault)	City (79), distance-weighted	WTP $\approx$ \$472/yr per 10% less
Pope & Pope (2012)	$\approx$3,000 U.S. zip codes	Zip-code fixed effects and IV, 1990s crime drop	Violent and property	Zip code	Elasticity −0.15 to −0.35
Sale prices, point-located events					
Linden & Rockoff (2008)	Mecklenburg Co., NC	DiD, neighborhood-year FE	Sex-offender arrival	0.1 mile	$\approx -4\%$
Pope (2008a)	Hillsborough Co., FL	DiD around offender arrival and departure	Sex-offender arrival	0.1 mile	-2.3%
Wentland, Waller & Brastow (2014)	Central Virginia	3SLS (price and time on market jointly)	Sex-offender proximity	0.1 mile	Price $\approx -7\%$; time on market +88 days ($\approx +80\%$)
Congdon-Hohman (2013)	Akron, OH	DiD, near vs. slightly farther	Methamphetamine-lab discovery	1/8 mile (vs. 1/8–1/4)	-10% to -19%
Dealy, Horn & Berrens (2017)	Linn County, OR	Spatial difference-in-differences	Meth-lab discovery and decontamination	0.1 mile (near vs. far)	-6.5% on discovery; +5% after decontamination
Gautier, Siegmann & van Vuuren (2009)	Amsterdam	DiD on asking prices	Murder (van Gogh)	Neighborhood (Muslim share $> 25\%$)	$\approx -3\%$ within 10 months

Continued on next page

Table A5, continued

Study	Setting	Design	Offense	Spatial unit of crime	Estimate as reported
Bhattacharya, Huang & Nielsen (2021)	Hong Kong	Unit FE (repeat sales), before/after death	Murder in unit	Unit, floor, building	Murder: unit ≈ −34%; floor −19%; building −10% (Table VI); all unnatural deaths: unit −20%
Tang & Le (2023)	Baltimore	Before/after offender arrival	Ex-gun-offender arrival	0.1 mile	−8.0% ($12,632); no rise in crime
Braakmann (2017)	England and Wales	Street FE, regional time trends	Six categories (anti-social behaviour, violent, other, burglary, robbery, vehicle)	Street	Violent −0.6 to −1.6%, anti-social −0.6 to −0.8% per case per 10 residents; robbery, burglary, vehicle ≈ 0 or positive
Dentler & Rossi (2021)	Rochester, NY	Discontinuity in time (±30–60 days) within a distance band	Robbery	0.3–0.6 km	−8 to −10%
			Aggravated assault		−7 to −9%
			Burglary		≈ −6%
			Shootings		Insignificant
Rents					
Rizzo (1979)	Chicago	Cross-section, IV	Total crime	Community area	Value elasticity ≈ −1.67†; rent elasticity −0.24†
Koirala, Chakraborti & Pradhan (2023)	North Carolina cities	Seemingly unrelated regressions of wages and rents	Violent and property rates	City	≈ $52/month net compensation
Donovan et al. (2024)	New Zealand	Cross-section of urban areas, spatial equilibrium, control-function IV	Victimizations per 1,000	Urban area (134)	Rent elasticity −0.15 to −0.44
Klimova & Lee (2014)	Sydney	Hedonic DiD	Murder	0.1–0.3 mile	Rents: no effect; values −3.9% within 0.2 mile
Carrillo & Yezer (2021)	Washington, DC	Rent and value hedonics, tract FE	Homicide, violent	0.25 mile	Not directly comparable
This paper					
This paper	Chicago	Repeat sales, near–far, parcel + tract×month FE	Homicide	0.05 mile vs. 0.20–0.30 mile	−3.0% per incident (−3.0 to −3.1%, 10–11 months)
			Burglary		−0.24% (−0.22 to −0.34%, 6–11 months)
			Narcotics		−0.29% (−0.29 to −0.38%, 8–12 months)
			Vandalism		−0.17% (−0.16 to −0.17%, 6–7 months)
			Sex offenses; disorderly conduct		−0.47%; −0.65% (12 months only; suggestive)
			Other offenses		No detectable discount
			Rents (burglary; sex offenses; liquor law)		−0.34 to −0.07% (6 of 12 windows); −0.32 to −0.16% (7 of 12, parcel×era); −0.30 to 0.09% (4 of 12, parcel×era)

Notes. Estimates are in the units of the original study and are not on a common scale. Prior estimates can be compared with this paper's estimates for the same offense in the final panel; ranges cover the windows significant at five percent (Table A3); rent ranges are the minimum and maximum across the twelve windows with the number of significant windows (Table A23). Distance bands for Pope (2008a) are inferred from the design. Dentler & Rossi (2021): Model (4) of their Table 3; their simpler models are mostly insignificant. Outcomes are sale prices except asking prices (Gautier, Siegmann & van Vuuren, 2009), a tract repeat-sales price index (Ihlanfeldt & Mayock, 2010), owner-reported values and rents (Rizzo, 1979), owner-estimated median values (Gray & Joelson, 1979; Burnell, 1988), sale prices and asking rents (Klimova & Lee, 2014), rents and wages (Koirala, Chakraborti & Pradhan, 2023; Donovan et al., 2024), rents and values (Carrillo & Yezer, 2021), and housing values (Tang & Le, 2023); Bishop & Murphy (2011) estimate willingness to pay from sale prices in a dynamic model. This paper uses sale prices, asking rents, and transaction margins. †Reported in a secondary source.

C Spatial and Temporal Structure

Spatial decay. Table A6 sets the near ring to each successive ring while holding the far ring at 0.20–0.30 miles. Widening burglary's near ring to 0–0.10 miles gives −0.14 percent (s.e. 0.07) at eight months, significant in 5 of 12 windows, with no significant post-sale placebo; the 0.05–0.10 ring alone is significant only at one month. Homicide discounts are significant in the 0.10–0.15 mile ring in seven of twelve windows (months three to nine), but not at ten months, where the estimate is −1.16 percent (s.e. 0.64). The post-sale placebo is not significant in that ring at any window, and homicide is not significant in the 0.05–0.10 ring at any window. For narcotics, the pre-sale estimate in the 0.05–0.10 ring is not significant at any window, but the post-sale placebo is significant there at windows four to eight.

Table A6: The Near–Far Discount Across Nine Ring Definitions

Near vs. far ring (miles)	Homicide	Burglary	Narcotics	Vandalism
0–0.05 vs. 0.20–0.30 *(baseline)*	-3.02 (1.38)	-0.24 (0.12)	-0.29 (0.14)	-0.17 (0.08)
0–0.05 vs. 0.25–0.35	-2.75 (1.39)	-0.24 (0.12)	-0.32 (0.14)	-0.16 (0.08)
0–0.05 vs. 0.30–0.40	-2.66 (1.35)	-0.27 (0.12)	-0.32 (0.14)	-0.15 (0.08)
0–0.10 vs. 0.20–0.30	-0.42 (0.67)	-0.14 (0.07)	0.01 (0.06)	-0.06 (0.05)
0–0.15 vs. 0.20–0.30	-0.90 (0.54)	-0.05 (0.05)	0.05 (0.04)	0.01 (0.03)
0–0.20 vs. 0.20–0.30	-0.74 (0.46)	-0.03 (0.04)	0.02 (0.03)	-0.02 (0.03)
0.05–0.10 vs. 0.20–0.30	0.55 (0.78)	-0.09 (0.07)	0.06 (0.07)	-0.02 (0.05)
0.10–0.15 vs. 0.20–0.30	-1.16 (0.64)	0.04 (0.06)	0.06 (0.05)	0.07 (0.04)
0.15–0.20 vs. 0.20–0.30	-0.45 (0.58)	-0.00 (0.05)	-0.02 (0.04)	-0.05 (0.04)

Notes: Each row is one near/far ring pair: the baseline (row 1), the far ring moved out (rows 2–3), the near ring widened (rows 4–6), and an outer ring in place of the near ring (rows 7–9). Windows: homicide 10, burglary 8, narcotics 10, vandalism 6 months. $N = 242{,}815$.

C.1 Recent versus Earlier Exposure

Estimated jointly, exposure two to six months before the sale and exposure eight to eleven months before carry different weights across offenses. The vandalism discount is concentrated in the recent block and absent from the late one, and the difference is significant ($p = 0.008$); the narcotics discount is concentrated in the late block, with the difference significant at ten percent ($p = 0.09$); for homicide and burglary the point estimates are larger in the late and early block respectively, but the two blocks are not statistically distinguishable ($p = 0.12$ and 0.20; Figure A1).

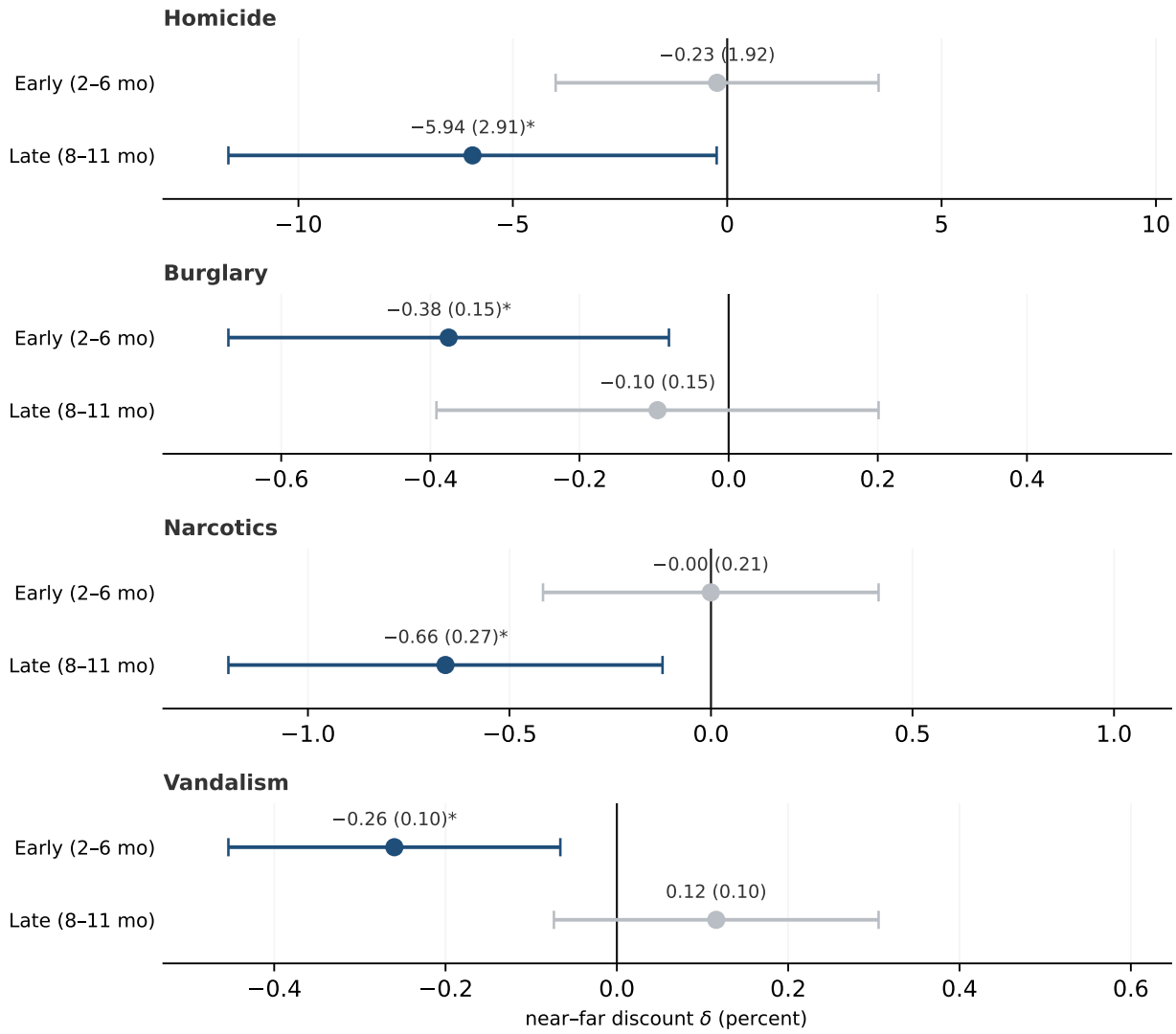


Figure A1: Early versus late exposure, estimated jointly. For each priced offense, the near–far discount δ (%) from one regression that enters an early block (incidents two to six months before the sale) and a late block (eight to eleven months before) as separate counts, each with its far-ring counterpart and separate controls for the omitted months (one, seven, twelve). Month one is omitted because it falls after the contract date for most sales: the median time from contract to close is 42 days, and for about three-quarters of sales it exceeds 30 days, so crime in the last month before the close could affect the agreed price only through renegotiation. Month seven separates the two blocks, and month twelve ends the window. A single late block of months seven to twelve, estimated alone, gives the same pattern (homicide −3.60%, s.e. 2.13; narcotics −0.47%, s.e. 0.20; burglary −0.03% and vandalism 0.06%, neither significant). The two points in each panel are the early-block and late-block estimates (significant at five percent in colour, otherwise grey), shown with 95% confidence intervals and the estimate and standard error beside each point; each panel has its own horizontal scale. The Wald test that the early and late coefficients are equal gives $p = 0.121$ (homicide), 0.200 (burglary), 0.090 (narcotics), and 0.008 (vandalism), so the early and late estimates are individually distinguishable only for vandalism. Parcel and tract×year-month fixed effects, clustered by census tract, $N = 242{,}815$.

D Observability and Recurrence

Table A7: Where the Discount Comes From: Private versus Public Incidents

Offense	Private (residential)	Public (street)	p: priv.=pub.	Windows sig.	N
Homicide	−0.41	−4.59***	0.137	1/12	242,815
	(2.39)	(1.68)			
Burglary	−0.25*	−0.93	0.289	0/12	242,815
	(0.14)	(0.65)			
Narcotics	−0.28	−0.38**	0.897	0/12	242,815
	(0.65)	(0.19)			
Vandalism	−0.56***	−0.01	0.018**	7/12	242,815
	(0.20)	(0.11)			

Notes. Near–far close-price effect (%) of each priced offense's near-ring count, split by whether the incident was recorded at a *residential* address (private: the dwelling and its garage, porch, yard, hallways, and stairwells), in *public space* (street and other public way), or at a commercial address; residential, street, and commercial incidents are entered jointly in one regression under parcel fixed effects, and the commercial estimates are reported in Table A9. Pooled 2007–2025, parcel and tract×month fixed effects, near 0–0.05 vs. far 0.20–0.30 mi; standard errors clustered by tract in parentheses. Estimates at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months); "Windows sig." counts, of the twelve exposure windows, how many reject private=public at five percent. $N = 242{,}815$.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A8: Residential versus Street Incidents by Exposure Window, Priced Offenses

Panel A						
	Homicide			Narcotics		
Window	Residential	Street	p	Residential	Street	p
1	0.19	−8.29*	0.227	−2.61	−0.65	0.447
	(5.62)	(4.29)		(2.63)	(0.81)	
2	3.47	−4.75	0.099	−1.10	−0.26	0.647
	(3.97)	(3.42)		(1.69)	(0.50)	
3	0.80	−2.23	0.526	−1.16	−0.26	0.520
	(3.36)	(3.47)		(1.25)	(0.46)	
4	−0.58	−2.67	0.624	−0.22	−0.30	0.948
	(2.92)	(2.93)		(1.09)	(0.35)	
5	−0.10	−1.72	0.678	−0.70	−0.22	0.647
	(2.64)	(2.65)		(0.92)	(0.30)	
6	0.81	−2.33	0.410	−0.34	−0.16	0.843
	(2.72)	(2.50)		(0.82)	(0.25)	
7	0.79	−2.83	0.297	−0.08	−0.24	0.853
	(2.50)	(2.17)		(0.75)	(0.21)	
8	1.93	−2.89	0.136	−0.14	−0.47**	0.679
	(2.47)	(1.95)		(0.71)	(0.22)	
9	0.35	−4.22**	0.106	−0.24	−0.39**	0.851
	(2.31)	(1.73)		(0.70)	(0.19)	
10	−0.41	−4.59***	0.137	−0.28	−0.38**	0.897
	(2.39)	(1.68)		(0.65)	(0.19)	
11	0.00	−5.04***	0.080	−0.32	−0.43**	0.879
	(2.48)	(1.64)		(0.62)	(0.17)	
12	1.16	−4.53***	0.046	−0.62	−0.35**	0.677
	(2.43)	(1.65)		(0.58)	(0.14)	
Sig. windows	0/12	4/12	1/12	0/12	5/12	0/12
Panel B						
	Vandalism			Burglary		
Window	Residential	Street	p	Residential	Street	p
1	−0.02	0.39	0.495	−0.21	1.72	0.216
	(0.53)	(0.27)		(0.41)	(1.47)	
2	−0.20	0.29	0.270	−0.30	0.36	0.520
	(0.38)	(0.19)		(0.27)	(1.00)	
3	−0.35	0.07	0.238	−0.19	−0.09	0.916
	(0.31)	(0.15)		(0.22)	(0.89)	
4	−0.37	−0.01	0.182	−0.22	−0.51	0.712
	(0.24)	(0.13)		(0.20)	(0.78)	
5	−0.45**	−0.03	0.090	−0.25	−0.33	0.925
	(0.21)	(0.12)		(0.18)	(0.74)	
6	−0.56***	−0.01	0.018	−0.37**	−0.55	0.810
	(0.20)	(0.11)		(0.16)	(0.72)	
7	−0.62***	0.05	0.002	−0.31**	−0.86	0.425
	(0.18)	(0.10)		(0.15)	(0.69)	
8	−0.61***	0.08	0.001	−0.25*	−0.93	0.289
	(0.18)	(0.09)		(0.14)	(0.65)	
9	−0.53***	0.09	0.002	−0.25*	−0.86	0.340
	(0.17)	(0.08)		(0.13)	(0.63)	
10	−0.51***	0.11	0.001	−0.28**	−0.92	0.317
	(0.16)	(0.08)		(0.12)	(0.64)	
11	−0.45***	0.11	0.002	−0.29**	−0.87	0.340
	(0.16)	(0.08)		(0.12)	(0.60)	
12	−0.47***	0.13*	0.000	−0.24**	−0.88	0.287
	(0.15)	(0.07)		(0.11)	(0.59)	
Sig. windows	8/12	0/12	7/12	5/12	0/12	0/12

Notes. Window-by-window version of Table A7: near–far close-price effect (%) per incident of residential and street (public-space) incidents, estimated jointly with commercial incidents in one regression, at every exposure window 1–12 months before the sale; standard errors clustered by tract in parentheses. p: Wald test that the residential and street effects are equal. "Sig. windows" counts windows significant at five percent. Pooled 2007–2025, parcel and tract×month fixed effects, N = 242,815. The pooled residential burglary estimate is matched by post-sale crime (Table 3 note); the frontage–interior split in Table 3 is the relevant test for burglary. Main exposure windows are 10 months for homicide and narcotics, 8 months for burglary, and 6 months for vandalism.
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Table A9: Incident-Location Split, All Estimable Offenses

Offense	Residential	Street	Commercial	p res=str	p str=com	p all eq.	N
Homicide	−0.41	−4.59***	−0.77	0.137	0.247	0.269	242,815
	(2.39)	(1.68)	(2.76)				
Burglary	−0.25*	−0.93	0.00	0.289	0.151	0.243	242,815
	(0.14)	(0.65)	(0.16)				
Narcotics	−0.28	−0.38**	−0.10	0.897	0.766	0.951	242,815
	(0.65)	(0.19)	(0.91)				
Vandalism	−0.56***	−0.01	−0.23	0.018**	0.338	0.054*	242,815
	(0.20)	(0.11)	(0.20)				
Theft	−0.03	0.04***	0.00	0.326	0.040**	0.105	242,815
	(0.07)	(0.01)	(0.02)				
Motor-vehicle theft	0.86**	−0.04	−0.62	0.042**	0.308	0.069*	242,815
	(0.42)	(0.09)	(0.56)				
Robbery	0.47	−0.06	−0.06	0.480	0.994	0.777	242,815
	(0.75)	(0.15)	(0.21)				
Assault (non-dom.)	−0.28*	0.01	0.03	0.110	0.849	0.219	242,815
	(0.16)	(0.08)	(0.07)				
Assault (dom.)	−0.26**	0.04	0.10	0.296	0.892	0.453	242,815
	(0.12)	(0.24)	(0.35)				
Disorderly conduct	−1.58*	−0.21	−0.85	0.145	0.625	0.340	242,815
	(0.84)	(0.41)	(1.09)				
Trespass	−0.43**	−0.24	0.09	0.478	0.210	0.071*	242,815
	(0.19)	(0.21)	(0.14)				
Prostitution	0.80	0.02	0.27	0.612	0.294	0.551	242,815
	(1.54)	(0.16)	(0.18)				
Fraud	−0.40***	0.24	0.02	0.010**	0.324	0.007***	242,815
	(0.15)	(0.24)	(0.08)				
Sex offenses	−0.64**	−0.05	−0.71	0.284	0.305	0.498	242,815
	(0.32)	(0.47)	(0.47)				
Forgery	−0.39	0.39	0.95***	0.155	0.289	0.044**	242,815
	(0.43)	(0.41)	(0.29)				
Weapons	−1.10	0.19	1.54	0.311	0.338	0.376	242,815
	(1.21)	(0.28)	(1.37)				
Arson	−1.90	2.21	2.52	0.072*	0.897	0.164	242,815
	(1.96)	(1.40)	(1.95)				
Stolen property	−0.82	−1.58	1.93	0.858	0.093*	0.231	242,815
	(3.81)	(1.60)	(1.51)				
Liquor	3.24	0.60	0.31	0.408	0.812	0.601	242,815
	(2.94)	(1.14)	(0.36)				
Gambling	−5.64	−1.69	−1.95	0.706	0.957	0.931	242,815
	(9.72)	(2.03)	(4.27)				
Bad checks	0.76	2.40	0.41	0.605	0.533	0.768	242,815
	(0.62)	(3.14)	(0.44)				
Family offenses	1.00	0.78	2.36	0.879	0.484	0.742	242,815
	(0.68)	(1.26)	(1.68)				

Notes. Three-category incident-location split (residential / street / commercial) for all 22 estimable offenses, at each priced offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6) and a 12-month window otherwise; entered jointly, pooled 2007–2025, parcel and tract×month fixed effects, standard errors clustered by tract. The last three columns are Wald p-values that residential equals street, that street equals commercial, and that all three are equal. This three-category detail underlies the private/public summary in Table A7. $N = 242{,}815$.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A10: Forecast of Future Near-Ring Crime by Incident Location, All Windows

	Homicide		Burglary		Narcotics		Vandalism	
Win.	Res	St.	Res	St.	Res	St.	Res	St.
1	0.000	0.000	0.032	0.015	0.007	0.099	0.020	0.021
2	0.000	0.001	0.069	0.024	0.014	0.156	0.050	0.049
3	0.000	0.002	0.100	0.029	0.023	0.186	0.079	0.073
4	0.000	0.003	0.126	0.033	0.029	0.205	0.104	0.093
5	0.000	0.004	0.149	0.035	0.036	0.218	0.127	0.109
6	0.000	0.005	0.169	0.037	0.042	0.232	0.148	0.125
7	0.000	0.006	0.189	0.038	0.049	0.241	0.167	0.141
8	0.000	0.008	0.208	0.037	0.054	0.251	0.185	0.155
9	0.000	0.009	0.226	0.036	0.060	0.258	0.200	0.168
10	0.001	0.011	0.240	0.034	0.066	0.263	0.214	0.178
11	0.001	0.012	0.253	0.033	0.071	0.267	0.227	0.188
12	0.001	0.014	0.265	0.031	0.077	0.271	0.238	0.196

Notes. Forecast R^2 from regressing future near-ring crime (0–0.05 mi, after the sale) on past surrounding crime (far ring 0.20–0.30 mi, before the sale), estimated separately for residential (Res) and street (St.) incidents, no fixed effects (the informational benchmark available to a buyer at sale), all geocoded closed MLS sales before the constant-parcel and repeat-sales filters, $N = 923{,}730$, at each exposure window (Win., months). For *vandalism*, residential and street incidents forecast future near-ring crime comparably at six months, yet only residential vandalism is priced, so the capitalization difference cannot reflect a difference in objective recurrence. Homicide and narcotics forecast from the street and burglary from residential incidents, as expected from where those offenses physically occur.

Table A11: Vandalism by Target: Damage to Property versus Vehicle

	Exposure window (months before sale)											
	1	2	3	4	5	6	7	8	9	10	11	12
Panel A. Parcel + tract×year-month fixed effects (baseline)												
Damage to property	0.85	1.06	0.80	0.22	0.03	−0.49	−0.89**	−0.95**	−0.91**	−1.03***	−0.92**	−0.89***
	(1.12)	(0.83)	(0.63)	(0.58)	(0.51)	(0.47)	(0.45)	(0.43)	(0.41)	(0.39)	(0.36)	(0.34)
Damage to vehicle	0.70	0.89	0.50	0.33	0.20	0.21	0.05	−0.07	−0.13	−0.08	−0.09	0.02
	(0.93)	(0.67)	(0.56)	(0.53)	(0.48)	(0.45)	(0.42)	(0.40)	(0.40)	(0.38)	(0.37)	(0.34)
Property = vehicle (p)	0.916	0.874	0.724	0.896	0.818	0.298	0.143	0.142	0.168	0.078	0.102	0.057
Panel B. Parcel×era + tract×year-month fixed effects												
Damage to property	−0.23	2.49*	1.17	0.02	0.33	0.11	−0.59	−0.65	−0.66	−1.22*	−0.99	−0.91
	(1.97)	(1.47)	(1.17)	(1.02)	(0.92)	(0.80)	(0.73)	(0.68)	(0.64)	(0.62)	(0.60)	(0.56)
Damage to vehicle	0.52	0.29	0.11	0.16	0.20	−0.09	−0.30	−0.37	−0.62	−0.80*	−0.86*	−0.81*
	(1.20)	(0.80)	(0.65)	(0.67)	(0.65)	(0.60)	(0.56)	(0.52)	(0.53)	(0.48)	(0.46)	(0.42)
Property = vehicle (p)	0.759	0.208	0.447	0.914	0.918	0.849	0.764	0.753	0.962	0.602	0.869	0.895

Notes. Full twelve-window version of Table 3, Panel B: vandalism frontage incidents split by target, damage to the property versus damage to a vehicle, each as its own near–far difference $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ in one regression, pooled 2007–2025, clustered by tract; standard errors below the estimates. The "Property = vehicle" row is the Wald p-value for equality of the two arms. *Panel A* is the baseline design (parcel and tract×year-month fixed effects, $N = 242{,}815$). *Panel B* absorbs parcel×era instead of parcel ($N = 123{,}133$), which drops parcels that do not resell within an era.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Forecastability and recurrence. Table A13 examines whether past crime in the far ring (0.20–0.30 miles) predicts future crime in the near ring (0–0.05 miles). In forecasting regressions without fixed effects, far-ring crime significantly predicts near-ring crime at the five-percent level for twenty-one of twenty-two offenses at six months—all except stolen property—and for all twenty-two at twelve months. Its incremental explanatory power is modest: after controlling for past near-ring crime, the far ring's partial R^2 is at most 0.09 for any offense at either horizon.

Table A12 reports recurrence coefficients from regressions of post-sale near-ring crime on pre-sale near-ring crime, with parcel and tract $\times$ month fixed effects. The pre- and post-sale windows have equal lengths, matching the window used to report each offense's price estimate. The coefficient ρ measures how strongly past incidents predict future incidents within 0.05 miles of the same parcel, conditional on these fixed effects. It does not identify the causal effect of an incident on subsequent crime.

The forecasting and recurrence exercises measure different aspects of predictability. The first assesses the information in surrounding crime, including persistent differences across locations; the second measures recurrence after absorbing parcel and tract-by-month differences. Neither directly measures the change in buyers' expectations after learning of an incident.

Table A12: Crime Recurrence and Capitalization

Offense	Recurrence ρ (parcel + tract FE)	ρ, no parcel FE	δ (Table 2) [mo]
Homicide†	-0.026^{**}	0.070^{***}	-3.02^{**} [10]
	(0.012)	(0.006)	(1.38)
Burglary†	0.067^{***}	0.362^{***}	-0.24^{**} [8]
	(0.009)	(0.011)	(0.12)
Narcotics†	0.397^{***}	0.685^{***}	-0.29^{**} [10]
	(0.038)	(0.013)	(0.14)
Vandalism†	0.093^{***}	0.389^{***}	-0.17^{**} [6]
	(0.018)	(0.013)	(0.08)
Theft	0.237	–	0.01 [12]
	(0.158)		(0.02)
Motor-vehicle theft	0.140^{***}	–	-0.02 [12]
	(0.031)		(0.09)
Robbery	0.038^{***}	–	-0.03 [12]
	(0.012)		(0.13)
Assault (non-domestic)	0.280^{***}	–	-0.04 [12]
	(0.029)		(0.04)
Assault (domestic)	0.118^{***}	–	-0.19^{*} [12]
	(0.012)		(0.10)
All crime	0.314^{***}	–	–
	(0.113)		

Notes. Own-recurrence ρ of near-ring crime: the 0–0.05 mi incident count over the window *after* the sale regressed on the count over the matching window *before*, standard errors clustered by tract in parentheses; $N = 242{,}815$. For the four priced offenses ρ is measured over the same exposure window as δ (homicide and narcotics ten months, burglary eight, vandalism six; shown in the δ bracket); for the remaining offenses ρ is measured over six months and δ is shown at the twelve-month window (Table A3). The first ρ column uses parcel and tract × sale-month fixed effects; the second omits parcel fixed effects and is shown for the priced offenses. Parcel fixed effects bias ρ toward zero in short panels, whereas the no-fixed-effect estimate includes time-invariant level differences across properties and overstates recurrence, so the two bracket the within-parcel persistence; for homicide the within-parcel estimate is slightly negative while the no-fixed-effect estimate is positive. $\rho > 0$ means the offense recurs at the same property. δ is each offense's near–far close-price discount at its main window (Table 2; window length in brackets), unpriced offenses at the twelve-month reference window. †Priced offenses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A13: Rational Forecast: The Discount Does Not Track Forecastable Risk

	0–6 months		0–12 months	
Offense	Joint F	Partial R^2	Joint F	Partial R^2
Homicide†	127.3^{***}	0.004	237.0^{***}	0.009
Burglary†	1580.8^{***}	0.047	1871.1^{***}	0.062
Narcotics†	411.1^{***}	0.038	251.5^{***}	0.037
Vandalism†	711.6^{***}	0.045	627.5^{***}	0.054
Theft	6.8^{***}	0.035	9.1^{***}	0.017
Motor-vehicle theft	789.0^{***}	0.047	721.1^{***}	0.052
Robbery	357.2^{***}	0.045	365.8^{***}	0.060
Assault (non-dom.)	103.6^{***}	0.030	99.3^{***}	0.025
Assault (domestic)	696.5^{***}	0.062	497.1^{***}	0.058
Arson	80.3^{***}	0.001	204.3^{***}	0.002
Bad checks	27.4^{***}	0.002	34.0^{***}	0.005
Disorderly conduct	129.5^{***}	0.012	132.0^{***}	0.020
Family offenses	109.4^{***}	0.002	148.8^{***}	0.004
Forgery	82.3^{***}	0.012	91.1^{***}	0.018
Fraud	102.1^{***}	0.087	75.4^{***}	0.075
Gambling	170.2^{***}	0.013	261.6^{***}	0.029
Liquor	30.3^{***}	0.001	34.5^{***}	0.003
Prostitution	18.0^{***}	0.007	10.9^{***}	0.007
Sex offenses	75.8^{***}	0.008	45.2^{***}	0.014
Stolen property	3.2^{*}	0.000	5.8^{**}	0.000
Trespass	22.4^{***}	0.019	20.2^{***}	0.022
Weapons	251.0^{***}	0.054	348.9^{***}	0.067

Notes. Estimated without fixed effects. Outcome: crime within 0.05 miles after the sale. Predictor: crime 0.20–0.30 miles away before the sale, controlling for the near ring's own prior crime. Windows: 0–6 and 0–12 months. F tests that the far-ring term is zero; partial R^2 is the share of the remaining variation the far ring explains. A predictive benchmark, not a causal estimate.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. † priced offense.

E Price Formation and Agent Experience

Additional transaction-outcome results. Burglary's asking-price estimate is −0.15 (s.e. 0.12). Disorderly conduct lowers asking prices across all listings, significantly at the five-percent level in four of twelve windows with parcel fixed effects, but in none with parcel-by-era fixed effects.

Homicide time-on-market estimates are predominantly positive under both specifications, but significant at the five-percent level only at seven to nine months with parcel-by-era fixed effects. Cumulative time-on-market estimates change sign across specifications and are never statistically significant (Table A15). Withdrawal estimates are positive in every window under both specifications, with significance only at two months (2.2 percentage points, s.e. 0.96).

For weapons violations, withdrawal estimates are significant at the five-percent level in eight windows with parcel fixed effects and three with parcel-by-era fixed effects. The post-off-market placebo is significant in at most one window (Table A16). Among listings that sell, asking-price estimates are negative in every window with parcel-by-era fixed effects and significant in three.

Table A14: Homicide and the Asking Price, Every Window

	Sold listings		All listings	
Window (months)	Parcel	Parcel×era	Parcel	Parcel×era
1	−10.28**	−11.69*	0.02	0.12
	(4.44)	(6.84)	(1.39)	(1.40)
2	−2.49	−5.32	1.05	0.73
	(3.55)	(5.60)	(1.12)	(1.17)
3	−0.90	−0.42	0.10	−0.33
	(2.97)	(4.22)	(0.99)	(1.03)
4	−2.38	−3.47	0.24	−0.09
	(2.43)	(3.29)	(0.87)	(0.92)
5	−1.70	−4.25	0.00	−0.25
	(1.97)	(2.72)	(0.79)	(0.90)
6	−0.95	−3.36	0.40	0.13
	(1.88)	(2.45)	(0.74)	(0.87)
7	−0.44	−2.38	0.49	0.41
	(1.77)	(2.36)	(0.68)	(0.79)
8	−0.89	−1.27	0.45	0.50
	(1.67)	(2.10)	(0.64)	(0.75)
9	−1.04	−1.05	0.16	0.41
	(1.49)	(1.86)	(0.60)	(0.71)
10	−0.31	−0.88	0.51	0.76
	(1.38)	(1.74)	(0.58)	(0.67)
11	−0.44	−1.63	0.38	0.51
	(1.24)	(1.59)	(0.57)	(0.65)
12	−0.35	−1.46	0.50	0.64
	(1.17)	(1.53)	(0.55)	(0.62)
Listings	235,030	119,425	985,586	820,301

Notes. Near–far estimate δ (percent) of homicide on the ln original asking price: incidents within 0–0.05 miles against 0.20–0.30 miles in the given number of months before the list date, with controls for incidents in the remaining months up to twelve. Sold listings ended in a sale; all listings add those withdrawn. Parcel: parcel and tract × list-month fixed effects; parcel×era: parcel-by-era and tract × list-month fixed effects. Standard errors clustered by census tract. Listings: estimation sample (largest across windows). Each listing counts separately, so a home relisted after a withdrawal contributes several listings; the sample is 1,140,708 listings (523,916 sold, 616,792 withdrawn). Parcel fixed effects identify from homes with more than one listing in the sample (singletons are dropped), and parcel×era from homes listed more than once within an era.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A15: Homicide and Time on Market, Every Window

	Days on market		Cumulative days on market	
Window (months)	Parcel	Parcel×era	Parcel	Parcel×era
1	6.46	−8.97	2.61	12.49
	(11.91)	(16.44)	(13.39)	(18.83)
2	8.97	5.74	2.54	19.55
	(8.87)	(12.42)	(10.09)	(14.64)
3	4.38	8.43	−2.97	9.66
	(7.79)	(10.95)	(8.63)	(12.35)
4	3.08	13.63	−3.13	7.21
	(6.49)	(9.35)	(7.71)	(10.56)
5	−2.45	9.70	−5.41	4.82
	(6.31)	(8.82)	(7.27)	(9.51)
6	1.99	13.51	−2.56	5.24
	(5.70)	(8.43)	(6.59)	(9.06)
7	3.43	17.33**	−3.35	4.40
	(5.52)	(8.13)	(5.98)	(8.74)
8	2.36	14.54**	−2.22	6.68
	(5.13)	(6.85)	(5.58)	(7.48)
9	1.64	11.77**	−2.18	3.18
	(4.74)	(5.95)	(5.17)	(6.63)
10	0.35	9.05	−1.86	1.67
	(4.64)	(5.49)	(4.94)	(6.47)
11	0.06	10.58*	−1.34	3.37
	(4.44)	(5.56)	(4.87)	(6.31)
12	−0.20	8.43	−1.55	1.38
	(4.35)	(5.43)	(4.74)	(6.19)
Listings	235,655	119,296	235,978	119,617

Notes. Near–far estimate δ (percent) of homicide on ln days on market, sold listings: incidents within 0–0.05 miles against 0.20–0.30 miles in the given number of months before the list date, with controls for incidents in the remaining months up to twelve. Days on market: the current listing (median 24 days; 16 in the parcel×era estimation sample); cumulative: summed across relistings of the same home (median 41; 27 in the parcel×era estimation sample). Parcel: parcel and tract × list-month fixed effects; parcel×era: parcel-by-era and tract × list-month fixed effects. Standard errors clustered by census tract. Listings: estimation sample (largest across windows). Each listing counts separately, so a home relisted after a withdrawal contributes several listings; the sample is 1,140,708 listings (523,916 sold, 616,792 withdrawn). Parcel fixed effects identify from homes with more than one listing in the sample (singletons are dropped), and parcel×era from homes listed more than once within an era.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A16: Weapons Violations: Every Window

Outcome	FE	1	2	3	4	5	6	7	8	9	10	11	12	Placebo
Ask (all listings)	Parcel	−0.13	−0.38	−0.26	−0.25	−0.19	−0.13	−0.16	−0.14	−0.11	−0.11	−0.09	−0.14	1/12
		(0.41)	(0.33)	(0.28)	(0.26)	(0.24)	(0.21)	(0.20)	(0.20)	(0.19)	(0.18)	(0.17)	(0.17)	
	Parcel×era	0.50	0.08	0.15	−0.02	0.02	−0.04	−0.09	−0.08	−0.04	−0.04	−0.06	−0.09	0/12
		(0.38)	(0.30)	(0.26)	(0.23)	(0.21)	(0.20)	(0.20)	(0.20)	(0.19)	(0.18)	(0.19)	(0.18)	
Ask (sold listings)	Parcel	−1.45	−0.73	−0.13	−0.30	−0.24	−0.15	−0.23	−0.18	0.19	0.37	0.11	−0.02	0/12
		(0.97)	(0.75)	(0.62)	(0.56)	(0.47)	(0.40)	(0.40)	(0.38)	(0.32)	(0.29)	(0.29)	(0.28)	
	Parcel×era	−2.88**	−1.85*	−1.29	−1.26*	−0.97*	−1.12**	−1.22**	−1.05*	−0.72	−0.34	−0.50	−0.54	0/12
		(1.23)	(1.05)	(0.81)	(0.67)	(0.57)	(0.57)	(0.60)	(0.58)	(0.51)	(0.46)	(0.48)	(0.45)	
Sale-to-list	Parcel	0.21	0.11	0.17	0.11	0.15	0.13	0.09	0.08	0.13	0.06	0.03	0.06	0/12
		(0.34)	(0.25)	(0.20)	(0.16)	(0.14)	(0.13)	(0.12)	(0.11)	(0.10)	(0.09)	(0.08)	(0.08)	
	Parcel×era	0.19	0.34	0.14	0.07	0.02	0.10	−0.01	−0.04	0.06	0.05	0.03	0.04	2/12
		(0.41)	(0.30)	(0.23)	(0.17)	(0.15)	(0.14)	(0.14)	(0.14)	(0.12)	(0.11)	(0.10)	(0.10)	
Price cut (pp)	Parcel	−2.56**	−0.99	−0.71	−0.55	−0.02	−0.09	0.02	0.15	−0.01	−0.03	0.05	0.05	0/12
		(1.25)	(0.89)	(0.73)	(0.62)	(0.61)	(0.54)	(0.48)	(0.45)	(0.43)	(0.41)	(0.41)	(0.37)	
	Parcel×era	−2.02	−1.17	−0.63	−0.28	0.06	−0.19	0.39	0.40	0.08	−0.07	0.00	0.02	0/12
		(1.61)	(1.19)	(0.93)	(0.76)	(0.66)	(0.57)	(0.56)	(0.52)	(0.49)	(0.43)	(0.43)	(0.40)	
Days on market	Parcel	−5.96*	−2.61	−2.12	−0.62	0.13	−0.78	−0.53	−0.52	−1.27	−1.10	−0.87	−0.79	0/12
		(3.21)	(2.64)	(1.94)	(1.59)	(1.28)	(1.18)	(1.08)	(1.04)	(0.95)	(0.87)	(0.79)	(0.79)	
	Parcel×era	−2.86	−0.33	0.33	0.98	1.32	−0.23	0.91	−0.12	−0.61	−0.67	−0.54	−0.91	2/12
		(3.90)	(3.52)	(2.28)	(2.08)	(1.72)	(1.67)	(1.74)	(1.54)	(1.43)	(1.39)	(1.22)	(1.17)	
Withdrawal (pp)	Parcel	0.94**	0.59*	0.66***	0.59***	0.69***	0.63***	0.50***	0.42**	0.34**	0.28*	0.24*	0.25*	1/12
		(0.41)	(0.31)	(0.24)	(0.21)	(0.20)	(0.18)	(0.17)	(0.18)	(0.17)	(0.15)	(0.15)	(0.14)	
	Parcel×era	0.91*	0.56	0.67**	0.44*	0.55**	0.46**	0.31	0.27	0.18	0.11	0.09	0.13	0/12
		(0.47)	(0.35)	(0.27)	(0.24)	(0.22)	(0.21)	(0.20)	(0.20)	(0.19)	(0.17)	(0.17)	(0.16)	
Close	Parcel	−0.47	0.07	0.11	−0.51	−0.62	−0.39	−0.18	−0.12	−0.17	0.04	0.09	0.10	0/12
		(0.92)	(0.71)	(0.58)	(0.48)	(0.47)	(0.44)	(0.40)	(0.37)	(0.35)	(0.31)	(0.30)	(0.28)	
	Parcel×era	−0.99	−0.38	−0.51	−0.93*	−1.11**	−0.72	−0.59	−0.32	−0.32	−0.10	0.02	0.12	0/12
		(1.20)	(0.84)	(0.69)	(0.55)	(0.47)	(0.45)	(0.40)	(0.35)	(0.33)	(0.30)	(0.29)	(0.27)	

Notes. Near–far estimate δ for weapons violations (incidents within 0–0.05 miles against 0.20–0.30 miles) in the w months before the list date (w = column), with controls for incidents in the remaining months up to twelve; percent for log outcomes, percentage points for the price cut and withdrawal. Close: repeat sales, crime before the sale date. Parcel: parcel and tract × month fixed effects; parcel×era: parcel-by-era and tract × month fixed effects; standard errors clustered by tract. Placebo: windows (of twelve) significant at 5% when crime after the outcome is fixed replaces crime before it (after the list date for the ask, after the close date for sale-to-list, price cut, days on market and close, after the off-market date for withdrawal). The price cut in Table 4 is a separate specification: crime during the listing (list date to close date), sold listings, with fixed effects for the number of months listed (one to twelve), parcel and tract × list-month fixed effects: 1.35** (0.53); its placebo, crime in the same number of months after the close date with tract × close-month fixed effects, is 0.49 (0.60).
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Table A19: Agent Local Experience, All Twenty-Two Offenses

		Listing agent, asking price				Buyer's agent, closing price			
		Local only		+ Total experience		Local only		+ Total experience	
Offense	Window	Gradient	Sig. windows	Gradient	Sig. windows	Gradient	Sig. windows	Gradient	Sig. windows
Priced									
Homicide	10	−0.97***	7/12	−0.79**	1/12	−1.61**	6/12	−1.99**	5/12
		(0.34)		(0.35)		(0.75)		(0.86)	
Burglary	8	0.00	0/12	0.02	0/12	0.09	1/12	0.15	2/12
		(0.04)		(0.04)		(0.10)		(0.12)	
Narcotics	10	−0.08***	7/12	−0.02	0/12	0.08	1/12	0.10	1/12
		(0.03)		(0.03)		(0.10)		(0.11)	
Vandalism	6	−0.03	3/12	−0.04	3/12	0.14**	2/12	0.15*	1/12
		(0.03)		(0.03)		(0.06)		(0.08)	
Suggestive									
Disorderly conduct	12	−0.07	0/12	−0.02	0/12	−0.47**	4/12	−0.64***	5/12
		(0.09)		(0.10)		(0.21)		(0.20)	
Sex offenses	12	−0.08	0/12	−0.04	0/12	0.03	0/12	−0.01	0/12
		(0.06)		(0.06)		(0.14)		(0.14)	
Precise nulls									
Theft	12	−0.01	0/12	−0.01	0/12	0.00	0/12	0.01*	1/12
		(0.00)		(0.00)		(0.01)		(0.01)	
Motor vehicle theft	12	0.02	1/12	0.00	0/12	0.00	4/12	−0.03	1/12
		(0.02)		(0.02)		(0.06)		(0.08)	
Robbery	12	−0.16***	10/12	−0.17***	9/12	0.03	0/12	0.12	1/12
		(0.04)		(0.04)		(0.07)		(0.09)	
Assault (non-domestic)	12	−0.07***	9/12	−0.05***	7/12	0.01	3/12	0.03	3/12
		(0.01)		(0.01)		(0.03)		(0.03)	
Assault (domestic)	12	−0.06**	8/12	−0.06**	7/12	−0.05	0/12	−0.10*	3/12
		(0.03)		(0.03)		(0.05)		(0.06)	
Criminal trespass	12	0.01	0/12	0.03	0/12	0.05	1/12	0.05	1/12
		(0.03)		(0.03)		(0.06)		(0.07)	
Fraud	12	−0.04**	9/12	−0.05**	9/12	0.01	0/12	0.03	0/12
		(0.02)		(0.02)		(0.04)		(0.05)	
Prostitution	12	−0.03	0/12	−0.06**	1/12	−0.13	1/12	−0.15*	0/12
		(0.03)		(0.03)		(0.08)		(0.09)	
Forgery	12	−0.14	0/12	−0.14	0/12	−0.21*	0/12	−0.09	0/12
		(0.09)		(0.09)		(0.12)		(0.13)	
Family offenses	12	−0.07	0/12	−0.01	0/12	0.11	0/12	−0.10	0/12
		(0.15)		(0.15)		(0.30)		(0.29)	
Bad checks	12	−0.15	2/12	−0.12	0/12	−0.25	0/12	−0.35	0/12
		(0.10)		(0.10)		(0.27)		(0.28)	
Liquor law	12	−0.13	0/12	−0.10	0/12	0.28	0/12	0.19	0/12
		(0.11)		(0.13)		(0.42)		(0.51)	
Inconclusive									
Weapons	12	0.11*	3/12	0.10*	0/12	0.11	0/12	0.07	0/12
		(0.06)		(0.06)		(0.08)		(0.17)	
Arson	12	0.05	0/12	0.03	0/12	1.93**	6/12	2.74***	7/12
		(0.29)		(0.30)		(0.84)		(1.01)	
Gambling	12	−1.00**	3/12	−0.65	0/12	−0.82	0/12	−0.65	0/12
		(0.49)		(0.49)		(1.80)		(1.91)	
Stolen property	12	−0.33	0/12	−0.41	0/12	0.12	0/12	0.48	0/12
		(0.39)		(0.43)		(1.16)		(1.39)	

Notes. Local-experience gradient, $(\text{near} \times E - \text{far} \times E) \times 100$, at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 months for other offenses), with standard errors clustered by tract in parentheses. E is the agent's prior listings in the same tract; "+ Total experience" adds the agent's total prior listings in any tract, interacted with near and far exposure. "Sig. windows" counts the exposure windows (of twelve) in which the local-experience gradient is significant at five percent, as in Table 2; every window is reported in Tables OA10 and OA11. Design as in Table 5: parcel, tract×month, and agent fixed effects; $N = 972{,}629$ listings (listing agent) and 186,510 sales (buyer's agent). Offense groups as in Table 2. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A17: Agent Experience Gradient on Price

Offense	Discount (mean agent)	Experience gradient (per SD) no agent FE	agent FE	Sig. windows (no FE/FE)
A. Asking price (listing agent)				
Narcotics	−0.01 (0.04)	-0.15^{***}	-0.09^{**}	9/7
		(0.05)	(0.04)	
Vandalism	0.02 (0.03)	-0.11^{***}	-0.07^{**}	8/3
		(0.03)	(0.03)	
Homicide	0.18 (0.59)	−0.82	-1.01^{**}	0/7
		(0.63)	(0.49)	
Burglary	−0.05 (0.10)	−0.04	0.02	0/0
		(0.11)	(0.08)	
B. Closing price (buyer's agent)				
Homicide	−0.94 (0.79)	−0.01	-1.14^{**}	0/6
		(0.95)	(0.53)	
Narcotics	−0.09 (0.12)	-0.29^{**}	0.08	3/0
		(0.14)	(0.09)	
Vandalism	−0.13 (0.11)	−0.02	0.06	0/0
		(0.10)	(0.09)	
Burglary	−0.07 (0.08)	0.14^{*}	0.07	0/0
		(0.08)	(0.07)	

Notes. Gradient per standard deviation of agent experience ($E = \ln(1 +$ prior same-agent same-tract listings), s.d. 1.20 for listing agents and 0.71 for buyer's agents with agent fixed effects, 1.20 and 0.67 without; each on its own estimation sample), each offense and panel at the window whose experience-gradient estimate is closest to the median of its estimates significant at five percent (of all twelve if none is): listing side homicide 6, burglary 2, narcotics 9, vandalism 12; buyer side homicide 10, burglary 9, narcotics 7, vandalism 2 months. Panel A: listing agent, asking price. Panel B: buyer's agent, closing price. Discount: near–far effect at mean experience. Gradients are estimated without and with agent fixed effects; Sig. windows counts significant windows of twelve, without/with. Sample: listings with an agent identifier.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A18: Agent Local versus Total Experience, All Twenty-Two Offenses

Offense	Window	Local experience gradient (s.e.)	Total experience gradient (s.e.)	Sig. windows local / total
Panel A. Listing agent, asking price (N = 972,629)				
Priced				
Homicide	10	−0.79** (0.35)	−0.45 (0.48)	1/12 / 3/12
Burglary	8	0.02 (0.04)	−0.06 (0.09)	0/12 / 0/12
Narcotics	10	−0.02 (0.03)	−0.21*** (0.05)	0/12 / 12/12
Vandalism	6	−0.04 (0.03)	0.04 (0.04)	3/12 / 0/12
Suggestive				
Disorderly conduct	12	−0.02 (0.10)	−0.22 (0.18)	0/12 / 0/12
Sex offenses	12	−0.04 (0.06)	−0.15* (0.09)	0/12 / 0/12
Precise nulls				
Theft	12	−0.01 (0.00)	0.00 (0.01)	0/12 / 0/12
Motor vehicle theft	12	0.00 (0.02)	0.04 (0.05)	0/12 / 0/12
Robbery	12	−0.17*** (0.04)	0.00 (0.06)	9/12 / 1/12
Assault (non-domestic)	12	−0.05*** (0.01)	−0.04* (0.02)	7/12 / 2/12
Assault (domestic)	12	−0.06** (0.03)	−0.01 (0.04)	7/12 / 0/12
Criminal trespass	12	0.03 (0.03)	−0.06 (0.05)	0/12 / 0/12
Fraud	12	−0.05** (0.02)	0.05 (0.04)	9/12 / 0/12
Prostitution	12	−0.06** (0.03)	0.30** (0.15)	1/12 / 5/12
Forgery	12	−0.14 (0.09)	−0.02 (0.08)	0/12 / 1/12
Family offenses	12	−0.01 (0.15)	−0.20 (0.31)	0/12 / 0/12
Bad checks	12	−0.12 (0.10)	−0.17 (0.22)	0/12 / 0/12
Liquor law	12	−0.10 (0.13)	−0.14 (0.22)	0/12 / 0/12
Inconclusive				
Weapons	12	0.10* (0.06)	0.04 (0.11)	0/12 / 0/12
Arson	12	0.03 (0.30)	0.03 (0.41)	0/12 / 1/12
Gambling	12	−0.65 (0.49)	−0.11 (0.73)	0/12 / 0/12
Stolen property	12	−0.41 (0.43)	0.33 (0.58)	0/12 / 0/12
Panel B. Buyer's agent, closing price (N = 186,510)				
Priced				
Homicide	10	−1.99** (0.86)	0.41 (0.46)	5/12 / 0/12
Burglary	8	0.15 (0.12)	−0.07 (0.05)	2/12 / 1/12
Narcotics	10	0.10 (0.11)	−0.05 (0.07)	1/12 / 0/12
Vandalism	6	0.15* (0.08)	−0.01 (0.05)	1/12 / 0/12
Suggestive				
Disorderly conduct	12	−0.64*** (0.20)	0.26* (0.16)	5/12 / 0/12
Sex offenses	12	−0.01 (0.14)	0.05 (0.11)	0/12 / 0/12
Precise nulls				
Theft	12	0.01* (0.01)	−0.01*** (0.00)	1/12 / 11/12
Motor vehicle theft	12	−0.03 (0.08)	0.03 (0.03)	1/12 / 0/12
Robbery	12	0.12 (0.09)	−0.08 (0.05)	1/12 / 0/12
Assault (non-domestic)	12	0.03 (0.03)	−0.01 (0.01)	3/12 / 0/12
Assault (domestic)	12	−0.10* (0.06)	0.06* (0.04)	3/12 / 0/12
Criminal trespass	12	0.05 (0.07)	0.00 (0.05)	1/12 / 0/12
Fraud	12	0.03 (0.05)	−0.04 (0.03)	0/12 / 0/12
Prostitution	12	−0.15* (0.09)	0.03 (0.24)	0/12 / 0/12
Forgery	12	−0.09 (0.13)	−0.15 (0.11)	0/12 / 0/12
Family offenses	12	−0.10 (0.29)	0.40 (0.43)	0/12 / 0/12
Bad checks	12	−0.35 (0.28)	0.20 (0.18)	0/12 / 3/12
Liquor law	12	0.19 (0.51)	0.13 (0.27)	0/12 / 1/12
Inconclusive				
Weapons	12	0.07 (0.17)	0.03 (0.09)	0/12 / 0/12
Arson	12	2.74*** (1.01)	−0.97 (0.66)	7/12 / 1/12
Gambling	12	−0.65 (1.91)	−0.74 (0.82)	0/12 / 1/12
Stolen property	12	0.48 (1.39)	−0.55 (0.64)	0/12 / 0/12

Notes. Local experience $E = \ln(1 + \text{prior same-agent listings in the tract})$ and total experience T (all prior listings) enter the same equation, each interacted with near and far exposure. Gradients are (near $\times E -$ far $\times E) \times 100$ and (near $\times T -$ far $\times T) \times 100$ per log unit of E and per standard deviation of T, at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 months for other offenses); last column counts windows significant at five percent. Design as in Table 5. Correlation of E and T: 0.36 (listing), 0.37 (buyer).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E.1 Agent Experience: Further Checks

Three further checks refine these patterns. First, the asking-price response to listing-agent experience is negative for homicide, narcotics, and vandalism in nearly every window, both with parcel fixed effects and with agent fixed effects added (Table A17). Second, for narcotics it weakens once agents listing in a tract for the first time are allowed to differ, and for vandalism it comes mainly from those first listings (Table OA16).

Third, controlling for the agent's total experience (all prior listings, in any tract), the buyer's-agent homicide result remains (-1.99, s.e. 0.86; significant in five of twelve windows), and total experience has no effect on the buyer side. The listing-agent homicide result is significant only at the main window (-0.79, s.e. 0.35), and total experience lowers the homicide asking price in three of twelve windows. For narcotics, the asking price falls with the listing agent's total experience (-0.21, s.e. 0.05; significant in all twelve windows), and local experience has no effect once total experience is controlled for (Table A18). Table A19 reports these tests for all twenty-two offenses, and Tables OA10 and OA11 report the estimates at every window.

F Robustness

Table A20: Priced Discounts with Predetermined Neighborhood and Housing-Market Controls

	Homicide	Burglary	Narcotics	Vandalism
δ, baseline (no neighborhood control)	-3.02^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
Adding each predetermined control alone (near & far, stock to 12 months before close):				
+ Building permits (investment)	-3.03^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
+ Vacant buildings (disinvestment)	-3.06^{**}	-0.23^{**}	-0.25^{*}	-0.17^{*}
	(1.40)	(0.12)	(0.15)	(0.08)
+ Business licenses (commercial)	-2.95^{**}	-0.24^{**}	-0.30^{**}	-0.17^{**}
	(1.39)	(0.12)	(0.14)	(0.08)
+ Tree canopy (amenity)	-3.02^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
+ 311 service requests, all types (disorder)	-3.00^{**}	-0.24^{**}	-0.30^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
δ, all five controls together	-2.97^{**}	-0.22^{*}	-0.26^{*}	-0.17^{**}
	(1.40)	(0.12)	(0.15)	(0.08)
Lagged housing-market controls (near & far, 12–24 months before close):				
+ Five market controls only	-2.96^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
δ, all ten controls together	-2.91^{**}	-0.22^{*}	-0.26^{*}	-0.17^{**}
	(1.40)	(0.12)	(0.15)	(0.08)

Notes. δ at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months). Row 1 is the baseline; the middle rows add one neighborhood control at a time; the next row adds all five. The market controls are the neighbors' repeat-sale price change, the distressed-sale share, the sale-to-list ratio, days on market, and new listings, each in the near and far rings over the 12 to 24 months before the close, with indicators for missing values; the last row adds all ten controls. Each control is a near–far cumulative stock of events recorded more than twelve months before the close, except tree canopy, which is the near–far difference in average canopy share two years before the sale; 311 data begin in 2011 and change composition in 2019 (Section 3).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A21: Hedonic Estimates on All Sales

Offense	δ (%)	(s.e.)	Window
Priced offenses			
Homicide	-1.86^{***}	(0.71)	10
Burglary	-0.47^{***}	(0.13)	8
Narcotics	-0.52^{***}	(0.09)	10
Vandalism	-0.62^{***}	(0.11)	6
No discount under repeat sales			
Theft	0.01	(0.08)	12
Motor-vehicle theft	-0.46^{**}	(0.20)	12
Robbery	-0.72^{***}	(0.20)	12
Assault (non-dom.)	-0.52^{***}	(0.09)	12
Assault (dom.)	-0.70^{***}	(0.10)	12

Notes. Cross-sectional hedonic near–far regression on all sold listings (N = 490,563), replacing the parcel fixed effect with observed characteristics (bedrooms, bathrooms, ln living area, ln lot size, age, stories, property type, new construction, and units, with missing indicators) plus tract×month fixed effects, clustered by tract. Regressors are the 0–0.05 and 0.20–0.30 mi counts in the w months before sale and the same-ring counts in months (w+1)–12. Each cell reports $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ at the offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 for the other offenses).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

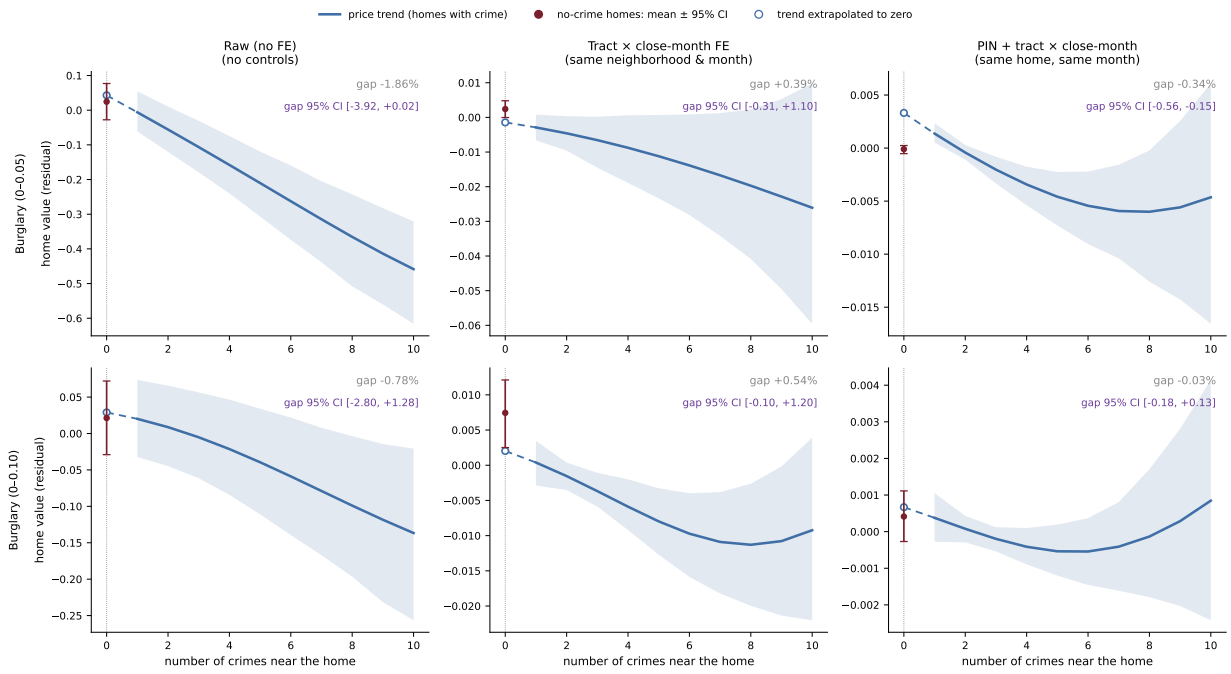


Figure A2: Burglary: Boundary Diagnostic at Two Near Rings

Notes. Top row: near ring 0–0.05 miles; bottom row: 0–0.10 miles. Columns add fixed effects from left to right; the right column is the baseline design (parcel and tract×close-month fixed effects). Blue: price trend for homes with nearby burglary; red: homes with none (mean and 95 percent interval); open circle: the trend extrapolated to zero. The gap shown in each panel is the mean residual price of no-crime homes minus the fitted trend extrapolated to zero; the formal test is the boundary-dummy coefficient in Table OA27 (−0.98, s.e. 0.32 at 0–0.05 miles; −0.06, s.e. 0.27 at 0–0.10 miles). Fewer than one home in ten (9 percent) has more than five incidents within 0.10 miles, so the trend beyond that point is imprecise.

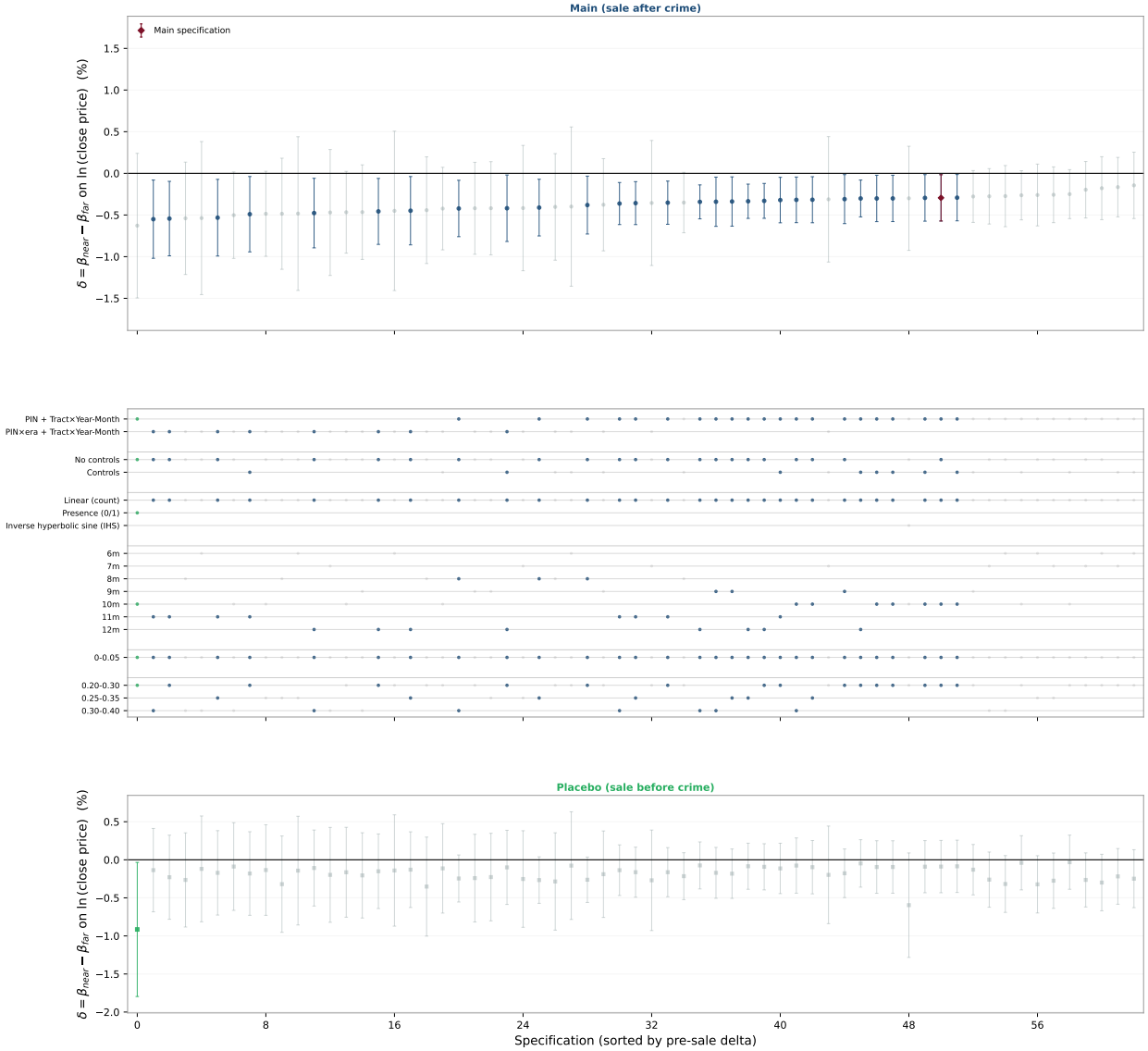


Figure A3: Narcotics: Sale-Price Discount Across Specifications

Notes. The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: δ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and a ten-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Narcotics: 29/63 significant at 5%, placebo 1/63.

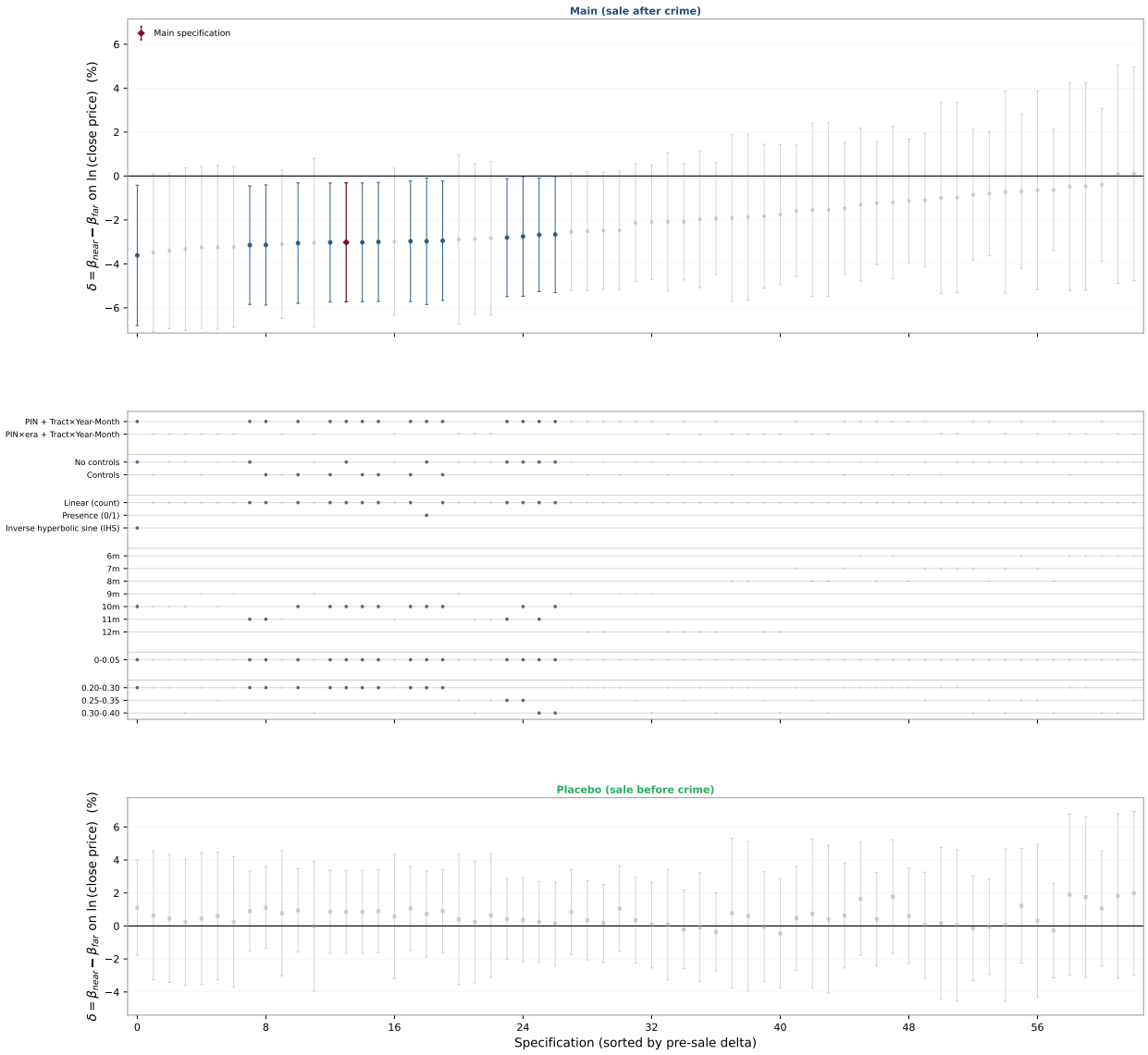


Figure A4: Homicide: Sale-Price Discount Across Specifications

Notes. The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: δ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and a ten-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Homicide: 15/63 significant at 5%, placebo 0/63.

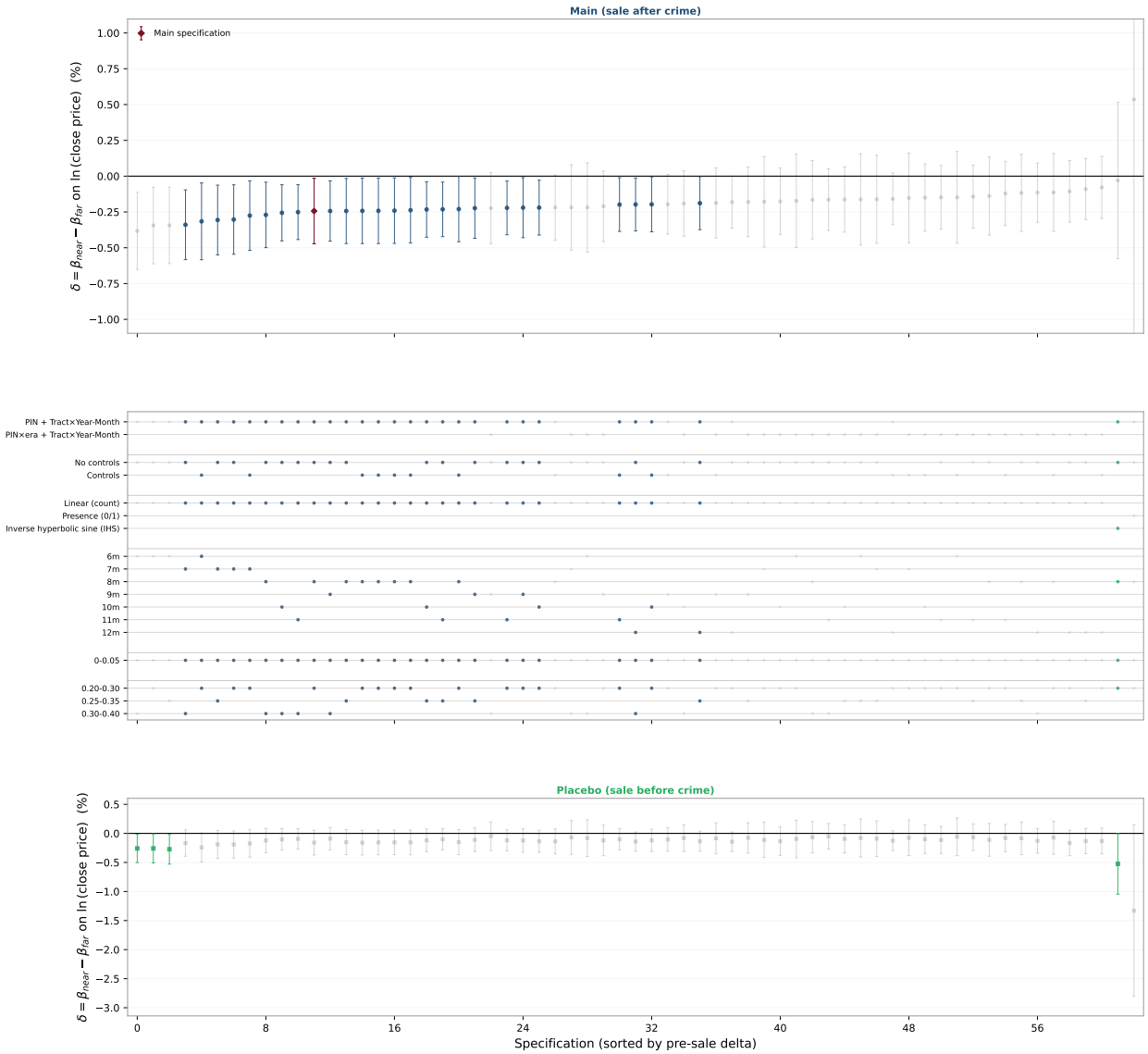


Figure A5: Burglary: Sale-Price Discount Across Specifications

Notes. The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: δ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and an eight-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Burglary: 29/63 significant at 5%, placebo 4/63.

G Rental Prices

I apply the same near–far design to lease listings, with the log asking rent as the outcome (Table A22). Three offenses lower the asking rent, but the estimates are sensitive to the fixed effects: burglary under parcel fixed effects (6 of 12 windows at five percent), sex offenses consistently under parcel-by-era fixed effects (7 of 12 windows), but rarely under parcel fixed effects (1 of 12), and liquor violations only under parcel-by-era fixed effects (4 of 12); all three survive the post-date placebo (0 of 12 windows). Narcotics leaves rents unchanged under parcel fixed effects, where its post-date placebo is clean, though the placebo fails under parcel-by-era fixed effects; homicide is estimated too imprecisely in the lease sample to sign, in line with Klimova and Lee (2014), who find no effect of nearby murders on Sydney rents. Narcotics instead appears in the leasing frictions of Panel B: it raises the probability a listing is a relisting (the unit had an earlier spell on the market) in 11 of 12 windows under each specification, and the result holds across all twenty-two robustness specifications (Figure OF2), with a null future-crime placebo. The full offense battery is in Table A23.

Table A22: The Rental Market

	PIN			PIN×era		
Offense	Min (%)	Max (%)	Sig. (plac.)	Min (%)	Max (%)	Sig. (plac.)
A. Asking rent (percent)						
Priced (asking rent lower; identifying FE in note)						
Burglary	−0.34*** (0.12)	−0.07 (0.06)	6/12 (0)	−0.30** (0.12)	−0.04 (0.07)	1/12 (0)
Sex offenses	−0.29* (0.15)	−0.21* (0.13)	1/12 (0)	−0.32 (0.27)	−0.16 (0.14)	7/12 (0)
Liquor law	−0.23 (0.26)	−0.01 (0.13)	0/12 (0)	−0.30*** (0.07)	0.09 (0.49)	4/12 (0)
Precise nulls (rent effect near zero, tightly estimated)						
Narcotics†	0.01 (0.06)	0.10 (0.08)	0/12 (0)	0.02 (0.17)	0.14 (0.09)	0/12 (4)
Vandalism	−0.06 (0.06)	0.05 (0.07)	0/12 (0)	−0.03 (0.04)	0.09 (0.07)	0/12 (1)
Prostitution	−0.16 (0.10)	0.00 (0.17)	0/12 (1)	−0.33*** (0.12)	−0.03 (0.14)	1/12 (0)
Imprecise / underpowered						
Homicide	−0.55 (0.61)	1.03 (1.20)	0/12 (0)	−0.59 (0.59)	1.16 (1.22)	0/12 (0)
B. Leasing frictions						
Narcotics, relisting (pp)	0.18** (0.08)	0.97*** (0.34)	11/12 (0)	0.27*** (0.08)	1.28*** (0.40)	11/12 (0)
Homicide, time on market (%)	1.01 (4.65)	12.19 (11.00)	0/12 (0)	1.54 (3.98)	19.63* (11.70)	4/12 (0)
Burglary, withdrawn (pp)†	−1.06** (0.46)	−0.02 (0.16)	3/12 (0)	−1.12** (0.53)	−0.08 (0.15)	5/12 (2)

Notes. Near–far estimates on lease listings, 2007–2025, tract × listing-month fixed effects, clustered by tract (118,849 listings with parcel fixed effects; 98,813 with parcel × era). For each specification, Min and Max are the most negative and most positive estimates across the twelve windows, followed by the number significant at 5%; in parentheses, windows with a significant post-listing placebo; †, placebo fails in two or more. Panel A: log asking rent. Panel B: relisting (the unit had an earlier spell on the market) and withdrawal (percentage points), time on market (percent). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA18 shows how the listing agent's experience in the tract moderates the effect of nearby crime on rental listing outcomes. Near narcotics incidents, more experienced agents lease units faster, with days on market about 0.6 percent shorter per standard deviation of experience, and set slightly lower asking rents; the average rental discount stays small. The tenant's agent's experience is unrelated to asking rents or days on market in any window.

Specification curves: rents The rental specification curves cover twenty-two specifications: eleven exposure windows (two to twelve months) under each of two fixed-effect designs (parcel, and parcel by era). The narcotics estimate on the probability a lease listing is a relisting (the unit had an earlier spell on the market) is positive and significant in all twenty-two, with no significant placebo (Figure OF2). The sex-offense estimate on the asking rent is significant in eight of the same twenty-two specifications, with no significant placebo (Figure OF3).

Table A23: The Rental Market: Full Offense Battery

	PIN			PIN×era		
Offense	Min (%)	Max (%)	Sig. (plac.)	Min (%)	Max (%)	Sig. (plac.)
A. Asking rent (percent)						
Priced (asking rent lower; identifying FE in note)						
Burglary	-0.34^{***} (0.12)	−0.07 (0.06)	6/12 (0)	-0.30^{**} (0.12)	−0.04 (0.07)	1/12 (0)
Sex offenses	-0.29^{*} (0.15)	-0.21^{*} (0.13)	1/12 (0)	−0.32 (0.27)	−0.16 (0.14)	7/12 (0)
Liquor law	−0.23 (0.26)	−0.01 (0.13)	0/12 (0)	-0.30^{***} (0.07)	0.09 (0.49)	4/12 (0)
No consistent rent effect (near zero; prostitution's parcel×era minimum is significant at one window)						
Narcotics†	0.01 (0.06)	0.10 (0.08)	0/12 (0)	0.02 (0.17)	0.14 (0.09)	0/12 (4)
Vandalism	−0.06 (0.06)	0.05 (0.07)	0/12 (0)	−0.03 (0.04)	0.09 (0.07)	0/12 (1)
Larceny/theft†	-0.01^{**} (0.00)	0.00 (0.01)	6/12 (2)	−0.02 (0.01)	0.00 (0.01)	0/12 (0)
Assault (non-dom.)	−0.06 (0.04)	−0.03 (0.02)	2/12 (0)	−0.05 (0.04)	−0.01 (0.02)	0/12 (0)
Motor-vehicle theft	−0.12 (0.08)	−0.01 (0.04)	0/12 (0)	−0.13 (0.09)	−0.01 (0.03)	0/12 (0)
Robbery	0.01 (0.04)	0.07 (0.06)	0/12 (0)	0.04 (0.05)	0.12 (0.11)	0/12 (0)
Assault (dom.)†	−0.03 (0.05)	0.02 (0.09)	0/12 (5)	−0.06 (0.11)	0.03 (0.08)	0/12 (0)
Fraud	−0.01 (0.04)	0.08 (0.06)	0/12 (0)	0.03 (0.04)	0.07^{***} (0.03)	0/12 (1)
Trespass	-0.17^{*} (0.10)	0.01 (0.09)	0/12 (0)	−0.13 (0.10)	0.05 (0.10)	0/12 (0)
Forgery	−0.08 (0.15)	0.12 (0.10)	0/12 (0)	0.06 (0.16)	0.22 (0.16)	0/12 (0)
Prostitution	−0.16 (0.10)	0.00 (0.17)	0/12 (1)	-0.33^{***} (0.12)	−0.03 (0.14)	1/12 (0)
Imprecise / underpowered (too sparse near leases to sign)						
Homicide	−0.55 (0.61)	1.03 (1.20)	0/12 (0)	−0.59 (0.59)	1.16 (1.22)	0/12 (0)
Disorderly conduct	−0.40 (0.30)	0.15 (0.25)	0/12 (0)	−0.25 (0.16)	0.40 (0.29)	0/12 (0)
Weapons	0.02 (0.26)	0.17 (0.22)	0/12 (0)	−0.08 (0.23)	0.22 (0.22)	0/12 (0)
Arson	−0.21 (0.47)	0.20 (0.47)	0/12 (0)	−0.27 (0.43)	0.54 (0.85)	0/12 (0)
Bad checks†	−0.03 (0.18)	0.82^{**} (0.37)	0/12 (3)	−0.16 (0.17)	0.67 (0.43)	0/12 (0)
Family (non-violent)	0.07 (0.22)	0.36^{*} (0.21)	0/12 (0)	0.11 (0.38)	0.68 (0.50)	0/12 (0)
Kidnapping†	-0.70^{*} (0.36)	−0.30 (0.27)	0/12 (3)	−0.62 (0.72)	0.05 (0.29)	0/12 (0)
Embezzlement	−0.39 (0.34)	0.96 (0.94)	0/12 (0)	−0.06 (0.48)	0.94 (0.83)	0/12 (0)
Child pornography/obscenity	−0.02 (0.56)	0.40 (0.74)	0/12 (0)	−0.34 (1.06)	0.34 (0.51)	0/12 (0)
Extortion†	-1.30^{*} (0.70)	−0.28 (0.33)	2/12 (3)	-1.13^{**} (0.51)	−0.19 (0.26)	2/12 (5)
Gambling†	−1.50 (1.44)	3.54 (2.25)	0/12 (4)	−1.25 (1.51)	2.88 (4.05)	0/12 (12)

Notes. Asking-rent near–far estimates on lease listings for the full offense battery, the near ring against the 0.20–0.30 mile far ring, pooled 2007–2025 with tract×listing-month fixed effects, clustered by tract (118,849 listings under parcel fixed effects; 98,813 under parcel×era). Under each fixed-effect specification, parcel (PIN) and parcel×era, Min and Max are the most negative and most positive estimate across the twelve exposure windows (standard error in parentheses), and Sig. counts windows significant at five percent; the parenthetical is the number of windows in which the post-date placebo (crime after the listing) is significant. A dagger (†) marks an offense whose placebo fails in two or more windows, shown only for completeness. Priced: asking rent lower, two or more significant windows under an identifying fixed-effect specification with a clean placebo. The compact version is Table A22. "Child pornography/obscenity" is predominantly child pornography (IUCR 1582). Non-domestic assault meets the window rule (2/12, clean placebo), but its estimates are small (about −0.06%) and it is not counted as priced.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

H Heterogeneity across Eras

The point estimates differ across macroeconomic eras, but the differences are not statistically significant (Table A24). Burglary and vandalism discounts are larger in the 2009–2013 foreclosure crisis, homicide's in the 2014–2025 recovery, and narcotics is negative in each era (crisis −0.33,

s.e. 0.23; recovery -0.52, s.e. 0.31). For no priced offense does a Wald test reject equal crisis and recovery discounts ($p = 0.41$ for homicide, 0.26 for burglary, 0.63 for narcotics, and 0.10 for vandalism). The crisis estimates for rare offenses rest on few within-crisis repeat sales and are imprecise, as their standard errors show.

Table A24: Crime Capitalization by Era

Offense	Identifying parcels	Pooled 2007–25	Crisis 2009–13	Recovery 2014–25
Priced				
Homicide	4,950	-3.02^{**}	-0.07	-4.12^{***}
		(1.38)	(4.73)	(1.47)
Narcotics	49,513	-0.29^{**}	-0.33	-0.52^{*}
		(0.14)	(0.23)	(0.31)
Burglary	76,323	-0.24^{**}	-0.49^{*}	-0.18
		(0.12)	(0.25)	(0.14)
Vandalism	88,765	-0.17^{**}	-0.54^{**}	-0.11
		(0.08)	(0.25)	(0.10)
Suggestive				
Disorderly conduct	6,315	-0.65	-1.77	0.50
		(0.63)	(1.21)	(0.74)
Sex offenses	30,103	-0.26	0.84	-0.27
		(0.25)	(0.83)	(0.24)
Precise nulls				
Larceny/theft	110,169	0.02	0.00	0.03^{***}
		(0.01)	(0.05)	(0.01)
Motor vehicle theft	83,626	0.07	0.00	0.09
		(0.09)	(0.27)	(0.10)
Assault (non-dom.)	64,538	-0.03	-0.21	0.02
		(0.09)	(0.26)	(0.09)
Family (non-violent)	12,632	1.49^{**}	0.99	2.05^{**}
		(0.60)	(1.08)	(0.86)
Assault (domestic)	77,543	-0.03	0.18	-0.02
		(0.12)	(0.28)	(0.14)
Trespass	9,395	-0.09	-0.49	0.05
		(0.38)	(0.93)	(0.39)
Robbery	63,513	0.01	0.01	-0.04
		(0.13)	(0.30)	(0.14)
Liquor law	5,417	0.51	0.19	0.40
		(0.36)	(0.80)	(0.31)
Parcel FE		Yes	Yes	Yes
Tract × year-month FE		Yes	Yes	Yes
Crime × era interaction		No	Yes	Yes

Notes. δ as in Table 2. Offense groups as in Table 2: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Crisis (2009–2013) and recovery (2014–2025) effects come from an era-interacted model with 2007–2008 as the base; the pooled column is the full-sample effect and need not lie between them. The priced offenses are shown at their main windows (homicide 10, burglary 8, narcotics 10, vandalism 6 months) and the other offenses at the window of their median estimate in Table 2 (the sixth-lowest of the twelve pooled estimates), held fixed across the three columns; identifying parcels (exposure varies within parcel) are counted at that window, whereas Table 2 counts them at twelve months. The crisis and recovery columns report per-era near–far levels, not deviations from the pooled column.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Supplementary Material

The *Online Appendix* contains additional tables, figures, and data details.